University of Pisa
Department of Physics
Master's Degree in Physics of Complex Systems

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and seeks a minimal stochastic transport model able to reproduce those statistics. As the work develops it will also examine how flying in competition changes the motion relative to solo flights.

This document reports the current state of the project. A first dataset has been acquired reproducibly—the archive of the FFVL *Coupe Fédérale de Distance*, in its paraglider and hang-glider disciplines—and Chapter 2 describes its acquisition, its coverage and the pre-processing that turns the raw logs into analysis-ready trajectories.

Chapter 3 measures the transport of the retained ensemble without segmenting a flight. Displacement from take-off turns out to measure the geometry of the launch site rather than the motion, and is retired as a source of exponents in favour of a within-flight estimator that annihilates any polynomial trend. Over 60 s to 2,000 s that estimator gives $\alpha = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders — strongly super-diffusive, and moved by no stratification the archive allows by more than its own uncertainty. Beyond the second moment, the spectrum of moments is straight, so the motion is monofractal and is *not* a Lévy walk; the propagator is nonetheless not Gaussian; and no leg duration is distinguished, which is a constraint on any segmentation attempted next. Chapter 4 sets out that segmentation and the modelling it enables, and is a plan rather than a result.

Contents

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from air currents in the atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is one of the settings in which *anomalous transport* arises. Let $\mathbf{r}(t)$ denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle|\mathbf{r}(t)|^2\rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. The continuous-time random walk (CTRW) is the standard modelling framework for such regimes [1]: the walker waits a random time τ, then makes a random displacement $\mathbf{x}$, and the joint density $\psi(\mathbf{x}, \tau)$ of the two fixes the transport. The Lévy walk is not a separate framework but the particular case in which the two are coupled through a constant speed v,

$$\psi(\mathbf{x}, \tau) \propto \delta(|\mathbf{x}| - v\tau)\, w(\tau), \tag{1.1}$$

with the direction drawn uniformly, so that each waiting time is spent travelling ballistically rather than standing still [2]. Anomalous transport of this kind has been reported in random-search strategies [3], in disordered media [4], and in turbulent flows [5].

The soaring cycle lies somewhere between the uncoupled and the fully coupled case, and it has three parts rather than two: a directed glide, then a search phase, then a localized climb. The glides are legs flown at a nearly constant speed, as in a Lévy walk; the search and the climb that end each leg are episodes of slow, localized motion, closer to the waits of an uncoupled CTRW. Whether those two phases can be collapsed into a single waiting phase is a modelling decision that the data have to support, and it is argued for where the quantitative model is built (Sec. 4.4). One feature is captured by neither limiting case: successive glides need not point in independent directions. The model this thesis works towards is therefore a persistent random walk in continuous time—one whose successive leg directions are correlated, with waiting times between legs.

Which of these ingredients the data actually require is settled by measurement, not assumed in advance, and the order of the measurement is the order of what it needs. Chapter 3 takes the whole trajectory, unsegmented, and asks what class of transport it is: an exponent, whether one exponent governs every moment, how long a heading survives, and whether any leg duration is distinguished. Chapter 4 then splits the flight into its phases, which is what the leg lengths and waiting times of the model require, and the minimal model is built on what both return. The persistence in particular is a working hypothesis about angular memory, which only the measured heading correlations can confirm or rule out. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [6]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [7], who analysed a large set of human soaring flights and found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$, i.e. $\langle |\mathbf{r}(t)|^2 \rangle \sim t^{2H}$, between diffusive $H = 1/2$ and ballistic $H = 1$, so that a value $1/2 < H < 1$ marks superdiffusive transport). That study also introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on it: it enlarges the dataset, revisits the cleaning and filtering that precede any analysis, and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues the following objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;
2. to clean and filter the raw GPS logs into analysis-ready trajectories, and to define, measure and analyze suitable transport observables—the test for anomalous transport is carried out on those observables, as soon as the trajectories exist;
3. to define a segmentation strategy for the transition–search–climb cycle and characterize the motion statistics within each phase;
4. to build a minimal stochastic model able to reproduce the empirical statistics;
5. to extend the dataset to competition flights and examine how flying in competition changes the motion relative to solo flights.

A first dataset—the archive of the *Coupe Fédérale de Distance* (CFD) of the French free-flight federation (FFVL), covering paraglider and hang-glider flights—has been assembled and verified and is described in Chapter 2; data collection continues and will be extended to further sources, in particular sailplanes and competition flights (Sec. 2.1). The dataset (Secs. 2.2–2.5) and its cleaning-and-filtering pipeline (Sec. 2.7) are described in Chapter 2, and the transport of the unsegmented trajectory is measured in Chapter 3; the segmentation, the modelling it enables (Chapter 4), and the extension to competition flights define the work ahead.

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. The statistics it quotes, and the tables and figures that carry them, are generated from the dataset by the analysis code rather than typed, so those numbers cannot drift from the data they describe. Chapter 2 follows the data end to end: its acquisition and the reproducibility of the pipeline (Sec. 2.1); the tracklog format and the flight catalog (Secs. 2.2–2.3); the glider categories and the coverage statistics (Secs. 2.4–2.5); the known data-quality caveats (Sec. 2.6); the pre-processing that turns the raw logs into analysis-ready trajectories (Sec. 2.7); a first characterization of the cleaned dataset (Sec. 2.8); and a one-map summary of the whole pipeline that closes the chapter (Sec. 2.9). Chapter 3 measures the transport of that ensemble on the whole trajectory, without segmenting a flight: the estimator and the exponent it returns (Secs. 3.2–3.4), and what a second moment cannot say (Sec. 3.5). Chapter 4 sets out the phase segmentation and the modelling it enables; none of it is implemented yet.

Throughout, the discussion keeps to the scientific and statistical reasoning behind each choice. Implementation-level detail (code modules, configuration values, sample sizes, computation times, reproduction commands) is collected in Implementation and Computational Details, from p. 82. That part is organised chapter by chapter to mirror the body, and the body links into it at each point where an implementation detail was trimmed out of the text, as “implementation details: Sec. X”.

Chapter 2

The dataset

This chapter follows the data end to end: how the raw archive is acquired, verified and kept reproducible (Sec. 2.1); what the records contain and what the collection looks like (Secs. 2.2–2.6); and the pre-processing that turns a raw tracklog into the analysis-ready trajectory the transport analysis consumes (Sec. 2.7), closed by a first characterization of the cleaned dataset (Sec. 2.8) and a one-map summary of the pipeline (Sec. 2.9).

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata—among them pilot, date, take-off and landing sites, scored distance, duration, and glider, with the full set of fields listed in Sec. 2.3—and, in most cases, the raw GPS tracklog in IGC format (the standard flight-recorder format, Sec. 2.2). A "season" is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large job: downloads are resumable and safe to interrupt, a response is accepted only once validated as real IGC content, and download requests are rate-limited so as not to load the server (implementation details: Sec. 2.1.1).

2.1.4 Reproducibility

The destination directory is not hard-coded: it is set per machine through an environment variable or a configuration file, of which the repository ships only a placeholder, so no machine-specific path is committed. Archiving the season XMLs verbatim keeps the full provenance (implementation details: Sec. 2.1.2).

2.1.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time (implementation details: Sec. 2.1.3). Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass the same content check applied at download — a readable file whose first record is an `A` header and which carries at least one `B` fix record (Sec. 2.2). These are the tracklogs actually fetched; they cover a subset of the flights that carry a track at all (Sec. 2.5).

2.1.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data are the main analysis target; the hang-glider data are collected in parallel and enable the cross-type comparison of Sec. 2.4 (the counts for both are given in Sec. 2.5). Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Sec. 2.4). A large sailplane source with a machine-readable interface is *WeGlide*, which exposes flight metadata and IGC tracks through an API. The default access is rate-limited (of order sixty requests per day), which rules out a bulk export of the archive; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. A comparable research-access request has been sent to XContest; as of 20 July 2026 neither WeGlide nor XContest has replied. All of these can be acquired with the same approach, each producing a catalog in the same format.

2.2 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [8]. Its layout is fixed by that standard, so a fix can be decoded from fixed character positions; every downloaded track is checked at acquisition to be IGC content — an `A` record followed by at least one `B` record (Sec. 2.1).

Each line is a *record* whose type is set by its first character – a code assigned by the standard, not chosen by the logger. The records relevant here are:

`A` the flight-recorder/manufacturer identifier (first line of the file);

`H` header records (flight date, pilot, glider, geodetic datum, . . .);

`I` the declaration of the optional extra fields appended to every fix;

`B` the GPS fixes – the body of the file.

A `B` record stores its fields at fixed character positions. Decoded against the standard, the example

`B 094329 4933044N 00005193E A 00095 00149`

reads as: UTC time `09:43:29`; latitude 49°33.044′ N and longitude 000°05.193′ E on the WGS84 datum; a fix-validity flag; and *two* altitudes in metres – the barometric (pressure) altitude (95 m) followed by the GNSS altitude (149 m).

The validity flag takes two values and concerns the *GNSS* solution only. `A` marks a fix whose GNSS altitude is usable; `V` marks a GNSS drop-out, or a fix for which the receiver could not resolve altitude, in which case the standard requires the GNSS-altitude field to be written as zero. A `V` flag therefore does not make the fix two-dimensional: the latitude and longitude stay usable and, on a logger with a working pressure sensor, so does the barometric altitude, which is the channel this analysis adopts (Sec. 2.7.1).

Decoding a `B` record from fixed character positions can succeed numerically while the result is not a valid encoding at all – a corrupted byte can still parse as a digit. The parser therefore checks, at decode time and independently of any flight-dynamics judgement, that the record is actually a legal encoding – a valid time, valid latitude/longitude fields, a valid hemisphere letter – dropping it otherwise, on the same footing as a record with the wrong length or a non-numeric field (implementation details: Sec. 2.3). This is a check of *format* validity, distinct from the *plausibility* checks of fix-level cleaning (Sec. 2.7.2, e.g. the inter-fix speed bound), which target fixes that decode validly but would be physically implausible given the rest of the trajectory.

The two altitude channels differ in reference and in noise; the analysis adopts the *barometric* one for the vertical dynamics, for reasons quantified in Sec. 2.7.1—a choice that matters because the vertical velocity derived from it feeds the phase segmentation. On older loggers one channel may be absent (recorded as zero).

The sequence of `B` records is thus a time-stamped three-dimensional trajectory. The sampling period is typically a few seconds, is not guaranteed to be uniform, and varies across recording devices – a point the analysis must handle explicitly.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	**Content / caveat**
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
`aile`	wing model name (raw XML field; French for “wing”)
`aile_class`	wing class, one system per discipline (Sec. 2.4)
flight type	the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, . . . (*caveat*: a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), with *one row per flight* across all seasons: the metadata above plus a `local_path` linking each flight to its track on disk and a `downloaded` flag. It is the working index of the project – a flight can be traced to its file, and back (through the identifier embedded in the file name) to its metadata and public page, without any external lookup. A smaller **season index** (`seasons_index.csv`), versioned in the repository, summarises each season by the number of flights declared, carrying a track, and downloaded; it is the source of the statistics below. Both tables are regenerated from the archived XMLs on demand (`build-catalog`), so the catalog itself need not be versioned.

2.4 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three broad categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio $\sim$8–11, tight turns), **hang gliders** (delta wing, spanning flexible framed wings and rigid wings; faster, glide ratio $\sim$ 10–15), and **sailplanes** (rigid aircraft; fastest, glide ratio $\gtrsim$25). They are, in effect, different aircraft—differing several-fold in cruise speed, glide ratio and turn radius—which is why the reference study [7] analyses each separately rather than pooling them.

In practice the category is set by the *data source*, since the public databases are organised by discipline. Because the FFVL competition is split by discipline (Sec. 2.1), the present dataset spans *both* glider types: the glider-type stratum has two levels, and the cross-type comparison of the reference study [7] is therefore already possible between paragliders and hang gliders. The third category, *sailplanes*, is collected by different portals and is not yet in hand (Sec. 2.1); until it is added the cross-type analysis is limited to the two FFVL disciplines.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems, collected in Table 2.2. Paragliders follow the EN/LTF certification ladder, which does track performance (higher classes fly faster and flatter) and is therefore the natural within-source stratification variable. The hang-glider scheme is *not* a ladder: the FAI classes are defined by the method of control, so they stratify the fleet by design rather than by performance, and any ordering imposed on them is an assumption to be checked rather than a property of the scheme.

The two ladders are not documented on equal terms, which governs how far each entry of Table 2.2 can be trusted. The FAI classes come from the Sporting Code [9], which the FAI publishes for free: anyone can download the current edition and check a definition against it. The EN classes come from a *European standard*, which is a different kind of document. Standards are drafted by the European Committee for Standardization (CEN) and then published and *sold* by the national standards bodies—AFNOR in France, DIN in Germany, BSI in the United Kingdom—as copyrighted documents, priced per copy at a few hundred euros. There is no public version: unlike a law or a sporting code, the authoritative text of EN 926-2 cannot be consulted without buying it, and it cannot be quoted here.

The EN entries of Table 2.2 are therefore condensed from Table 1 of the *final draft* of the standard, FprEN 926-2:2012, the version submitted to formal vote, which the CEN working group circulated publicly [10]. That draft is the closest thing to the source text that is openly available, and the class descriptions in it are unlikely to have moved in the published EN 926-2:2013, though that has not been checked against the published text. Nothing in this thesis depends on the wording: the class field is used as a stratification label, and what the analysis needs from it is that the labels are distinct and ordered, not what each one certifies.

Two caveats bear on any use of `aile_class`. First, the field must be normalised before it can serve as a stratum: normalisation here means merging the label variants of the same class onto one canonical value per discipline and, where possible, recovering a blank or placeholder entry from the wing model name. Second, the class certifies the wing, not the pilot. The

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. The EN ladder is ordered by increasing demand on the pilot, and that ordering is all the stratification needs from it; the FAI classes are *not* a ladder, being defined by the method of control. Sources: the FAI Sporting Code, Common Section 7 [9] (§§1.2, 1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the standard, on the caveat explained in the text.

Class	Notes
Paragliders — EN/LTF ladder	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI *Standard* sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI *Sport* sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

normalisation is done at the pre-processing stage (Sec. 2.7.4) (implementation details: Sec. 2.5); the second point no processing can remove, and it bounds what a class-stratified result can be taken to mean.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.6)

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.6)

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the "With GPS" column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, with the channel actually used recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries have placeholder dates, written as the sentinel value `0000-00-00` at acquisition. The sentinel makes them recognizable, and such flights are discarded rather than carried along: a flight whose date is unknown cannot be assigned to a season, and no downstream step needs it. The cut is negligible, and measured rather than assumed to be: 8 of the 203,343 paraglider entries carry the sentinel (0.0039 %) and 0 of the 9,259 hang-glider ones. The per-season breakdown is checked too, and not only the total, because a cut that is negligible archive-wide could still empty a single early season, which is where the smallest flight counts sit: the worst-affected season loses 0.17 % of its entries (2001-2002), so no season is materially thinned.

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude – with both altitude channels retained (implementation details: Sec. 2.7). The pipeline then proceeds in a fixed order:

(i) choose the altitude channel (Sec. 2.7.1);

(ii) clean individual fixes (Sec. 2.7.2);

(iii) trim the ground phases (Sec. 2.7.3);

(iv) filter out unfit whole flights (Sec. 2.7.4);

(v) convert to a local Cartesian frame (Sec. 2.7.5);

(vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);

(vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output is a tidy, analysis-ready representation in a growing `soaring.analysis` subpackage, written to three tables on the data disk: one row per fix, one per segment, one per flight *attempted*. What each file holds and how the disk is organised is set out in Sec. 2.2.

Steps (i)–(iv) act on the *raw geographic* coordinates, since the only geometric quantities they need (inter-fix speed and spatial extent) are scalar distances, which the great-circle (haversine) formula of Eq. (2.1) gives directly from latitude and longitude, with no coordinate conversion. The conversion (v) to the metric East–North–Up frame (Sec. 2.7.5) is then done on the cleaned, trimmed track, with the origin at the first fix that survives the trimming, so the tangent-plane frame is built from good data.

Steps (vi)–(vii) come last for a different reason: unlike the scalar checks above, they produce *vector* quantities—the components of the 3D position, velocity and acceleration—which require Cartesian axes to be expressed in, and therefore cannot run before the frame of step (v) exists.

Every step below is governed by a handful of numerical thresholds. They are collected once, here, in Table 2.5, so that the value in force at any point of the chapter can be read off a single page instead of being hunted through the text; each subsection then argues for its own entries rather than restating them. All of them are *working values*: fixed a priori, on physical scales rather than on the transport observables, and audited a posteriori on the fraction of data each rule removes (Sec. 2.7.2). The table is typeset from the configuration file itself, so a value changed there changes here.

Table 2.5. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; every value is generated from `configs/preprocessing.yaml`, and the per-parameter rationale is given in the section listed. Config keys and implementation notes: Sec. 2.7.2 onward.

Symbol	Value	What it controls
Fix-level cleaning (Sec. 2.7.2) — removes corrupt fixes, keeps every genuine one		
k	5	robust-σ multiplier of the Hampel test, Eq. (2.3)
w	20 s	half-width of the Hampel window
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	$45\,\mathrm{m\,s^{-1}}$ (para), $55\,\mathrm{m\,s^{-1}}$ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$\lvert v_z\rvert^{\max}$	$13\,\mathrm{m\,s^{-1}}$	absolute climb/sink bound
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude

continued on the next page

Table 2.5 (continued)

Symbol	**Value**	**What it controls**
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration of a run before it counts as frozen
—	95 %	smallest share of a flight's fixes carrying a non-zero pressure altitude for the barometric channel to count as present (Sec. 2.7.1); below it the flight falls back to GNSS
—	30 m	smallest whole-flight barometric range below which the channel is treated as absent (Sec. 2.7.1)
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3) — keeps only the airborne stretch		
v_0	$5\,\mathrm{m\,s^{-1}}$	ground-speed threshold separating ground from flight
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		
—	40 min	minimum airborne duration
—	20 km	minimum flown path length
—	16 h	maximum duration, above which the record is not a single flight
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)
—	$v_{xy}^{\max}$ (above)	reach bound: the extent of the flight may not exceed what that same speed would cover in its duration (Sec. 2.7.4)
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		
$g_{\max}$	$\min\big(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\Delta t)\big)$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained *segment* may have had to interpolate
—	90 s	shortest segment kept: below this a segment cannot carry the smoothing window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s	correlation time setting the window, on all three channels. Per-channel keys exist—E/N, and the altitude split by `alt_source`—for the noisy GNSS minority, and all currently hold this same value, because the spectral knees of the three coincide (Sec. 2.7.7)

continued on the next page

Table 2.5 (continued)

Symbol	Value	What it controls
—	5 samples	floor on the window, one sample more than a cubic determines. It is this floor and not τ_c that fixes the smoothing scale for every logger slower than 1 Hz: in seconds the window is $\max(\tau_c,\, 5\,\Delta t)$

2.7.1 Choice of altitude channel

The analysis adopts the *barometric* altitude rather than the GNSS altitude for the vertical dynamics. The reason is one of signal quality in the frequency band that matters. The IGC format writes both altitude channels as whole metres, so the resolution of either is fixed at 1 m by the file itself; what distinguishes the barometric channel is that it carries very little high-frequency noise. Its errors are slow: a constant offset from the ICAO reference, tens of metres up to order 100 m (visible in Fig. 2.1a as the gap between the two channels; most of that gap is the barometric offset, though the GNSS altitude contributes a smaller bias of its own), and a drift that already reaches tens of metres within the half-hour window of Fig. 2.1b, and can grow further over a multi-hour flight. Both have the same origin: the barometric altitude is a pressure reading converted through the ICAO standard atmosphere, so it is offset by the difference between the day's actual sea-level pressure and the ICAO reference of 1,013.25 hPa, and it drifts as that pressure changes over the hours of the flight and as the real temperature profile departs from the standard one. Drift of the sensor itself is a third and much smaller contribution.

Slow errors are annihilated by the differencing that every dynamical quantity involves (a vertical velocity, a displacement increment, the autocorrelation of a velocity). The GNSS altitude, by contrast, is referenced to a fixed geometric surface (the WGS84 ellipsoid), so it carries no atmospheric offset, but it is vertically noisier: the satellite geometry resolves the vertical coordinate less well than the horizontal one[1] (the *vertical dilution of precision*): the GPS performance standard commits to 8 m horizontal against 13 m vertical error at 95 % globally, and 15 m against 33 m at the worst site [11], and *multipath*—the direct signal arriving together with replicas delayed by reflection off terrain or the pilot's own equipment—biases the range on timescales of minutes rather than adding independent noise from fix to fix [12].

This is why free-flight variometers derive the climb rate from the pressure sensor rather than from GNSS.

Empirical confirmation. The data confirm this picture, with one qualification. Figure 2.1 compares the two channels over a subsample sized for a robust spectral estimate (Appendix 2.A): 3,000 flights per discipline, a size chosen on the stability of the median spectrum itself and justified in Sec. 2.7.1 (implementation details: Sec. 2.7.1). At low frequency the two agree, tracking the real climbs and glides. The high-frequency behaviour is where they differ. For the *typical* flight the two are comparable: both approach the same flat high-frequency plateau—the level at which the spectra in panel c flatten towards the Nyquist frequency—set by the metre resolution to which the IGC format rounds *both* channels, so the median spectra (panel c, solid lines) nearly coincide.

What separates the channels is the *spread* across flights, the shaded bands. The barometric band stays tight (that channel is uniformly clean), whereas the GNSS band fans out by one to two orders of magnitude towards the Nyquist frequency: a substantial minority of flights carry

[1] The asymmetry is geometric. A receiver sees satellites only above the horizon, so the ranging directions bracket it in the horizontal plane but all arrive from the same side vertically; the vertical coordinate is correspondingly the worst-determined one. This is what the *vertical dilution of precision* measures.

large GNSS high-frequency noise, from receiver tracking noise amplified by a poor vertical dilution of precision—not from multipath, whose error is correlated over minutes and so raises the low-frequency end of the spectrum rather than the floor—while most do not. Since the vertical velocity is a finite difference, which amplifies exactly that band (differencing scales a spectral component of frequency f by $2\sin(\pi f\,\Delta t)/\Delta t$, an amplification that grows with f: it acts as a high-pass filter), and since one cannot know in advance which flights have poor GNSS reception, the uniformly clean barometric channel is the safe choice. This *departs from the reference study* [7], whose pipeline derives the vertical coordinate and velocity from the GNSS altitude; the divergence is deliberate, and one of the points re-examined here.

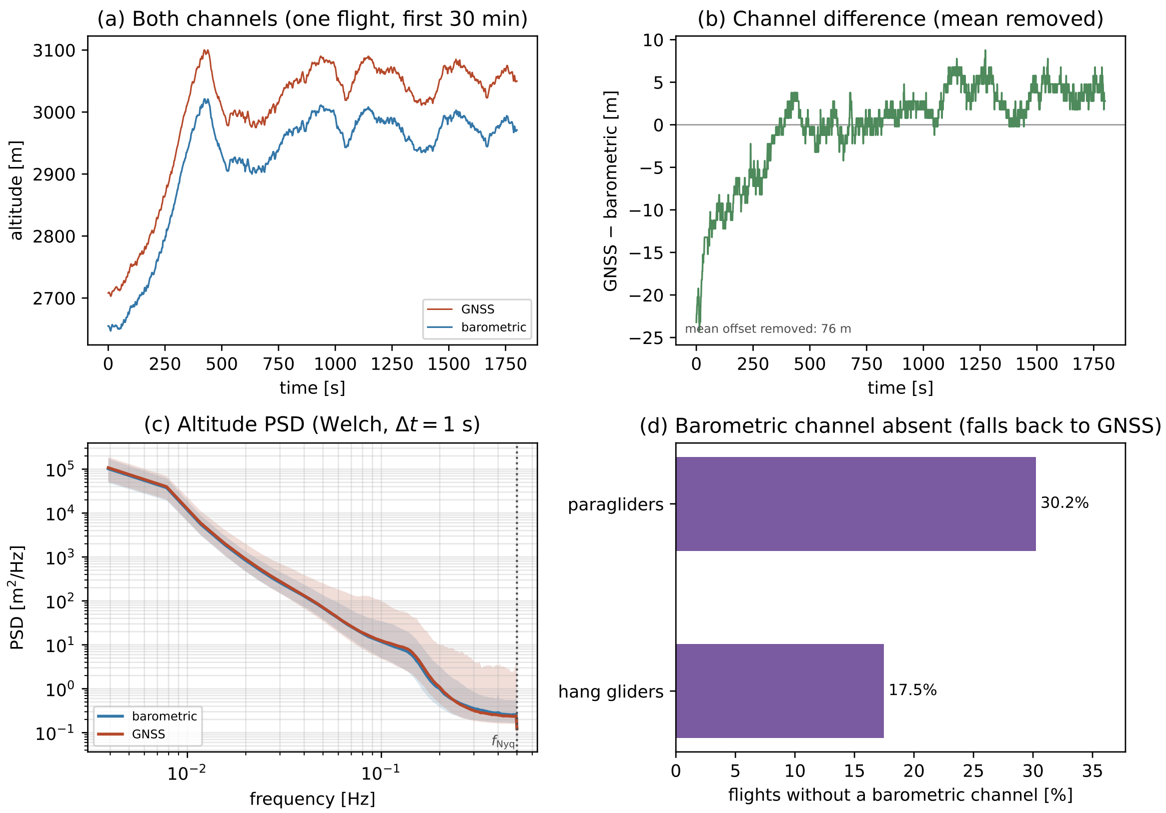

Figure 2.1. Barometric vs GNSS altitude noise. (a) One representative flight over a 30 min window, the two raw channels on a shared axis. The vertical gap between the curves is the offset between their reference surfaces (pressure vs. ellipsoidal). (b) Their difference, mean removed: the offset is not constant but drifts by tens of metres already within this 30 min window. (c) Per-frequency *median* Welch spectrum of each channel (Appendix 2.A), pooled over the two disciplines, with the 10 to 90 percentile band across flights; the medians nearly coincide, while the GNSS band fans out towards the Nyquist frequency $f_{\mathrm{Nyq}} = 1/(2\Delta t)$ (dotted). The median is used because the per-flight spectra are heavy-tailed (Sec. 2.7.1). (d) Fraction of flights whose barometric channel is absent, per discipline. (implementation details: Sec. 2.7.1)

Several practical points follow.

One channel per flight. A trajectory uses a single altitude channel end-to-end; the barometric and GNSS values are never spliced within a flight, since a mid-flight switch between two channels offset by tens of metres would inject a spurious jump.

Source recorded. Which channel a flight uses is recorded as a per-flight attribute of the processed dataset, so the two source groups remain separable in every downstream analysis;

the raw IGC files are never modified (implementation details: Sec. 2.7.1).

Missing fixes. A flight is used with its barometric channel when it has one, and the few individual fixes with a missing barometric value are dropped rather than back-filled from GNSS. Only 4.2 % of barometric paraglider flights (2.1 % of hang gliders) miss any barometric value at all, with a median of 3 (10) missing fixes among them—against $\sim 10^4$ fixes in a typical track—and a worst case of 1,397 (455), bounded by the presence threshold itself: a flight whose channel covers less than 95 % of its fixes is not treated as barometric at all (Sec. 2.7.1). Whether the dropped fixes are scattered or consecutive, the hole they leave is handled downstream like any other gap (Sec. 2.7.6).

Absent or stuck sensor: the GNSS fallback. A logger with *no* pressure sensor writes the whole barometric channel as zero (the presence is essentially all-or-nothing, not a scattering of gaps), so a flight counts as barometric only when the channel covers at least 95 % of its fixes—high on purpose, since a partially filled channel is worse for the vertical analysis than a uniformly sourced GNSS one (Sec. 2.7.1); such a flight falls back to the GNSS altitude, unfiltered – a non-negligible minority of flights (Figure 2.1d). The mere presence of a non-zero barometric channel is not enough: a stuck sensor writes a constant non-zero value and would feed the segmentation a vertical velocity of identically zero, so a barometric channel whose total range over the flight stays below 30 m is treated as absent, and the flight falls back to GNSS. The fallback rate is measured on the full archive: 30.2 % of the 186,025 paraglider and 17.5 % of the 6,716 hang-glider flights scanned carry no barometric channel (implementation details: Sec. 2.7.1).

The liveness test as written catches only a sensor that is dead for the *whole* recording, and it should be read as that and no more. The statistic is a single whole-flight max – min, so a barometer that records normally for the first ten minutes and then freezes keeps a range of hundreds of metres, is judged alive, is adopted, and delivers a vertical velocity of identically zero over most of the flight—which is precisely the outcome the test is here to prevent. Nothing downstream closes the gap: the frozen-lock rule of Sec. 2.7.2 needs the *position* to be frozen, and a cruising wing never trips it; the flight-level altitude-activity cut consumes the same extremum difference and inherits the same blindness. A test that did close it would be a *local* one—a moving-window range, or a longest-constant-run bound—and that is a new threshold with its own failure mode against a wing genuinely holding altitude, so it is not adopted here on the strength of an argument alone. What the present rule guarantees is stated exactly: a channel that is constant throughout is refused, and one that dies partway is not.

Mixed altitude sources. The mix is acceptable, for two reasons. The *horizontal*, two-dimensional transport analysis—the main subject of this work, of which the MSD and its scaling are one observable among several—does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables; there the two source groups are kept distinct through the per-flight altitude-source record introduced above. The compatibility check therefore has to be run on the segmentation itself rather than on the altitude statistics: both groups are segmented with the same rules and compared, per discipline, on the fraction of airborne time each phase accounts for. The two estimates of each fraction are compatible when they agree within their uncertainties; a disagreement in any phase is the signal that the noisier channel is biasing the phase assignment. One number per phase per group is enough for it: the fraction of time per phase is what the vertical channel feeds (Sec. 2.8). If the noisier GNSS-only flights were to bias a vertical statistic, that analysis is restricted to the barometric flights – still the large majority – while the horizontal analyses keep the full sample. Forcing GNSS on *all* flights, to avoid the mix, would be the wrong trade: it would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

2.7.2 Fix-level cleaning

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i), before the Cartesian conversion (Sec. 2.7.5), so distances between fixes cannot be Euclidean; they are great-circle (haversine) distances [13] on a sphere of mean radius $R_\oplus = 6{,}371\,\mathrm{km}$. For two consecutive fixes,

$$\Delta s_i = 2R_\oplus \arcsin\sqrt{\sin^2\frac{\varphi_{i+1}-\varphi_i}{2} + \cos\varphi_i\cos\varphi_{i+1}\,\sin^2\frac{\lambda_{i+1}-\lambda_i}{2}}\,, \tag{2.1}$$

from which the horizontal speed between the fixes is $v_{xy,i} = \Delta s_i/(t_{i+1}-t_i)$ and, from the barometric altitude, the vertical speed is $v_{z,i} = (h_{i+1}-h_i)/(t_{i+1}-t_i)$. The same distance also yields two whole-flight quantities used below: the *path length* $L = \sum_i \Delta s_i$, the total distance flown along the track, and the *extent* $D = \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i)$, the farthest any fix lies from the first. The spherical approximation is ample here. Replacing the WGS84 ellipsoid by a sphere perturbs an inter-fix distance by at most a fraction of the order of the ellipsoid's flattening, $\lesssim 0.3\,\%$[2]—about 3 cm on a 10 m step—whereas the horizontal position error of consumer GNSS receivers is of the order of metres (the GPS performance standard commits to 8 m at 95 % globally [11]); the correction is therefore far below the noise on the very same quantity. Working on the raw coordinates also avoids converting them before the trajectory has been cleaned.

Design principle. Corrupt or physically impossible fixes must be removed without discarding the flight around them. At the same time, every removed or altered fix modifies the trajectory whose statistics are about to be measured, and an over-aggressive cleaning biases the result as much as dirty data would. The cleaning therefore assumes as little as possible about the motion: only smoothness at the sampling scale, never a model of the transport itself.

The two possible errors have different costs. A spurious fix that survives the tests below lies at worst tens of metres from the true track—at most the paraglider speed bound of Table 2.6 times the native step, $\approx 45\,\mathrm{m}$ at 1 Hz—and typically far closer, within the absolute floor $\varepsilon_{\min}$ of the outlier test defined below in Eq. (2.3): a displacement the smoothing absorbs. A genuine fix removed from a slow, horizontally localized phase (a search, a climb) erases a real trapping event, and nothing downstream can restore it. The cleaning therefore defaults to keeping the data: a fix is removed only on positive evidence of a defect, and no detector deletes a fix that still carries a usable horizontal position (at most, a corrupt *altitude* channel is marked missing and restored at resampling, Sec. 2.7.6; a corrupt horizontal position is never reconstructed—the fix carrying it is deleted outright).

Three classes of defect. Past the format-validity check made at decode time (Sec. 2.2), which rejects any record that is not a legal encoding and is already implemented and unit-tested, a surviving fix can still be wrong in one of three ways, summarized in Table 2.6.

[2] An arc length on a curved surface is proportional to the radius of curvature used to compute it, so the relative error made by using the mean radius $R_\oplus$ in place of the ellipsoid's local radius is of the order by which the two differ. On WGS84 the radii of curvature range from the polar $b^2/a = 6{,}335.4\,\mathrm{km}$ to the equatorial $a^2/b = 6{,}399.6\,\mathrm{km}$, a spread of the order of the flattening $f = (a-b)/a = 1/298.257 \approx 3.4 \times 10^{-3}$, hence the 0.3 % quoted here [14].

A fix *off the trend* – a GPS position spike, or a jump out and back within one step – is found by a robust[3] local-outlier test, the *Hampel identifier* [15,16]. Let the window of fix k be the set of fixes within $\pm w$ of t_k, and let $\tilde{\varphi}_k = \mathrm{med}\{\varphi_j : |t_j - t_k| \le w\}$ and $\tilde{\lambda}_k = \mathrm{med}\{\lambda_j : |t_j - t_k| \le w\}$ be the component-wise window medians. The residual of fix k is its great-circle distance from that median point,

$$r_k = \Delta s\big((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)\big), \tag{2.2}$$

with Δs as in Eq. (2.1), and the robust local scale of the residuals is $\mathrm{MAD}_k = \mathrm{med}\{r_j : |t_j - t_k| \le w\}$. The fix is flagged when

$$r_k > \max(k\,\sigma_k,\ \varepsilon_{\min}), \qquad \sigma_k = 1.4826\,\mathrm{MAD}_k, \tag{2.3}$$

Here σ_k is the normalized MAD, the median absolute deviation rescaled to estimate a Gaussian standard deviation [17]: the flag requires a departure of more than k robust standard deviations *and* more than an absolute floor $\varepsilon_{\min}$ of a few times the GPS noise (working values in Table 2.5; step-by-step definition and rationale in Sec. 2.7.2). A flag does not delete. What deletes is the absolute v_{xy} bound: a fix is removed when it is unreachable from the last accepted fix at that bound *and* the removal of the block it belongs to lets the track rejoin within it. The identifier's role is to flag and to attribute—to say which fix of a neighbourhood departs from it—and to supply the per-flight anomaly count; the reason it cannot also be required is argued below.

A genuine but under-sampled tight turn is kept either way. It is what a naive speed-only rule would get wrong and what the identifier was originally there to protect. The Hampel test measures a departure from the local median track, and a tight thermalling circle produces exactly such a departure: at a 10 s cadence a circle of a few tens of metres in diameter puts consecutive fixes on opposite sides of their own window median, and the residual clears $\varepsilon_{\min}$ on geometry alone. Those fixes are the trapping events the transport analysis exists to measure. What separates them from a spike is not the flag but the speed: a circling wing flies at an ordinary one, whereas a position spike of the same size implies an inter-fix speed no aircraft in this dataset can sustain. The protection is therefore already in the bound that does the deleting, which is where it came from.

The rule was originally a conjunction—flagged *and* impossible—and running it over the archive exposed a limit in the *flag* that is the reason it no longer is. The Hampel identifier has a breakdown point of 50 %: up to half a window may be arbitrarily wrong without displacing the median, which is what makes it robust—and past that half the median moves onto the corruption instead. The archive contains the case. Some loggers write the null position $(0, 0)$ for scattered runs of a few seconds each; where several such runs fall inside one 20 s window they exceed half of it, the median follows them, the corrupt fixes score a residual of *zero* and go unflagged, and it is the good fixes around them that are flagged instead. Requiring the flag there does not make the rule conservative, it makes it blind: on one such flight the identifier caught 17 of 99 null fixes, and every survivor left a step of some 5,000 km in the record.

The rule adopted is therefore the *impossibility gate on its own.* A fix is deleted when it is unreachable from the last accepted fix at the v_{xy} bound *and* the removal of the block it belongs to lets the track rejoin within that bound. Both halves of the argument above survive the change. A genuine tight circle is still never a candidate, because it flies at an ordinary speed—so the protection the flag was there to give is given by the bound itself, which is where it came from in the first place. And the attribution the flag supplied—which endpoint of an impossible step is the wrong one—is supplied instead by the rejoining test, which names the block whose removal restores the trend. What the flag stops being is a *necessary* condition for

[3] *Robust* has its technical sense here: the median and the median absolute deviation have a breakdown point of 50 %, so up to half the fixes in the window may be arbitrarily wrong without displacing the estimate. A mean and a standard deviation have breakdown point zero—a single spike drags both, inflating the scale by just enough to hide itself.

deletion; what it remains is the per-fix record and the per-flight anomaly count. The change bites only where the identifier is outside its own stated range of validity, and that is precisely where the previous rule failed.

A fix *on the trend* agrees with its neighbours yet is still wrong as a record – a duplicated or backward timestamp, or a run of frozen positions from a lost GNSS lock; its residual is near zero, so it escapes the outlier test and needs a structural rule instead.

A value *out of range* concerns the vertical channel: an impossible altitude or climb rate is caught by an absolute bound that needs no context.

An unreturned vertical step. The $|v_z|$ rule recognises two shapes and there is a third. An *out-and-back* spike is censored; a coherent *run* is a sustained manoeuvre and is counted and left (Sec. 2.7.2). A single super-threshold step that is never undone—a barometric re-reference, or one corrupt record—is neither. It was therefore censored by nothing *and counted by nothing*, and since it is a discontinuity rather than an outlier the smoothing stage differentiated it: the written table carried vertical velocities of up to 5,117 $\mathrm{m\,s^{-1}}$. It is not a curiosity. On a sample of the archive, 8.3 % of paraglider flights and 6.2 % of hang-glider flights carry at least one.

What is done here is the part that is not a judgement: the shape is detected and counted, per flight, as `n_alt_level_shift`, so that it is visible and auditable rather than silently present. What to *do* about it is left open, deliberately. The three candidate treatments—split the flight at the step, on the same footing as an impossible horizontal step; re-reference the altitude after it, restoring continuity; or invalidate a window around it and let the resampling bridge it—each change a rule argued elsewhere in this chapter, and none of them follows from the specification. The vertical channel feeds the phase segmentation of Chapter 4 and not the transport measurement of Sec. 3.1, which is horizontal, so the choice belongs with that analysis rather than ahead of it.

The vertical channel is treated differently from the horizontal one on purpose, and the asymmetry is worth stating, because the obvious symmetric design—a Hampel test on z too, corroborated by the climb-rate bound—is the wrong one here. Three things differ.

First, an absolute bound is *sufficient* on z and insufficient on xy. The vertical envelope is narrow and known: no wing in this dataset sustains $|v_z| > 13\,\mathrm{m\,s^{-1}}$, and no altitude outside $[-100, 6000]$ m is sensor-plausible, so a spike in altitude is almost always out of range on its own. Horizontally there is no such luck: a 50 m spike is an impossible 50 $\mathrm{m\,s^{-1}}$ at a 1 s cadence but an unremarkable 5 $\mathrm{m\,s^{-1}}$ at 10 s, so the same defect passes or fails the bound depending on the logger. The horizontal case needs a local, scale-free test precisely because its absolute bound is not decisive; the vertical case does not.

Second, a Hampel test on z would flag the physics. The residual from a local median is largest where the signal turns fastest, and on the altitude channel that means thermal entries and exits—where v_z swings by several $\mathrm{m\,s^{-1}}$ within seconds (Sec. 2.7.6). Those transitions are what the phase segmentation is built on. A horizontal circle, by contrast, can be told from a horizontal spike by its speed, so the local test there has a corroborating witness that the vertical one lacks: the quantity that would have to arbitrate a suspected altitude spike *is* the climb rate, which is the quantity under suspicion.

Third, the costs are not symmetric. A false vertical flag marks one altitude missing and it is interpolated back at resampling; a false horizontal flag would delete a genuine sample of the path. But an interpolation laid across a thermal entry erases the transition rather than restoring it, so a cheap-looking vertical false positive is expensive exactly where it is most likely. Keeping the vertical rule to a bound that only fires on the physically impossible avoids that failure mode altogether.

The thresholds are set in two ways. The absolute bounds are placed on data: the v_{xy} and $|v_z|$ bounds sit where the per-fix distributions of Fig. 2.2 stop decaying, and the altitude window is a sensor-plausibility band audited on the same distributions (lower bound negative for the reference-offset reason in the caption of Table 2.6). The remaining parameters—those of the Hampel test and of the frozen-lock rule—are working values fixed a priori. All of them, with their values, are in Table 2.5; their reasonableness is then checked a posteriori on the fraction of fixes each rule removes (paragraph *Validating the cleaning* below).

Removal policy. A speed bound alone cannot attribute a defect: the speed belongs to the segment *between* two fixes and does not identify which endpoint is wrong. The attribution comes from the bounded-removal test—the block whose removal restores the trend is the corrupt one—so an isolated spike is removed without touching its neighbours, and a short burst is removed as one block, the smallest that works. The local-outlier residual, being a property of a single fix, says the same thing where it is valid, and is recorded for that reason; it is not what decides.

What happens to a defective fix depends on *which* of its values is corrupt. Two cases arise—a corrupt altitude, and a corrupt horizontal position or timestamp—and they are handled asymmetrically. The asymmetry follows from a single criterion: keep every genuine horizontal sample, and never hide a hole in the horizontal record. In either case the corrupt value itself is simply discarded—nothing is estimated in its place at this stage; whatever needs reconstructing is reconstructed once, at the resampling step (Sec. 2.7.6).

If the corrupt value is the *altitude*, the fix is kept and only its altitude is marked missing. In practice the fix keeps its row in the trajectory table and its altitude entry is set to a missing value, with a per-fix boolean recording that the value was removed by the cleaning rather than never written by the logger; the two are indistinguishable downstream otherwise, and only the first is a statement about data quality. The horizontal position and the timestamp are intact, so the fix still samples the path—exactly the information the transport analysis measures—and discarding it would throw that genuine sample away. The missing altitude is restored by interpolation at resampling, where the filled point is flagged as interpolated in turn; the barometric channel is smooth enough that this costs less than the sensor noise (Sec. 2.7.1).

If the corrupt value is the *horizontal position or the time*, the whole fix is deleted. One could imagine the symmetric treatment—keep the fix and mark the horizontal position missing—but that would be dangerous: the resampling step decides whether to bridge a hole or to split the flight by looking at the *time gap* between surviving fixes. Ten consecutive kept-but-positionless fixes would look like ten samples of a continuously recorded stretch; the gap rule would never fire, and the resampler would lay an invented straight line across an interval that may hold a full thermalling circle. Deleting the fixes leaves the hole visible, and the gap rule then does its job: a short hole is bridged, a long one splits the flight (Sec. 2.7.6). The altitude sample lost with the deleted fix is the cheap part, for the same smoothness reason as above.

The IGC validity flag needs no rule of its own here: a `V` flag certifies a degraded GNSS altitude (Sec. 2.2), so on a barometric flight the fix is untouched, and on a GNSS-fallback flight it reduces to the corrupt-altitude case above.

An impossible step is a boundary, never flown path. The two actions above—delete the block, or split at the step—cover between them what the scan can resolve. What must hold whatever it resolves is a weaker statement, and it is the one the rest of the pipeline leans on: *no step between two surviving fixes may break the v_{xy} bound.* Where a block is kept because nothing could rejoin it, the step by which the track returns from that block is itself impossible, and left alone it enters a segment as flown motion—in the archive, a jump of tens of kilometres inside one second, carried thereafter into every average taken over that segment. Such a step is a transition whose course is unknown, which is what a long gap is, so it is made a segment boundary on the same footing (Sec. 2.7.6). It is stated as a property of the output rather than as one more case of the scan, so that it holds whatever the scan enumerated. How often it acts is counted per flight, as `n_boundaried`: a high count would locate the fault in the scan rather than in the data.

A corrupt block that opens the record. The blind spot is real and it matters more than its rarity suggests. A scan that carries the last accepted fix as an anchor never questions its own first anchor; and the first fix of the trimmed track becomes the origin of the local frame (Sec. 2.7.5), so a corrupt one displaces every coordinate of the flight rather than one of them, and the error does not average out with length of record. Five paraglider flights left the first full-archive run sitting some 4,500 km from their own origin. Since the ensemble MSD is an average of $|\mathbf{r}(t)|^2$ over flights, five records out of 156,017 raised it by seven orders of magnitude at the shortest lags.

No fix-level rule attributes it, and that absence is a decision rather than an omission. An impossible step is a statement about a *pair* of fixes and says nothing about which of the two is the wrong one. Away from the ends of a record the bounded-removal test settles the question by evidence—it asks which removal restores the trend—and at the very first fix there is no trend behind the step to ask. Two rules were nevertheless written, and both were wrong, in opposite directions. One fired on the first impossible step, which accuses the earlier end always: applied to an ordinary position spike three seconds into a record it deleted the good fixes and promoted the spike to origin, the same failure with the sign reversed. The other fired on a discontinuity the scan itself declares near the start, which sounds like evidence—a discontinuity survives only where no bounded removal after the step restored the trend—but the split is placed at the first fix of the block that could not be rejoined, and that block lies *after* the step whenever the corruption simply outlasts w. A 50 s corrupt run five seconds in therefore deleted five good fixes, kept fifty bad ones, and left the origin 5,039 km out.

So the discontinuity is left as a discontinuity, and the guarantee is taken one stage later, where it can be stated without attributing blame to either end: a record out of reach of its own first fix is refused whole (Sec. 2.7.4). That bound never depended on a fix-level rule. What such a rule would have bought is the recovery of two flights in 156,017, against a failure mode now demonstrated twice—and the invariant the whole of Chapter 3 rests on, $|\mathbf{r}(t)| \le v_{xy}^{\max} t$, holds either way, checked as such on the written tables (Sec. 2.2).

Frozen-position runs (GNSS lock loss). When the receiver loses its GNSS lock, many loggers repeat the last position for tens of seconds while the clock keeps running. Such a run passes every test above: horizontal speed zero, altitude in range, valid timestamps. If kept, it enters the analysis as a long *waiting time* and contaminates the tail of the waiting-time density $\psi(\tau)$, the distribution of the durations of the slow, localized phases. If the detector is too aggressive, however, it splits tight thermalling climbs, which are also slow and *horizontally* localized, and destroys real trapping events. The two errors enter the same tail with the same weight: a kept freeze inserts a false long wait, a cut climb removes a real one. They differ in what can be repaired downstream: a kept run with a flat barometric trace is still flagged suspect and swept in the $\psi(\tau)$ sensitivity checks (Sec. 2.7.3), while an erased trapping event is unrecoverable. A run is therefore cut only when three independent signatures agree, combined as in Eq. (2.4) below:

(i) *Spatial diameter.* A lost lock collapses the whole run into the logger's quantisation: the run's *bounding diameter* – the largest great-circle distance between any two of its fixes – stays below δ_{xy}, a tolerance of this rule set a few times above the GPS noise. It is written δ_{xy}, and not ε, to keep it apart from the Hampel floor $\varepsilon_{\min}$: the two carry the same value at present but answer different questions—one is the smallest departure from a local median worth calling an outlier, the other the largest spread a run may show and still count as motionless (Table 2.5). The run is not a fixed-length window: a candidate run is the maximal stretch of consecutive fixes whose bounding diameter stays below δ_{xy}, and the rule then asks how long that stretch lasts. A circling climb instead traces a disc of diameter $2R \approx 30\,\mathrm{m}$ to $100\,\mathrm{m}$, larger than δ_{xy} by at least a factor of two (quantified below), and clears the test on the diameter alone.

(ii) *An independent witness – the barometric altitude.* A GNSS lock-loss freezes only the GNSS position; the pressure sensor is a *separate* instrument. A wing that is really thermalling keeps gaining height on the barometric channel, whereas a true freeze shows a flat barometric trace. A horizontally motionless run whose barometric altitude is still changing at a plausible rate is a (tight) climb, and is *kept*. One signature, and only one, may overrule this witness: coordinates repeated *byte-identically*, defined in (iii) (the structure of Eq. (2.4) makes this explicit: on a barometric flight the witness is a flat barometer *or* a byte-identical repeat, so nothing else can cut a run whose barometer climbs).

(iii) *Recorder declarations.* Two signatures declare a defect directly, and they carry different weight. A *byte-identical* repeat of the coordinates – the latitude and longitude fields of consecutive `B` records written character for character the same – is conclusive on its own: no moving receiver rewrites the very same bits, so the positions are invented whatever the barometer says. The `V` flag and the zeroed GNSS altitude are weaker: they certify a degraded *GNSS altitude*, not a frozen horizontal fix (Sec. 2.2), so they decide only where no barometric witness exists, i.e. on a GNSS-fallback flight.

A run is classified as a loss of signal only when it is spatially collapsed, *witnessed*, and sufficiently long,

$$\underbrace{\mathrm{diam(run)} < \delta_{xy}}_{\text{(i) collapsed}} \;\wedge\; \underbrace{W}_{\text{witness (ii)/(iii)}} \;\wedge\; \underbrace{\Delta t_{\mathrm{run}} \geq \tau_{\mathrm{freeze}}}_{\text{duration}},$$
$$W = \begin{cases} |\Delta z_{\mathrm{baro}}| < \delta_z \vee \text{byte-identical} & \text{barometric flight,} \\ \mathtt{V}/\text{zero-alt} \vee \text{byte-identical} & \text{GNSS-fallback flight.} \end{cases} \tag{2.4}$$

The run is then marked as a gap and the trajectory is *split* there rather than interpolated across, so that no motion is invented over an interval whose true path is unknown; the segments keep their parent's origin and clock (Sec. 2.7.6).

The witness W is hierarchical. On a barometric flight none of the recorder's own declarations—the `V` flag, a zeroed GNSS altitude—can overrule a climbing barometer; the one exception is the byte-identical repeat, which is conclusive on its own. Coordinates repeated bit for bit are invented positions even when the height gain is real, so such a run is cut even while the barometric altitude climbs.

Quantitatively, a real thermalling climb fails the cut on two independent instruments at once. Geometrically, a circling wing turns at $R = v^2/(g \tan\phi)$[4]: the tightest paraglider circles ($v \approx 10\,\mathrm{m\,s^{-1}}$ at a bank of $\phi \approx 35°$, ordinary values for tight thermalling) give $2R \approx 30\,\mathrm{m}$, and hang gliders, which fly faster, turn wider ($v \approx 13\,\mathrm{m\,s^{-1}}$ at the same bank gives $2R \gtrsim 50\,\mathrm{m}$). Against the working value of δ_{xy} (Table 2.5), test (i) therefore misses by a factor of at least 2 for the tightest paraglider circles and $\gtrsim 3$ for hang gliders, more for anything wider—ratios to be rescaled if δ_{xy} moves. Barometrically, a climb of order $1\,\mathrm{m\,s^{-1}}$ sustained over $\tau_{\mathrm{freeze}} = 60\,\mathrm{s}$

[4] The coordinated-turn relation: in a steady banked turn the lift L balances gravity vertically, $L\cos\phi = mg$, while its horizontal component provides the centripetal force, $L \sin\phi = mv^2/R$; dividing the two gives $\tan\phi = v^2/(gR)$, i.e. $R = v^2/(g\tan\phi)$.

moves the barometer by $\gtrsim 60\,\mathrm{m}$ against the flatness tolerance δ_z, which is set well below that scale (Table 2.5).

The duration threshold itself is set on the motion's timescale: τ_{freeze} is about two thermalling periods and three times the 20 s gap cap of Sec. 2.7.6, so a run long enough to be cut spans at least two full circles of any genuine climb, whose diameter test (i) then sees in full. Cutting a genuine climb on a barometric flight therefore requires two failures at once: the geometric margin of test (i) and the barometric witness of test (ii). The only alternative is a byte-identical repeat, which no genuinely moving receiver writes.

The margins make that conjunction unlikely rather than merely improbable in principle: the diameter test is missed by a factor of two or more, and the barometric one by an order of magnitude, so both would have to fail well outside their designed range at the same moment and on the same run. On a barometric flight the rule is, in this sense, conservative by construction—it errs towards keeping.

The duration condition is the one place where the rule deliberately abstains. A run that is spatially collapsed and witnessed but lasts less than τ_{freeze} is *kept*, and that is the intended behaviour, not an oversight: at durations of a few tens of seconds a collapsed run and a genuine slow, horizontally localized stretch—a search, the core of a tight climb—are not distinguishable by these signatures, and cutting on them would remove the very events the analysis measures. The cost of the abstention is bounded by the threshold itself. A kept short freeze lengthens one waiting time by at most τ_{freeze}, which is small against the tail of $\psi(\tau)$ where the anomalous exponent is read, whereas a wrongly cut climb removes a waiting time outright. Short freezes that do get through are still visible downstream: they leave a run of interpolated-looking, near-identical positions, and the $\psi(\tau)$ sensitivity checks sweep τ_{freeze} precisely to bound how much of the tail depends on this choice.

Residual corner cases. The rule assumes a functioning barometric channel, and weakens where that channel is absent. This is not in tension with adopting the GNSS altitude for the *dynamics* of fallback flights (Sec. 2.7.1): there, GNSS is used because it is the only altitude available; here, the question is different—whether an instrument can *witness* a suspected GNSS lock loss—and the GNSS altitude cannot, because it comes from the very receiver whose lock is in doubt and freezes together with the position it would be checking. On GNSS-fallback flights the witness slot of Eq. (2.4) is therefore filled by the recorder's declarations alone (the `V` flag or zeroed GNSS altitude, or a byte-identical repeat); the diameter and duration conditions still apply unchanged, so "alone" refers only to the witness, not to the whole rule. The residual error cases are then:

- *Wrongly cut* (GNSS-fallback only): a circling climb tight enough to stay within δ_{xy} whose fixes carry a `V` flag—with no barometer to defend it, the declarations cut it.
- *Wrongly kept* (GNSS-fallback only): an undeclared *jittering* freeze. Some receivers, having lost the lock, do not repeat the last position byte-for-byte but re-estimate it each second, so consecutive positions differ by a metre or two of receiver noise: the run is genuinely frozen, yet neither byte-identical nor declared, and it is kept.
- *Kept, at bounded cost* (barometric flights): a jittered freeze under a *climbing* barometer is kept by construction. The cost is bounded, but it is not nil, and it is worth naming exactly. Every position recorded during the run sits inside the δ_{xy}-ball of the last true fix, so the run enters the analysis as a wait. If the wing did move during it—wind drift, or a genuine glide under a climbing barometer—that motion goes unrecorded: the path length is under-counted by the distance actually covered, and a waiting time of up to the run's duration is inserted where there was none. What is *not* lost is the displacement across the run, because the first position after re-acquisition is correct again; the quantities that

read a position at a given elapsed time (the MSD, the propagator) therefore see the right endpoint, and it is the path-accumulating quantities—path length, phase durations—that pay.

The first two cases—*wrongly cut* and *wrongly kept*—are expected to be rare: each needs a lock loss, a fallback flight, and a run whose bounding diameter happens to sit just on the wrong side of δ_{xy}, all at once. Their frequency, however, is not something the pipeline can report, and it would be wrong to claim otherwise: a rule that fails does so silently, and counting how often a detector *fires* says nothing about how often it was right to. What the audit below measures is the firing rate; what measures the error rate is the injected-defect calibration of check (d), where defects of known size are planted in clean tracks and the recovered fraction is counted directly. The default to keep is what makes the two asymmetric in the meantime: an unfired rule leaves real dynamics intact.

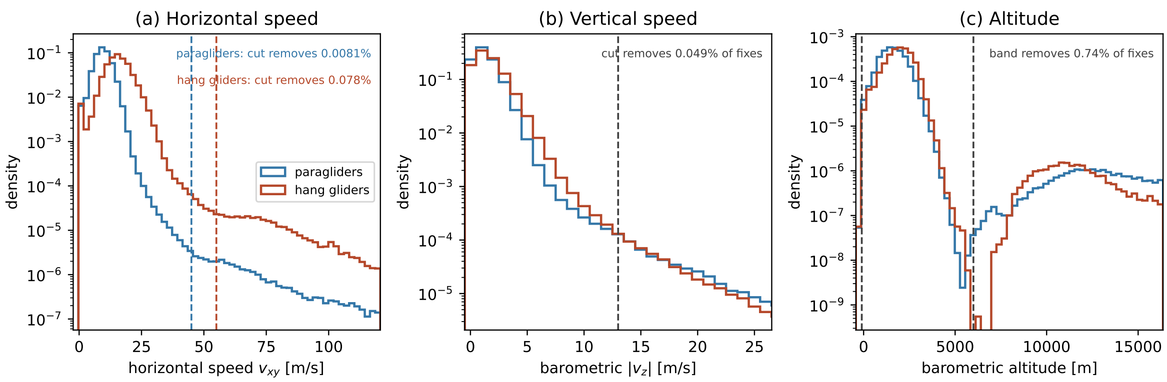

Figure 2.2. Fix-level cleaning: auditing the absolute bounds. Per-fix distributions of the quantities the absolute bounds test—bounds that serve both the out-of-range rule and, through the v_{xy} bound, the impossibility gate that deletes an off-the-trend fix (Sec. 2.7.2) – (a) horizontal speed v_{xy}, (b) barometric $|v_z|$, (c) barometric altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 2.6); the y-axis is logarithmic so the sparse tail the bounds act on is visible. In panel (a) each discipline shows the flat plateau of sustained speeds with no credible flight explanation that places its own bound: $45\,\mathrm{m\,s^{-1}}$ to $55\,\mathrm{m\,s^{-1}}$ for paragliders, $55\,\mathrm{m\,s^{-1}}$ to $70\,\mathrm{m\,s^{-1}}$ for hang gliders. Binning choices and sample sizes: paragraph *Diagnostic sample* of Sec. 2.7.2 (p. 97). This figure audits the *absolute* bounds only; the Hampel and frozen-lock detectors have no per-fix distribution to read a threshold off, and are validated instead by the removal-fraction sweep described below.

Completeness of the resampled trajectory. The segmentation (Ch. 4) needs a defined horizontal position *and* altitude at every time, since the vertical velocity is its discriminant. Cleaning does not guarantee this fix by fix – it may delete a fix (a corrupt horizontal position) or mark a channel missing (a barometric spike, a `V` fix on a GNSS-fallback flight) – because the guarantee is established one step later, at resampling (Sec. 2.7.6). A *segment* is a maximal stretch of the flight between two splits—equivalently, between long gaps. It is the unit of analysis in the sense that every quantity accumulated along the path—a path length, a phase duration, a smoothing window—is computed inside one segment and never across the gap that ends it, since the trajectory between two segments is unknown (Sec. 2.7.6). The guarantee that the segmentation needs is therefore not “complete (x, y, z) everywhere” but “complete (x, y, z) *within* a segment”, not across a split, established by the single audited interpolation of Sec. 2.7.6.

Defect	Detected by	Action
Off the trend — *the horizontal position itself is corrupt*		
Position spike, out-and-back jump	impossibility gate: unreachable from the last accepted fix at the absolute v_{xy} bound ($45/55\,\mathrm{m\,s^{-1}}$, para/hang), and its block's removal lets the track rejoin. The Hampel identifier [15, 16] flags and attributes; it does not gate	delete the fix; the short gap is bridged at resampling
Position discontinuity (re-acquisition offset)	impossible step that no bounded removal rejoins	split at the step, like a long gap
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — *record structurally wrong, horizontal position agrees with its neighbours*		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	diam $< \delta_{xy}$ for $\geq \tau_{\mathrm{freeze}}$, witnessed (Eq. 2.4): baro. flat or byte-identical; `V`/zero-alt on GNSS-fallback	mark as gap, *split* there
Out of range — *horizontal position intact, only the altitude value is impossible*		
Missing / out-of-band altitude, or $\lvert v_z \rvert$ spike	altitude channel absent, outside $[-100, 6000]\,\mathrm{m}$, or baro. $\lvert v_z \rvert > 13\,\mathrm{m\,s^{-1}}$	drop the altitude only; the fix stays
Invariant: a usable horizontal position is never discarded. A fix is deleted only when its *own* horizontal position is corrupt or invented (frozen lock); when only the altitude is wrong, the fix is kept and that channel is marked missing, to be restored at resampling (Sec. 2.7.6).		

Table 2.6. Fix-level cleaning, grouped by how each defect presents; in the order of the table, the three groups map onto a robust local test, a structural rule, and an absolute bound. The speed and altitude bounds are audited on the per-fix distributions of Fig. 2.2; the robust-test and frozen-lock parameters (k, w, $\varepsilon_{\min}$, δ_{xy}, δ_z, τ_{freeze}) are working values, listed in Table 2.5 and argued in Sec. 2.7.2. The v_{xy} bound is per-discipline because the speed envelopes differ (a `sailplanes` entry follows once that source is in). The altitude lower bound is $-100\,\mathrm{m}$, not 0: the barometric altitude is referenced to the ICAO standard atmosphere rather than to the day's sea-level pressure, so on a high-pressure day a sea-level take-off can read a few tens of metres negative (the offset scale is visible in Fig. 2.1a). (implementation details: Sec. 2.7.2)

Flight-level integrity gate. Cleaning acts fix by fix, but it also keeps count of how much it had to do. The counters are taken over the airborne window: defects in a ground phase that trimming removes anyway must not condemn the flight. If cleaning has removed or reconstructed more than a small fraction of a flight's airborne fixes, the flight is dropped whole.

Validating the cleaning. All the detector parameters are validated by one procedure, run once the cleaning routine is built (Ch. 4):

(a) *Freeze the working values.* The thresholds are fixed a priori at the values of Table 2.5; nothing is tuned on the transport observables themselves.

(b) *Audit the removal fractions.* The cleaning is run over the full archive with per-rule counters (the same machinery as the integrity gate): for each rule and each discipline, the fraction of fixes flagged and the fraction removed. The expectation is that each rule touches only a small fraction of fixes; this is the criterion to be verified, not a fact assumed in advance.

(c) *Sweep each threshold.* Each parameter is varied around its working value and the removal fractions are recomputed. "Well placed" is made quantitative rather than left to the eye. Write $\rho(\theta)$ for the fraction of fixes a rule removes at threshold θ, and read the local logarithmic sensitivity

$$S(\theta) = \left| \frac{\mathrm{d} \log \rho}{\mathrm{d} \log \theta} \right| .$$

A threshold sits on a plateau when $S \lesssim 1$ over a decade centred on the working value: halving the threshold then at most doubles what it removes, which is the behaviour of a rule biting into a sparse tail of defects. $S \gg 1$ means ρ is growing like a power of the threshold with a large exponent—the rule has reached the bulk of the distribution and is removing dynamics, not defects—and the working value is reconsidered. Reporting $\rho(\theta)$ and S at the working value, per rule and per discipline, is what the sweep produces. The same sweep covers the split cap of Sec. 2.7.6, monitored on its split fractions rather than on removed fixes.

(d) *False negatives.* Steps (b)–(c) measure what the cleaning removes, not what it misses—the corrupted fixes that pass undetected. Nothing in an audit of removals can supply that number, so this check is run as a matter of course and not only on suspicion: defects of known size (spikes, frozen runs, time glitches) are injected into clean tracks in known quantity, the cleaning is re-run, and each rule is scored against each defect type on the four counts—defects recovered and missed, clean fixes wrongly removed and correctly kept. The result is a confusion table per rule per defect type, with the recovered fraction as a function of defect size, and it is the only direct measurement of the error rates that the firing counts of step (b) cannot give.

(implementation details: Sec. 2.7.2)

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis. Detection uses the *horizontal* speed v_{xy} alone (from (2.1)) – the vertical speed plays no role here. With $v_0 = 5\,\mathrm{m\,s^{-1}}$ and $T_0 = 30\,\mathrm{s}$ (Table 2.5), the retained airborne segment is $[t_{\mathrm{on}}, t_{\mathrm{off}}]$ with

$$t_{\mathrm{on}} = \min\{t : \; v_{xy} > v_0 \text{ on } [t,\, t + T_0]\}, \qquad t_{\mathrm{off}} = \max\{t : \; v_{xy} > v_0 \text{ on } [t - T_0,\, t]\}. \tag{2.5}$$

Take-off is thus the *first* time the speed stays above v_0 continuously for T_0, and landing the *last* time it has done so: the same sustained-speed rule read forward and time-reversed. Reading the landing rule backward from the end matters. A rule of the form "cut at the first sustained drop below v_0" would fire in the middle of the flight, because a wing soaring into wind can hold a ground speed near zero while genuinely flying. The persistence T_0 prevents a gust or a GPS glitch on the ground from being mistaken for take-off. A flight with no sustained stretch at all has no airborne segment and is discarded, with the reason recorded.

T_0 is set between two scales rather than tuned. From below, it must outlast the artefacts it exists to reject: a ground gust that briefly pushes a laid-out wing, or a position glitch that fakes a fast step, lasts a few seconds at most, and $T_0 = 30\,\mathrm{s}$ is several times that. From above, it must cost little when it is wrong, and it can misfire only by at most T_0 at each end of the flight—negligible against the 40 min minimum duration a retained flight has to reach anyway (Sec. 2.7.4). Between those bounds the result is insensitive to the exact value, which is the reason it can be fixed a priori.

One caveat is structural. The criterion is a *ground* speed, so a slow phase that happens to sit right at the start or end of the flight—a launch straight into ridge lift, say—can be clipped along with the ground phase. The fraction of each flight that trimming removes is recorded per flight and its distribution over the archive reported as a histogram once the pipeline runs in full. That histogram bounds the *exposure* to this failure—how much of a flight the rule can have taken—but it does not detect the failure itself, since a correctly trimmed ground phase and a wrongly clipped ridge-lift launch produce the same trimmed fraction. What separates them is inspection at the tail: the flights in the upper percentiles of the trimmed-fraction distribution are the candidates, and they are few enough to be examined one by one against their altitude trace, which rises during a ridge-lift launch and does not on the ground. Both v_0 and T_0 affect that fraction, so they are made explicit (implementation details: Sec. 2.7.3).

Interior ground stints (mid-flight landings). The rule above trims only the *outer* ground phases: a mid-flight landing and relaunch recorded in one log (a top-landing, say) would survive as an interior near-zero-speed stint and read downstream as a false long wait in the tail of $\psi(\tau)$ (Sec. 2.7.2). An interior-ground detector is therefore specified, built like the frozen-lock rule: it cuts only on corroborated evidence, because its dangerous error is the opposite one – the soaring-into-wind failure mode above. A stint is cut only when *both* of the following hold:

- $v_{xy} < v_0$ continuously for at least $T_{\text{ground}} = 10\,\text{min}$—far beyond any search phase, so no genuine slow phase reaches this length;
- the adopted altitude channel flat over the whole stint, the central span of its detrended values staying within $\delta_z^{\text{gr}} = 5\,\text{m}$ once the slow barometric drift has been removed—a wing at zero ground speed in real air still moves vertically, the same independent-sensor argument as Eq. (2.4).

The detrending is not a detail. The barometric reference itself wanders by tens of metres over an hour (Sec. 2.7.1), which over a stint of T_{ground} is already of the order of δ_z^{gr}; testing the raw range would let a perfectly stationary pilot fail the flatness condition on a pressure change alone. What the condition has to measure is the *residual* vertical motion after that drift is taken out, and δ_z^{gr} is a tolerance on that residual.

That the spread is the central one is not cosmetic, and the alternative was in force until it was measured. A full range is a maximum statistic: over zero-mean noise it grows without bound with the number of samples, as $2\sigma\sqrt{2\ln n}$. At the metre-scale barometric noise of this archive it reads 4.6 m over a 60-sample stint and 7.1 m over 3,000, so against a 5 m tolerance the same motionless pilot passed the test on a one-minute stint and failed it on a ten-minute one—on noise alone. Since an interior stint must last T_{ground} to be considered at all, it failed on every stint the rule was ever offered, and the guard could not fire. The central span carries the same units and the same reading and sits at 3.3σ whatever the stint's length, so δ_z^{gr} means the same thing on a stint of any duration; a genuine climb of hundreds of metres exceeds it either way.

One consequence is stated rather than hidden. *On a GNSS-fallback flight the guard abstains.* The intended rule is that the vertical witness there uses the GNSS channel with a tolerance widened to match its noise, and in doubt the stint is kept; the widening is not implemented, because the factor it needs is a measured ratio between the two channels' short-timescale noise and a first measurement on the archive did not reproduce the ordering Fig. 2.1 reports. Until that is settled, the single barometric tolerance is applied to both channels, which on a noisier one simply means the flatness condition is not met and the stint is kept—the conservative half of the intended rule, and the safe direction, since the error this guard exists to avoid is excising a genuine climb. What it costs is that a mid-flight landing on a GNSS-fallback flight is neither excised nor flagged.

A stint that is cut is excised, and the flight is split at the excision: the parts before and after are treated from then on as separate segments of the same parent flight, exactly as at a long gap (Sec. 2.7.6). The segment rules apply unchanged—a phase interrupted by the cut is censored rather than counted as complete, and the flight-level cuts of Sec. 2.7.4 are not re-applied to the individual segments.

The order of the two steps is worth stating explicitly, because it is the reverse of the order in which they are described. Trimming runs first (step (iii) of the pipeline) and the flight-level cuts second (step (iv)): a duration or a path length is only meaningful once the ground phases are gone, so the cuts have to see the trimmed track. When trimming has split a flight, those cuts are still evaluated on the *parent*—its airborne duration is the sum of its segments' durations, its path length the sum of their path lengths—and not on each segment separately. The reason is stated in Sec. 2.7.6: a segment is short because a gap happened to fall there, which says nothing about whether the flight is a genuine cross-country one, and re-applying a whole-flight criterion to a fragment would throw away exactly the data recorded around gaps. Segments face only the one requirement that is physically theirs—being long enough to carry the smoothing window (90 s, Table 2.5).

Stints shorter than T_{ground} are not cut but *flagged*: a wait with sustained $v_{xy} < v_0$ and a flat barometric trace is marked suspect whatever its length, and the $\psi(\tau)$ fits are re-run with and without the suspect waits as a sensitivity check – the same flag-without-deleting pattern as the outlier identifier. Flagged waits are reported, not merely used internally: their count, and the share of the total waiting-time mass they carry, are given per discipline alongside the two $\psi(\tau)$ fits, so the reader can see how much of the tail rests on them. T_{ground} and δ_z^{gr} are parameters of this guard and sit in the `trimming` block of the configuration file, with the values of Table 2.5.

2.7.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. The step is delicate: too aggressive a cut biases the transport statistics – a high minimum duration, for instance, preferentially keeps successful long flights and inflates the apparent MSD exponent at long lags, since the flights that survive such a cut are the ones that kept going. The flight-level filtering therefore uses a *minimal-threshold strategy*: for each criterion, the threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little as possible. In practice "minimal" is read off the data, not guessed: for each criterion the full-archive distribution of its quantity is drawn (Fig. 2.3, top row), together with the fraction of flights that the cut alone would retain as the threshold is moved (bottom row).

The non-genuine population does not announce itself the same way in both criteria. In the path length it forms a distinct cluster at the low end, separated from the genuine flights by a sparse tail, and the threshold can be placed just beyond it. In the duration histogram no such cluster is visible; what locates the cut there is the retained-fraction curve alone, which is flat over a wide stretch before the bulk begins. The threshold is placed in either case on the stretch where the retained-fraction curve is flat—there, moving the cut changes almost nothing, which is precisely the statement that it removes junk and not signal. Five track criteria gate a flight, plus a metadata stratifier: Two of them are sums over the flight, and what they are summed over is part of the rule rather than an implementation detail: the duration and the path accumulate *within* a block and never across the boundary that ends it, since the trajectory there is unknown and a straight line drawn over it was never flown.

A boundary is either a marker an earlier stage left—an excised frozen-lock run, a re-acquisition offset, a mid-flight landing—or an inter-fix gap wider than the $g_{\max}$ the resampling will split at (Sec. 2.7.6). The second kind is easy to forget, because nothing marks it: a marker records what a *stage* decided, and a hole in the log is nobody's decision. Forgetting it counted ten minutes of missing record as ten minutes of flight, and the six kilometres of straight line across it as six kilometres flown. Using the same $g_{\max}$ here as there is what makes a flight measured over the intervals it will actually be analysed over.

(1) *Recorded duration* $T \geq 40\,\text{min}$. A cross-country flight spans many climb–glide cycles; shorter logs are sled runs or aborted attempts. An upper sanity bound $T \leq 16\,\text{h}$ removes loggers left running past the flight: the census holds 15 paraglider "flights" of 16 h to 166 h, with paths up to 10,059 km (and none among the hang gliders), which would poison the long-lag analyses generally, the MSD among them.

(2) *Flown path length* $L \geq 20\,\text{km}$, with $L = \sum_i \Delta s_i$ from the great-circle steps of (2.1), a floor against degenerate logs. On the census, above 40 min the cut catches 236 of 184,583 paraglider flights (median extent 5 km) and 7 of 6,638 hang-glider flights: a fraction of order 10^{-3} of either ensemble. Duration already clears tows and sled runs.

(3) *Altitude activity.* The total range of the adopted altitude channel over the airborne track—the difference between its maximum and its minimum—must reach 600 m. The threshold is a positive statement about what the flight did, not merely a guard against a dead sensor: a cross-country flight of at least 40 min lives by climbing, and several climbs of several hundred metres each are what keep it airborne, so a record spanning less than 600 m over that time was not soaring—a vehicle log, a tow, a stuck channel. The cut is nonetheless cheap, which is what makes it safe: moving it from 75 m to 600 m changes the retained share of barometric paraglider flights by under two points, so the two values sit on the same plateau and the population between them is thin. Like the other two, it is a *whole-flight* criterion, evaluated on the parent before any splitting; an individual segment may well span less, and that is accepted for the same reason the duration cut is not re-applied per segment—a segment is short because a gap fell there, not because the flight was not a flight. One limitation belongs with it: the cached census records only the *barometric* extremes, so on a GNSS-fallback flight it reads a range of zero whatever the flight actually did, and the diagnostic of Fig. 2.3c,f is drawn over the barometric flights alone. Auditing this cut on the fallback minority needs the GNSS extremes in the scan, which is a full re-read of the archive rather than a re-query of the cache. They are counted and dropped with the reason recorded.

(4) *Plausibility of the cleaned record.* The mean ground speed of the retained track, L/T, must not exceed the discipline's own fix-level bound $v_{xy}^{\max}$. This is a sanity bound of the same kind as the upper duration cut above, and it is here for the same reason: a small

number of logs are corrupt beyond what fix-level cleaning can repair—runs of the null position $(0, 0)$ scattered through the track—and every jump that survives adds thousands of kilometres of path that were never flown. Asking a *whole-flight average* to stay under the bound that already declares a single step impossible is the weakest statement that catches them, and it is deliberately weak: no genuine flight comes close, since the median mean speed is under $10\,\mathrm{m\,s^{-1}}$ and the bound sits at $45\,\mathrm{m\,s^{-1}}$. On the present archive it removes *no* flight at all—0 of either discipline—and that zero is a result rather than a redundancy. The criterion was written when a corrupt jump still entered the flown path; it no longer does, since an impossible step is made a segment boundary at the fix level and a long gap is excluded from the sum here, so by the time the path reaches this test it contains only motion. What such a record now fails is the reach criterion below. The cut is kept because it is the cheapest possible check on that chain: if a future source, or a change upstream, ever lets an impossible length back into the path, this is where it surfaces, and its count stops being zero. No new configuration value is introduced—the bound is the one already set for the fix level, read at the scale of a whole flight.

(5) *Reach.* No fix may lie further from the first than the bound could have carried it in the time elapsed since: $\Delta s(\mathrm{fix}_0, \mathrm{fix}_i) \le v_{xy}^{\max}\, t_i$ for every i, because the flight started there. It is stated per fix, and deliberately not as the extent $D = \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i)$ against the whole duration T: that weaker form licenses a fix 100 km out ten seconds into the record on the grounds that the flight went on for three hours, and it is not the statement the written table is checked against (Sec. 2.2). One criterion, one form, in both places. Where the previous criterion bounds the path, this one bounds the displacement, and the difference is exactly what makes it necessary. A flight whose *first* fix is the corrupt one has no impossible step at all once that step is made a boundary and its length is no longer summed into L: it passes on duration, on path, on altitude and on mean speed, and it simply sits thousands of kilometres from where it says it began. Five paraglider flights in the first full-archive run did (Sec. 2.7.2), and since that first fix becomes the origin of the local frame (Sec. 2.7.5), the displacement they carried was not one bad step but every coordinate of the record. On the archive the criterion removes 228 paraglider and 7 hang-glider flights, of order 10^{-3} of either ensemble; on a sample of 4,000 per discipline it touches nothing else. Like the previous one it introduces no configuration value, and it is checked again on the written tables as an invariant of the output (Sec. 2.2), where it is the statement $|\mathbf{r}(t)| \le v_{xy}^{\max} t$ that the whole of Chapter 3 assumes.

(6) *Glider class* (metadata). For class-stratified analyses, flights whose `aile_class` cannot be normalised onto the discipline's system (Sec. 2.4) are dropped from the class-resolved sub-analyses but kept in the pooled one. Stratification by glider *type* (paraglider vs hang glider) is already possible.

Every count quoted above is a generated macro, recomputed against the thresholds in force from a cached full-archive scan; how, and by which code, is in Sec. 2.7.4. The thin tail of very slow loggers is handled by the Δt stratification (Sec. 2.7.6); a separate minimum-fix-count cut would be redundant, since above the duration cut even a 10 s logger records hundreds of fixes. A further criterion, sampling regularity, is applied later, where a flight's time base is examined (Sec. 2.7.6) (implementation details: Sec. 2.7.4).

Figure 2.3 shows these diagnostics for the three criteria that carry a tunable threshold, (1)–(3), and both disciplines, computed on the full census (no subsampling). The duration and path cuts sit far out in the tail, so the retained fraction is insensitive to their exact value.

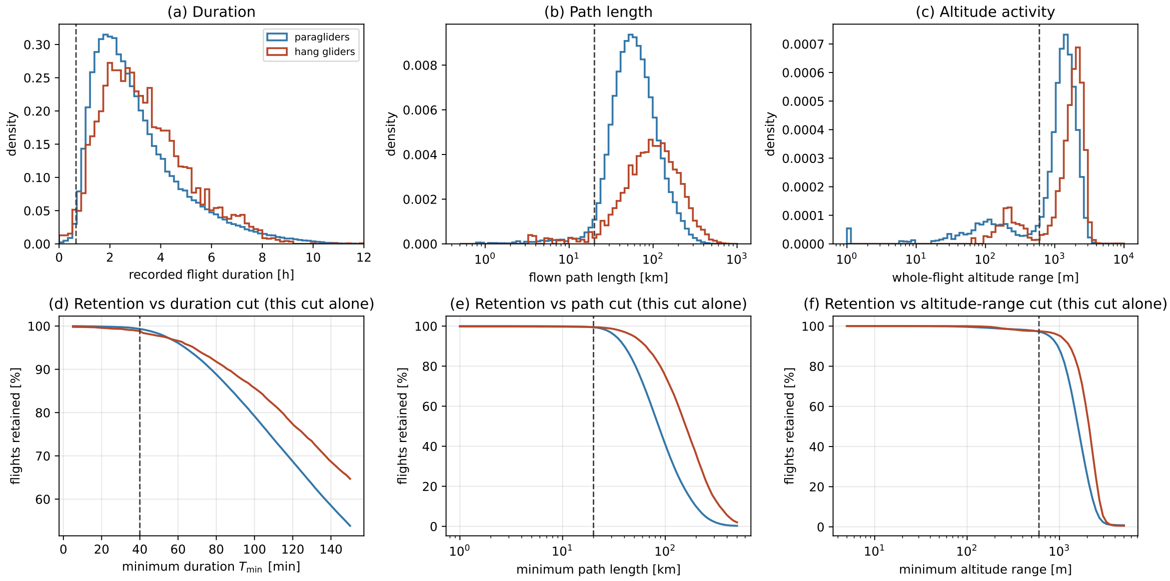

Figure 2.3. Flight-level filtering diagnostics, per discipline. Top: distributions of (a) recorded flight duration, (b) flown path length and (c) whole-flight altitude range. Bottom: the fraction of flights retained versus (d) the duration cut, (e) the path-length cut and (f) the altitude-range cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ($T \geq 40$ min, $L \geq 20$ km, range ≥ 600 m). The duration and path cuts sit far out in the tail. Panels (c) and (f) are drawn over the *barometric* flights only: the cached scan stores no GNSS altitude extremes, so a fallback flight would enter them at a range of zero for a reason unrelated to how it flew (Sec. 2.7.4). (implementation details: Sec. 2.7.4)

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates. The two remaining steps are different in kind (Sec. 2.7): resampling and smoothing operate on the coordinates *componentwise*—and the smoothing returns their derivatives in the same componentwise way (Sec. 2.7.7)—and therefore need actual Cartesian components to act on. Latitudes and longitudes cannot play that role—they are angles on a curved surface, and differencing them componentwise does not yield displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137$ m, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$) [14, 18],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad \begin{aligned} X &= (N + h) \cos\varphi \cos\lambda, \\ Y &= (N + h) \cos\varphi \sin\lambda, \\ Z &= (N(1 - e^2) + h) \sin\varphi, \end{aligned} \tag{2.6}$$

where $N(\varphi)$ is the prime-vertical radius of curvature and h the ellipsoidal height, measured along the ellipsoid normal; the transform is derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. The latitude φ in (2.6) is the *geodetic* latitude: the angle between the equatorial plane and the ellipsoid *normal* at the point (Figure 2.4). It must not be confused with the geocentric latitude ψ, the angle to the line joining the point to the Earth's centre O. On an ellipsoid the two differ, because the normal does not pass through O: the gap $\varphi - \psi \approx (e^2/2) \sin 2\varphi$ (derived in Appendix 2.B) peaks at $\varphi = 45°$, where $\sin 2\varphi = 1$, and there reaches $\approx 0.19°$ (about $11.5'$), which if the two were confused would misplace a fix by up to ~ 20 km of ground distance.

We use the geodetic latitude throughout, and no conversion is needed: WGS84 – hence the GPS receiver and the IGC `B` records (Sec. 2.2) – reports geodetic latitude, and (2.6) with $N(\varphi)$ is precisely the geodetic-to-ECEF transform that consumes it. These are also the (φ_0, λ_0) that orient the local frame of Step 2 (Figure 2.5).

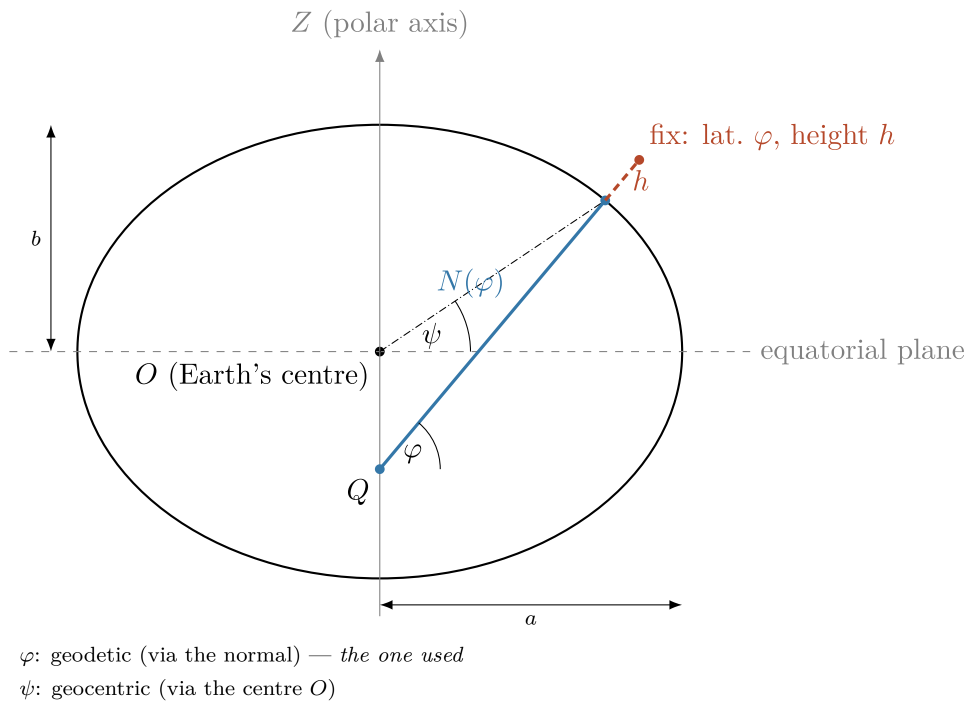

Figure 2.4. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b; the geodetic latitude φ, the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O, so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.5.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin—the first fix of free flight, which is close to but not the take-off point on the ground, since the ground phase has been removed (Sec. 2.7.3), the ECEF displacement $\Delta\mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is *rotated* into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point—the word is exact, not loose: the origin shift has already been taken out by forming $\Delta\mathbf{X}$, and what is left is an orthogonal matrix of unit determinant, so lengths and angles are preserved and only the axes change,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \Delta\mathbf{X}. \tag{2.7}$$

with the rotation matrix derived in Appendix 2.B. The ENU frame is the local Cartesian frame in which the whole analysis is carried out: E points east, N north, and U along the local vertical (the ellipsoid normal at the origin). The ECEF coordinates of Step 1 are only an intermediate representation of the computation—no observable is ever expressed in them. Over the spatial extent of a single flight the tangent-plane approximation is accurate to well below the GPS noise for the horizontal coordinates: projecting onto the tangent plane shortens a distance d along the surface by $\approx d^3/(24R_\oplus^2)$, about 1 cm at $d = 20$ km and 13 cm at 50 km, against metre-scale GPS noise (Appendix 2.B).

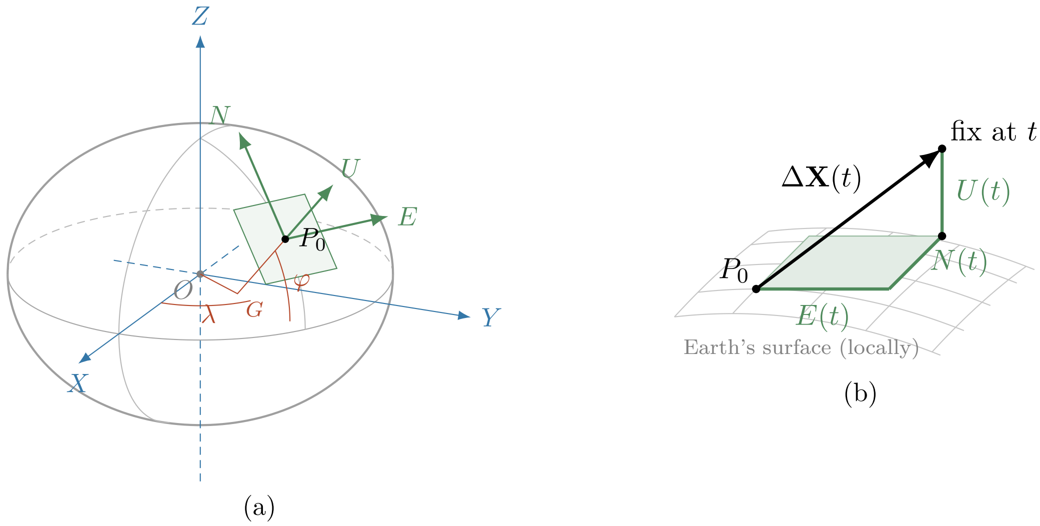

Figure 2.5. The ENU rotation: it turns a vector's global ECEF components into local East, North, Up ones. (a) The global ECEF frame (X, Y, Z) in blue, centred at the Earth's centre O; the two red angles that locate the origin fix P_0 (the first fix of the trimmed track, Sec. 2.7.5): longitude λ, measured in the equatorial plane from the prime meridian, and geodetic latitude φ, the tilt of the ellipsoid normal, measured at the point G where the normal meets the equatorial plane; and the local ENU frame in green, tangent at P_0. Flattening exaggerated as in Figure 2.4: U follows the normal G–P_0, not the radius from O. The green arrows are the rows of the rotation matrix, as unit vectors. (b) Close-up of the tangent plane: a later fix's ECEF displacement $\Delta\mathbf{X}(t)$ is the space diagonal of the box whose edges are its ENU components $E(t)$, $N(t)$, $U(t)$.

The altitude h that enters this transform is the adopted channel of Sec. 2.7.1. The horizontal coordinates are insensitive to that choice, and the mechanism is visible in Eq. (2.6): h enters the horizontal ECEF components only through the radius factor $(N + h)$, so switching channel—which changes h by the inter-reference offset, at most $\sim$100 m (Fig. 2.1a)—rescales them by a relative $\delta h/(N + h) \lesssim 2 \times 10^{-5}$. Across the full extent of a flight ($\sim$50 km) that is under a metre, below the GPS noise on the same coordinates.

For the vertical, the analysis does not use the rotation's U component as its working coordinate: it keeps the adopted altitude channel itself (Sec. 2.7.1; barometric for most flights, GNSS on the fallback minority), $z(t) = h(t)$, at its measured value—the altitude of the first fix is *not* subtracted. Nothing is gained by re-zeroing: every vertical quantity that matters downstream (the vertical velocity, the climb/sink discrimination of the segmentation) depends on increments, which are invariant to the reference, while keeping the measured value preserves the absolute height, an observable in its own right (potential energy, convective-boundary-layer structure) (Sec. 2.7.8).

2.7.6 Uniform sampling within a flight

The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight's time base is examined next. Loggers record at a fixed nominal rate – paragliders mostly at 1 s (69 % of flights), hang gliders spread across several rates from 1 s to 10 s with a thin tail beyond (Fig. 2.7) – but the realised series can carry jitter and drop-outs. We keep each flight at its *native* rate; a common cadence is unnecessary, because the derivatives are not taken by finite differences across flights but read off the local polynomial that the smoothing fits to each flight separately, and the flight's own Δt enters only as the factor that turns that polynomial's coefficients into physical units (Sec. 2.7.7). What matters is uniformity *within* a flight, not across flights.

Per flight, let Δt be the median inter-fix interval. Three cases arise:

Uniform. All intervals equal Δt to tolerance: the series is used as is.

Mildly irregular. Isolated jitter or single drop-outs – no gap beyond the split bound $g_{\max} = \min(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\,\Delta t))$ (its two scales are motivated in the paragraph *The split bound has two scales* below) and less than 10 % of the grid missing (Table 2.5): the flight is resampled onto the uniform grid at its native Δt, interpolating across the small gaps only, and the interpolated fraction is recorded.

Broken by a long gap. A gap longer than $g_{\max}$ – native, or opened where cleaning removed a frozen-lock run (Sec. 2.7.2), or a position discontinuity such as a re-acquisition offset, where the track jumps and stays: both sides are genuine and only the transition between them is unknown, so it is handled exactly like a long gap – is not bridged: interpolating across it would fabricate dynamics the smoothing could not tell from real motion. The flight is *split* at the gap into independently analysed segments, each carried forward on its own; what a split means for the analysis is stated below.

The missing-fraction check runs here, and not among the flight-level cuts of Sec. 2.7.4, because it needs the time base and the splits to exist first. It needs the time base because "missing" is only defined once there is a grid to be missing from: the quantity counted is the share of points of the uniform grid at the flight's own Δt that carry no recorded fix, and neither the grid nor its spacing exists before this step. It needs the splits because without them a single long gap—which is not going to be interpolated at all—would dominate the count and condemn a flight whose segments are each perfectly well sampled. It is evaluated *per segment*, after splitting: a segment whose residual missing fraction (the grid points that would have to be interpolated) exceeds 10 % is dropped alone, and the flight is lost only if no segment survives. A flight with no long gap is a single segment, so a gap-free flight missing more than 10 % of its grid is dropped whole by the same rule.

The split bound has two scales. The relative part, $10\,\Delta t$, tolerates isolated dropped fixes in proportion to the cadence and caps the number of interpolated grid points at ten. Alone, however, it would let a slow logger bridge very long holes: 30 s at $\Delta t = 3$ s, 100 s at $\Delta t = 10$ s. The harm of a bridged gap is set by the motion's own timescales, not the logger's: an interpolated stretch is a straight line, so a gap approaching the thermalling period (~ 30 s) could replace a full circle, and the segmentation downstream would read the invented straight stretch as a glide. The absolute cap of 20 s is pinned by three constraints:

- at least 2× the slowest common cadence (10 s, Fig. 2.7), so a single missed fix never splits a flight;
- at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle;
- well below $\tau_{\text{freeze}} = 60$ s, for consistency: an excised frozen-lock run of that length is split, so a native gap of comparable length cannot be bridged.

One refinement keeps the first constraint true at *every* cadence: the cap is floored at $2\,\Delta t$, i.e. $g_{\max} = \min(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\,\Delta t))$, so the thin tail of very slow loggers (Fig. 2.7) also tolerates a single missed fix. Which of the three terms sets $g_{\max}$ depends on the cadence: at the dominant 1 s rate it is the relative bound, $g_{\max} = 10$ s; from $\Delta t = 2$ s to 10 s it is the absolute cap, $g_{\max} = 20$ s; above that it is the $2\,\Delta t$ floor.

On the census, the two-scale bound $g_{\max}$ just defined leaves 15.3 % of paraglider and 40.5 % of hang-glider flights with at least one split; had the relative part $10\,\Delta t$ acted alone, without the absolute cap, the shares would be 11.8 % and 14.5 %. The increase sits at the slow cadences, where a 20 s hole is only a few missed fixes (generated macros, Sec. 2.7.6). A split is cheap for the ensemble observables (below) but costly for the *phase* statistics: every split censors the phase in progress when the gap opens, so splitting too eagerly preferentially removes long phases from the duration distributions—the same tail the cleaning protects. For this reason the absolute cap is treated as a working value like the detector thresholds: it is swept in the validation procedure of Sec. 2.7.2 and frozen where the split fractions plateau. The sweep is reported as a figure, not only as a number: the fraction of flights with at least one split, and the fraction of phases censored, plotted against the cap for each discipline, with the adopted value marked. That plot is what shows the plateau, and therefore what justifies the choice.

The bound is the same for every channel, altitude included. One might expect the altitude to tolerate wider gaps, since the vertical speed is about an order of magnitude smaller than the horizontal ($v_z \sim 1\,\mathrm{m\,s^{-1}}$ against $v_{xy} \sim 10\,\mathrm{m\,s^{-1}}$); it does not, for two reasons. First, the interpolation error over a gap is set by the *curvature* of the signal, not by its speed: a fast but straight motion is interpolated exactly, a slow but turning one is not, so a slower channel is not automatically a smoother one. Second, and more decisively, the altitude feeds v_z, the segmentation's discriminant. A long altitude gap interpolated linearly returns one average slope, which erases any climb–glide transition inside it—and transitions are exactly where v_z swings fastest: a paraglider's typical glide sink is 1–1.5 $\mathrm{m\,s^{-1}}$ and its climb in a working thermal 2–4 $\mathrm{m\,s^{-1}}$ (hang gliders sink faster, at 1.5–2 $\mathrm{m\,s^{-1}}$, and climb over the same range); both sit well inside the bulk of the $|v_z|$ distribution of Fig. 2.2b, far from the tail where the absolute bound is placed, so a thermal entry swings v_z by several $\mathrm{m\,s^{-1}}$ within a couple of seconds, the worst case for interpolation. A real asymmetry between the channels does exist, but it concerns *isolated* missing samples, not gap width: a single missing altitude is reconstructed far more reliably than a single missing position, because the barometric signal is much smoother—that is what justified "drop the altitude only" at cleaning (Sec. 2.7.2); it does not justify a wider gap bound for z.

Consequences of a split. The observables of Ch. 3 are affected differently by a split. In practice a split is bookkeeping rather than a new trajectory: the conversion to ENU (step v) runs before this step, so every fix already carries coordinates and elapsed time referred to the parent origin, and the split only assigns a segment identifier. **The key fact is that a segment keeps its parent's origin and clock** (Sec. 2.7.8): a GPS position is absolute, so the positions after the gap remain exactly as valid as those before it. Ensemble quantities (the MSD, the propagator) need the horizontal position at a given elapsed time, not the path travelled in between, so a split costs them nothing—the flight simply contributes no data at the lags that fall inside the gap. Quantities that instead *accumulate along the path*—the path length, a phase duration, a step length, any time-averaged window—cannot be evaluated across a stretch whose true course is unknown: sums and windows for these quantities are always taken inside one segment, never straddling a gap. Three cautions follow.

- *Censoring at the boundary.* A phase in progress when the gap opens is truncated: its duration is a lower bound, not a measurement. Boundary-truncated durations are flagged as censored and simply *excluded*—from the $\psi(\tau)$ fits and from the phase-duration histograms alike—rather than counted as complete, which would let every split masquerade as a spuriously short wait. The censored phases are identified, then, exactly as a survival analysis would identify them; what is not done is to feed them back in through a censoring-aware estimator—a Kaplan–Meier survival curve, or a likelihood with a right-censored term, which would use the information that the true duration exceeds the observed one instead of discarding the observation. As long as the censored fraction stays small (it is recorded per discipline), plain exclusion keeps the estimators transparent at negligible cost, and the recorded fraction tells us if that assumption ever fails—if it does, the machinery to switch is the standard one just named.

- *A segment is not a fresh flight.* For observables referenced to the time since the flight began—the time-averaged MSD stratified by duration, for instance (Ch. 3)—a segment starting at parent time $t_0 > 0$ is not re-zeroed into a new flight that starts at $t = 0$. It is used for what it is: an observation window $[t_0, t_1]$ onto a process already t_0 old. Re-zeroing would mix statistics of young and aged trajectories, which an aging analysis must keep apart.
- *No second pass of the flight-level cuts.* The duration and path cuts of Sec. 2.7.4 were passed by the parent flight and are not re-applied to its segments. Re-applying them would be biased in a specific way: gaps concentrate in particular conditions (weak reception late in long flights, for example), so discarding every sub-40 min segment would throw away precisely the data recorded around gaps. A segment has to satisfy only what the downstream machinery physically requires—enough duration to support the smoothing window and the statistics computed within it. That minimum has a value and a place in the pipeline, and it belongs here rather than only in the configuration: a segment shorter than $T_{\mathrm{seg}} = 90\,\mathrm{s}$ (Table 2.5) cannot carry a smoothing window of $\tau_c = 5\,\mathrm{s}$ with enough points on either side of it, and is dropped alone; the rest of the flight is retained.

Interpolation across short gaps. The scope of this step is set by the split bound: a gap longer than $g_{\max}$ splits the flight and is never interpolated, so what is filled here are only the *short* holes, up to $g_{\max}$, that remain inside a segment. The fill is done per channel, where a channel is one of the three scalar series on the uniform grid: the horizontal coordinates $E(t)$ and $N(t)$ and the adopted altitude $z(t)$.

The sentence above is true of the *time base* and not of a channel, and the two come apart in exactly one place. $g_{\max}$ bounds the interval between two consecutive *fixes*; it says nothing about an interval over which a fix exists but carries no value. That case is not hypothetical and is not rare: the vertical channel can be absent while the horizontal record is perfect—a logger that writes no barometric value, or a value the fix-level cleaning removed (Sec. 2.7.2, where the whole point of the asymmetry is that such a fix is *kept*). No time gap opens there, so no split is declared, and the linear fill spans the hole whatever its length.

Two things follow, and both are decided here rather than left implicit. First, such a hole does *not* split the flight. The horizontal trajectory, which is what Chapter 3 measures, is intact across it; cutting it because the vertical is missing would discard good data for a defect that does not touch it. Second—and this is the correction the check demanded—the reconstruction must be visible. It was not: a grid point sitting on a real fix whose altitude had to be invented was reported as *measured*, because the `interpolated` flag is a statement about the time base alone. On a test flight missing 500 s of altitude the filled z departed from the truth by up to 540 m, the smoothing differentiated the straight line into a vertical velocity that was pure invention, and the record read `frac_interpolated` = 0 with `was_resampled` false—the flight declaring itself uniform as recorded. The vertical therefore carries its own per-fix flag, `z_reconstructed`, with the per-segment share `frac_z_reconstructed` and, per flight, the longest unbroken run `z_gap_max_s`. The run length is reported next to the fraction because a fraction cannot separate the two cases that matter: a hundred holes of two seconds, which a straight line bridges well, from one hole of twelve minutes, which it does not. Any analysis of the vertical—the climb/sink discriminant of the segmentation above all—excludes on that flag; the horizontal analyses are untouched by it, and that is the reason the flag is per channel rather than one for all three. The two horizontal coordinates are interpolated independently, which is the componentwise interpolation of the horizontal position; the barometric and GNSS altitudes are never mixed here—only the flight's adopted channel is filled (Sec. 2.7.1).

The horizontal coordinates, which the transport analysis measures directly, are interpolated with a *monotone piecewise cubic* (a shape-preserving Hermite interpolant): one cubic per interval, whose slopes are chosen so that the interpolant reproduces the monotonicity of the samples themselves—where the data increase, the curve increases, and it does not turn back on itself between two points. The construction needs no derivative data: the slope at each sample is estimated from the neighbouring samples and then limited to whatever value keeps the monotonicity condition satisfied, which is what distinguishes it from an unconstrained cubic spline and is the whole reason for using it here. A spline free to choose its slopes can swing outside the range of the points it passes through, and such an excursion would be indistinguishable, to the smoothing step, from a real manoeuvre. The altitude is interpolated linearly[5]. The linear error over a hole of width g is bounded by $\max|\ddot{z}|\,g^2/8$; holes are bridged inside a phase far more often than across a transition between phases, and within a phase the barometric signal has a nearly constant slope—a steady climb or a steady glide—so $|\ddot{z}|$ is small there (the low high-frequency content of the barometric channel, Fig. 2.1c, is the same statement in the frequency domain). So the error is at the metre scale even at the 20 s cap and far smaller for typical holes—transitions, where $\ddot{z}$ is large, are handled by the flag below.

The altitude is interpolated linearly and the horizontal coordinates by a monotone cubic, and the asymmetry follows from what each channel is used for. The altitude is differentiated once more than the horizontal coordinates before it is read—v_z is the segmentation's discriminant—so an interpolation artefact in z propagates into a phase assignment, whereas the same artefact in E or N is absorbed by the smoothing. Linear interpolation cannot overshoot, and its error over a hole of width g is bounded by $\max|\ddot{z}|\,g^2/8$; a cubic is more accurate where the second derivative is smooth and worse exactly where it is not, which is at the transitions a phase assignment turns on.

The result is precisely the completeness invariant promised at cleaning (Sec. 2.7.2): *within a retained segment, every grid time carries a defined* (E, N, z), so the segmentation, and the analysis more generally, is never handed a missing value. The altitude values dropped at cleaning (a barometric spike, a `V` fix on a GNSS-fallback flight) are the ones restored here. Every filled point carries a flag—`interpolated` where the time base had no fix, `z_reconstructed` where a fix was there but its altitude was not—with a concrete consequence: downstream statistics can distinguish measured from reconstructed samples, and a phase boundary that the segmentation places inside a bridged stretch is treated like one at a segment boundary—the true transition could lie anywhere inside the hole, so the adjoining phase durations count as censored and are excluded from the duration fits, exactly as at a split.

The gap bound $g_{\max}$ is the criterion called "sampling regularity" in the flight-level filtering (Sec. 2.7.4); Figure 2.6 makes the fraction of flights it leaves whole auditable. One caveat carries downstream: a few observables are intrinsically Δt-dependent, the turning angle above all. The dependence is in the observable's own definition: a turning angle is the heading change over one step, so two loggers recording the *same* motion at 1 s and 10 s produce systematically different distributions of it. The velocity is a partial exception, because it is not formed as a finite difference here but read off the smoothing polynomial (Sec. 2.7.7); what remains Δt-dependent in it is the effective width of that polynomial's window, which is set in seconds and therefore spans a different number of samples at each cadence. Pooling all flights would blend this instrument artefact into the physics; where such observables are compared across

[5] The bound follows from the standard error formula for linear interpolation. On an interval of width g the difference between a C^2 function z and the chord through its endpoints is $z(t) - p(t) = \frac{1}{2}\,\ddot{z}(\xi)\,(t - t_0)(t - t_0 - g)$ for some ξ in the interval; the quadratic factor is largest at the midpoint, where it equals $g^2/4$, giving $|z - p| \le \max|\ddot{z}|\,g^2/8$.

flights, the ensemble is therefore stratified by rate or restricted to the dominant one.

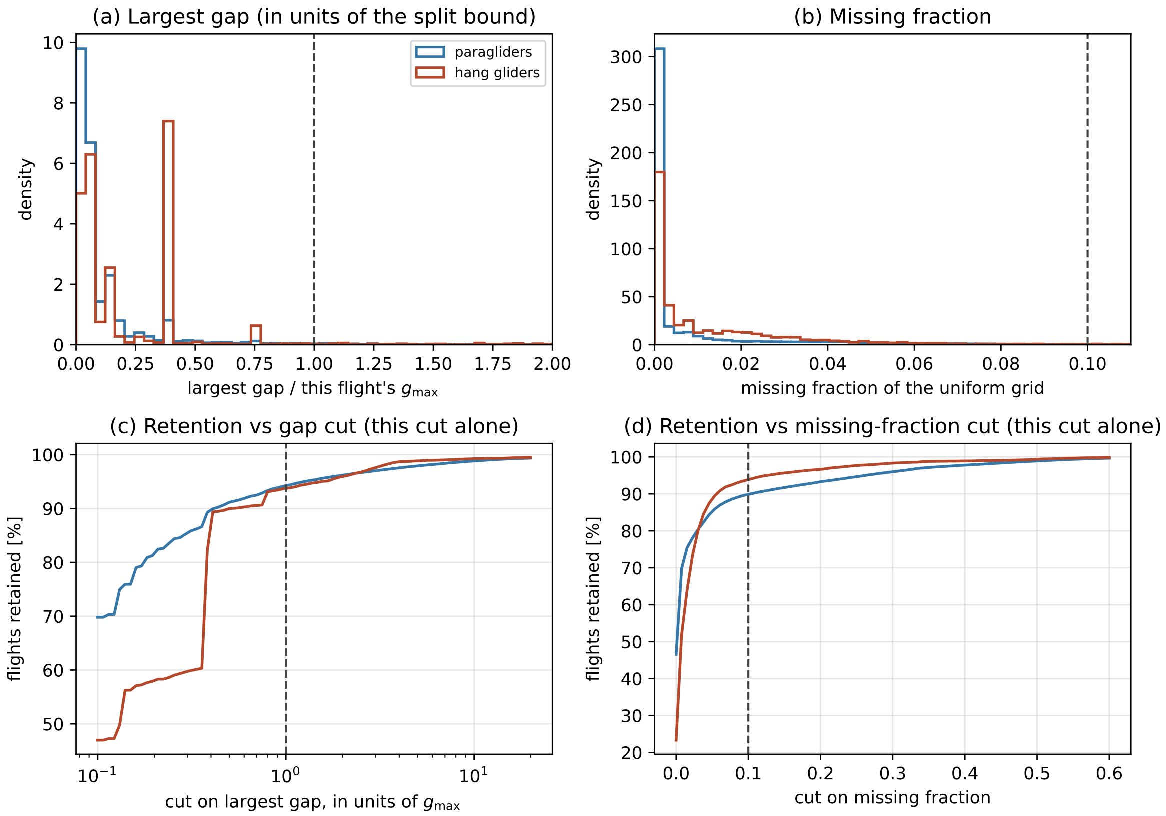

Figure 2.6. The two sampling-regularity criteria and their retention curves, per discipline. Computed on the same full track scan as Figure 2.3—every parsed flight, not only those passing the flight-level cuts, because a cut must be audited on the population it acts on, before its own effect hides part of the picture—what would hide it is the flight-level cleaning applied first, not this cut. (a)/(b) Distributions of the two tested quantities: the largest single gap of the flight and the missing fraction of a uniform grid at its native interval. The gap is plotted in units of *that flight's own* $g_{\max}$, not of Δt, so the criterion actually in force sits at 1 (dashed) for every flight; because $g_{\max}$ is the two-scale bound and not a fixed multiple of Δt, a plot against Δt alone would draw a cut that binds at no cadence but 1 s. The distribution is discrete below the cut, since a gap is usually a run of wholly missed fixes and hence an exact multiple of the native interval. (c)/(d) The fraction of flights each cut alone would retain as the threshold is moved: *marginal* curves, i.e. each cut applied by itself to the full population, not after the other cuts. Both panels are recomputed once the fix-level and flight-level stages run in full; the census behind them predates that pipeline. (implementation details: Sec. 2.7.6)

2.7.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS coordinate series amplifies the high-frequency noise characterised in Sec. 2.7.1. A **Savitzky–Golay** filter [19] solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial, or its derivatives, at the window centre, returning smoothed position, velocity and acceleration in one pass per component.[6] The fit is done in units of samples, so its coefficients come out per sample; the velocity and acceleration

[6]Concretely: in the window of w samples centred on t_i, the coefficients of the polynomial $P(t) = \sum_{k=0}^{p} a_k\,(t-t_i)^k$ minimize $\sum_j \left(x_j - P(t_j)\right)^2$; the smoothed value is a_0, the velocity a_1, the acceleration $2a_2$. Because the grid is uniform, the a_k are fixed linear combinations of the window samples, so the whole filter is a convolution with precomputed coefficients [19].

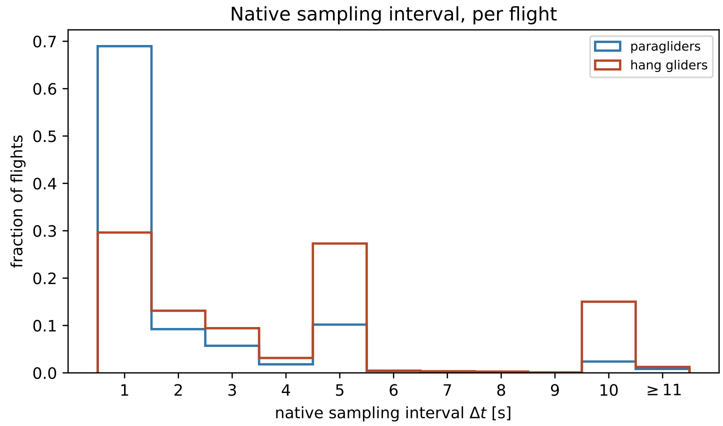

Figure 2.7. Native sampling interval Δt (median inter-fix time) per flight, on the full dataset per discipline. Δt is in practice discrete – it lands on an exact whole second for nearly every flight (Sec. 2.7.6) – so bins are one second wide, centred on each integer; each bar is the fraction of flights at exactly that rate. Paragliders concentrate at 1 s; hang gliders spread across several rates, with a thin tail beyond 11 s (last bin, aggregated). A thin tail is present in both disciplines; what differs is the bulk, with the 10 s rate far commoner among hang gliders than among paragliders. Each flight keeps its own Δt (Sec. 2.7.7). (implementation details: Sec. 2.7.6)

in physical units are $a_1/\Delta t$ and $2a_2/\Delta t^2$, with the flight's own Δt (Sec. 2.7.6). This is the only place the cadence enters, and it is why flights of different cadences need no common grid.

The window never crosses a segment boundary. Near a boundary, where a centred window no longer fits, the polynomial is fitted on the nearest full window inside the segment and evaluated off-centre (the standard boundary treatment of the filter). An off-centre evaluation is an extrapolation of the fitted polynomial rather than an interpolation, and the variance of a least-squares polynomial grows towards the ends of its fitting interval, so the first and last few samples of a segment carry a larger uncertainty than the interior ones. They are marked as such—the same per-sample flag mechanism as the interpolated points—so that an observable sensitive to it (an acceleration extreme, say) can be computed on interior samples only. Whether the exclusion is worth making is an empirical question: with windows of five to a few tens of samples and segments of at least 90 s, the affected samples are a small fraction of a segment, and the check is simply to recompute the observable with and without them.

The filter runs on the resampled grid of Sec. 2.7.6, so it sees a uniform time base with the short holes already filled. The filled points carry no independent information—each is a smooth function of its neighbours—so within a window they mildly correlate the samples; over holes this short that introduces no bias, and their fraction is recorded per flight. Flights that needed a great deal of reconstruction never reach this step: the interpolated share of a segment is capped at 10 % at resampling (Sec. 2.7.6) and the airborne share a flight may lose to cleaning at 10 % at the integrity gate (Sec. 2.7.2); what is *recorded* here, and not merely thresholded, is the per-flight value of that share, so a downstream statistic can be stratified on it.

The filter has two hyperparameters, the window length w (odd, in samples) and the polynomial order p, fixed from the measured noise spectrum rather than by eye:

(i) From the ensemble Welch power spectrum of the raw ENU components on a sample of flights (the diagnostic of Sec. 2.7.1 applied to E, N, U), read the frequency f_c of the knee where the dynamical signal meets the flat GPS noise floor. Its reciprocal is a smoothing timescale $\tau_c = 1/f_c$: dynamics slower than τ_c are signal, faster fluctuations are noise. Then set the window per flight from that timescale. Making w span τ_c is a *choice*—the natural one, since it puts the filter's cut-off at the measured knee—not a result: w is $\tau_c/\Delta t$ rounded up to the nearest odd integer (odd, so that the window has a centre sample to evaluate at), with a floor at $w = 5$, the smallest window on which a cubic fit is meaningfully determined. At the dominant 1 s cadence the two rules coincide: $w = 5$ samples spans exactly τ_c, and the smoothing acts at the measured noise scale. At every slower cadence the floor is the binding rule (already at $\Delta t = 2$ s, where $\tau_c/\Delta t < 5$), and the five-sample window then spans *more* than τ_c—about 10 s for a 2 s logger, 25 s for a 5 s one—so slow loggers are smoothed somewhat more than the noise scale requires. The cost is accepted: at those cadences the noise band sits at or beyond the Nyquist frequency $1/(2\Delta t)$, so there is little to remove at the noise scale in the first place, and these flights are in any case excluded from the Δt-sensitive observables by the rate stratification (Sec. 2.7.6).

(ii) Fix p as the lowest order that works. An acceleration requires curvature in the fit, so $p \geq 2$. One order more is needed for the sake of the *velocity*: on a symmetric window, adding an even-order term leaves the odd-order coefficients untouched (a parity property of least squares on a symmetric grid), so the velocity of a $p = 2$ fit is identical to that of a straight-line fit. What $p = 3$ adds is the cubic coefficient a_3, which is proportional to the third derivative—the rate of change of the acceleration—and which shifts a_1: the velocity of a cubic fit is the first one that responds to the acceleration *varying* across the window, rather than assuming it constant. Going higher only lets the fit follow more noise. Hence $p = 3$.

(iii) Treat the vertical separately from the horizontal, and condition it on the altitude *source*. The a priori expectation was asymmetric: the smooth barometric channel would need a *shorter* window than the horizontal coordinates, and the noisy GNSS vertical a *longer* one than them. The measurement said otherwise: the three knees coincide at $f_c \approx 0.2$ Hz ($\tau_c = 5$ s). This is a measured value and it carries the status of one: it comes from the per-channel Welch spectra of Sec. 2.7.7, read on a sample of the *present* archive, and the plot that shows the three knees falling together is the evidence for it. Two qualifications belong with the number. It is read on a sample, not on one flight, because a single spectrum is too noisy to locate a knee; and it is read on data that have not yet passed the fix-level and flight-level stages, so it is provisional—the spectra are recomputed, and τ_c re-read, once the cleaned ensemble exists. The reason is trivial: the flat high-frequency floor of every channel comes from the rounding the IGC format applies when the file is written—altitudes to whole metres, coordinates to thousandths of an arc-minute—rather than from receiver noise, and rounding noise flattens the spectrum at the same frequency whatever channel it sits on. What distinguishes the noisy GNSS minority is how *high* its floor sits, not *where* the knee is (Sec. 2.7.7). So the answer to whether E, N and z get different smoothing timescales is currently no: the per-channel machinery exists, for the noisy GNSS minority, but the adopted windows are equal.

The choice is then checked on a held-out sample—flights not used to place f_c—with three acceptance criteria: the residual spectrum (raw minus smoothed) must be flat above f_c, showing that what was removed is the noise band and nothing below it; the smoothed kinematics must stay within physical bounds—the acceleration first, since it is the highest derivative and therefore the most sensitive to the window, but the speed and the climb rate on the same footing, against the same envelopes the fix-level bounds use (Table 2.5); and the results must be stable when w is changed by one step (implementation details: Sec. 2.7.7).

A fourth criterion, on the observable itself. The three above are properties of the filter. A fourth is a property of what the filter is for: the smoothing must not remove the displacement the transport analysis is about to measure. The concern is not visible in τ_c alone, because τ_c is not what sets the window for most of the archive. The window is floored at 5 samples—one more than a cubic needs, so that it stays a least-squares fit rather than an interpolation—and that floor binds for every logger slower than 1 Hz. In *seconds* the window is therefore $\max(\tau_c, 5\,\Delta t)$: 5 s at a 1 s cadence, 50 s at 10 s, 150 s at 30 s—which reaches the lower end of the range over which Chapter 3 fits its exponent.

Measured rather than argued, on synthetic trajectories of known exponent passed through the adopted filter and read the way the pipeline reads them—displacement from the frame origin, which is the *raw* first fix (Sec. 2.7.5), not the smoothed one—the answer is that the filter is transparent to the motion. For a superdiffusive trajectory ($\alpha = 1.8$, the regime Chapter 3 finds) and no measurement noise, the filtered MSD sits within 2.3 % of the true one at a lag of two samples and within 1 % at every lag from three samples upwards. The reason is structural rather than fortunate: a cubic Savitzky–Golay filter reproduces any locally cubic trajectory *exactly*, and a strongly correlated trajectory is locally close to cubic, so it passes through untouched.

What does move the short-lag MSD is the opposite of smoothing, and in the opposite direction. Independent measurement noise of standard deviation σ per component adds $\approx 2\sigma^2$ to the MSD at every lag; where the true displacement is comparable to that, the measured curve is a noise floor rather than motion. With $\sigma = 3$ m the same experiment gives a measured-to-true ratio of 8.1 at one sample, 1.26 at five, 1.05 at twelve and 1.003 at sixty-four—and the smoothing is what pulls it down, the one-sample ratio falling from 8.1 to 3.6 when the window goes from 5 samples to 11. So the filter does not erase the observable, it de-biases it; the floor it leaves is set by $2\sigma^2 \approx 18\,\mathrm{m}^2$, which the measured MSD exceeds by orders of magnitude everywhere in the fitted range. The consequence for reading Fig. 3.3 is that its first decade is instrument, not transport, and the fit does not begin there. A second effect points the same way and belongs beside it: a flight answers a lag with the position at *its own* nearest fix, so the curve's tread at short lags is that flight's Δt rather than the 1 s the lag grid is snapped to. For every logger slower than 1 Hz the short-lag curve is a staircase, and the local slope of a staircase alternates between zero and a spike. Both effects die out well below the lower end of the fit range, which is why they are a caveat on reading panels (a) and (c) at short lag rather than on the exponent.

2.7.8 Notation

Throughout, $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$ denotes the *horizontal* position in this local frame and $z(t)$ the adopted altitude channel (Sec. 2.7.1), kept at its measured value (Sec. 2.7.5). By construction each trajectory starts at the horizontal origin, $\mathbf{r}(0) = \mathbf{0}$, with the clock reset to $t = 0$ at the start of free flight (the first fix of the trimmed track, Sec. 2.7.3), so t is the elapsed flight time (a segment created by a split at a long gap, Sec. 2.7.6, keeps its parent flight's origin and clock). The vertical is *not* re-zeroed: the measured height at the start of free flight is retained (Sec. 2.7.5). The horizontal velocity is $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; the vertical velocity is $v_z = \dot{z}$. The horizontal displacement over a lag t is $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$. A terminological rule holds throughout the document: when a statement concerns $\mathbf{r}$ it says *horizontal position* explicitly, and when it concerns the full three-dimensional point it says *3D position*; a bare "position" is avoided.

2.8 Preliminary characterization

This section concerns the dataset that the analysis actually consumes: what the whole of Sec. 2.7 leaves behind. Three of its stages discard: the fix-level cleaning, through the integrity gate; the flight-level filtering, where a flight is lost outright against a stated criterion; and the resampling, which is easy to overlook because it is a *re-expression* of a trajectory rather than a cut, and which is nonetheless responsible for the single largest loss in the cascade—a flight whose every segment fails the sparsity or the minimum-duration rule leaves nothing behind (Table 2.7). The conversion and the smoothing drop nothing. Its point is to establish the basic statistics on that retained ensemble—which is why, for instance, the native-rate distribution reappears here: Figs. 2.6 and 2.7 were computed on *every* parsed flight, as a filtering diagnostic, whereas the analysis needs the same quantities on the flights actually used. None of the checks below requires the phase segmentation, so all of them run as soon as the pipeline does; two of them constrain what the transport analysis of Chapter 3 is allowed to do, and both come out against the simpler option.

Trajectory statistics. For each retained flight: airborne duration T, total horizontal path length, number of valid fixes and native sampling interval Δt. Their distributions (Fig. 2.8) describe the dataset the results rest on, expose any outlier that survived the filtering, and, through Δt, fix the per-flight smoothing window (Sec. 2.7.7). Over the 155,788 retained paraglider flights the median airborne duration is 2.63 h (interquartile range 1.78 h to 3.93 h), the median flown path 87 km (57 km to 134 km) and the median record 7,129 fixes; 69.8 % of them log at 1 Hz. The 6,132 hang-glider flights are longer and faster—3.07 h and 151 km in the median—and slower-logging, with a median Δt of 3 s against 1 s and only 27.6 % at 1 Hz.

These are not the raw-archive figures of Sec. 2.5: the retained median duration exceeds the recorded median 2.6 h because the duration floor and the altitude-activity floor remove short and inactive records, and the retained median path is close to the recorded 86 km because the 20 km path floor bites on few flights (Table 2.7). Nothing in the retained ensemble is an outlier in the sense the filtering was meant to catch: the table carries 0 non-finite coordinates and 0 non-monotone clocks, and the fastest step between two *measured* samples is 49 m s^{-1} against the fix-level bound of 45 m s^{-1}—an excess of under a tenth, on 32 steps out of 1.4×10^9, which Sec. 3.1 accounts for.

What the cascade costs, and whether it costs it evenly. Table 2.7 reads the discards as a cascade, and one criterion dominates: 21,089 paraglider flights, 11.9 % of those still standing, leave nothing behind after resampling. That is four times the next largest cut, it is a rule about record quality rather than about flight, and it is not a threshold Sec. 2.7 argues for at all — so it deserves a check the others do not.

It fails that check in one respect and passes it in another. Across native cadences the sparsity rule is markedly uneven: it removes 0.1 % of the flights at its most forgiving cadence and 22.6 % at 2 s, so the retained ensemble is enriched in the loggers whose grids are easiest to fill. Across time it is even: over the 20 seasons carrying at least a thousand flights the retention rate stays between 70.3 % and 91.6 % (Fig. 2.10b), so two decades of changing logger technology have not left the pipeline eating one era of the archive. The seasons below that size are the ladder's first years, when it carried tens of flights, and their rates are noise rather than trend.

Where the flights are. The take-off sites need three frames rather than one, because the CFD is a French ladder flown partly outside France. Figure 2.9 gives them, on coastlines and borders rather than on bare axes: metropolitan France, La Réunion, and the rest of the world. Of the retained paraglider flights 94.7 % launch in metropolitan France, 3.2 % (4,939 flights) on La Réunion, and 2.2 % (3,359) elsewhere—the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil. The hang gliders are more domestic still, at 96.7 %. The overseas and foreign flights are kept: they are the same discipline flown under the same rules, and the geographic stratification below is where a difference between terrains would show up.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,332 distinct take-off sites at a hundredth of a degree, and 58.8 % of it launches inside the Alpine box alone, against 7.8 % in the Pyrenees, 8.2 % in the Massif Central and 19.8 % elsewhere in France. The density is drawn on a 0.15° cell with the count on a logarithmic colour scale, since at 155,788 flights individual markers saturate over the Alps and stop answering the question the map is drawn for; La Réunion, seventy kilometres across, gets a cell fifteen times finer. The orographic groups are longitude and latitude boxes, drawn on Fig. 2.9a and listed with their edges in the mirror of this section (implementation details: Sec. 2.8); they are used below to define strata, and the map itself is not the test.

Isotropy, and what it costs. The transport analysis would like to study the one-dimensional marginal of the propagator in place of the full two-dimensional density (Sec. 3.1), a substantial simplification. The reduction is legitimate only if the process is isotropic, and isotropy cannot be assumed here. The CFD flights are not races flown together—each is a solo cross-country flight, later entered into a scoring ladder—but a single flight is strongly directed: a chosen route, the orography, the valley winds. Isotropy can therefore hold at best at the level of the *ensemble*, and the check is run there.

The primary test is the per-component variance ratio: for an isotropic process $\langle E(t)^2\rangle = \langle N(t)^2\rangle = \mathrm{MSD}(t)/2$ at every t, so the ratio $\langle E(t)^2\rangle/\langle N(t)^2\rangle$ should be identically 1. Figure 2.11a shows it is not. Over the lags the transport analysis reads it sits at 0.80 for paragliders, between 0.70 and 0.86 and never touching unity: the north–south mean square exceeds the east–west by about a quarter, steadily and across three decades. The hang gliders are anisotropic the other way, at 1.19. A Kolmogorov–Smirnov comparison of the empirical distributions of $E(t)$ and $N(t)$ at 4 lags agrees, with distances from 0.038 to 0.125; at this sample size the associated p-values are numerically zero and carry no information, which is why the distance is quoted and the test is not.

The consequence is operative. The two marginals $P(E(t) = x)$ and $P(N(t) = x)$ are analysed *separately* rather than pooled, and the pooled 1-D marginal is not used as a stand-in for the 2-D propagator. That is the natural fallback and is informative in itself: unequal scaling exponents $H_E \neq H_N$, or equal exponents with different scaling functions, signal geometrically anisotropic transport, which for flights channelled along Alpine valleys is the expected outcome rather than a nuisance. The anisotropy is also a reason to read the ensemble MSD with the care Sec. 3.1 gives it, since $\langle|\mathbf{r}|^2\rangle$ sums two components that do not grow alike.

Compatibility across strata. The dataset spans more than two decades, two glider types and a range of performance classes (Sec. 2.4), from recreational EN A/B wings to CCC racing wings, whose kinematics differ substantially. Before treating the flights of one discipline as draws from a single statistical process, the subpopulations are checked. The test is the horizontal MSD computed within each stratum and read against the pooled curve over the same lags; the two disciplines are never pooled, and a stratum below 2,000 flights is not drawn as if it were a measurement. The class normalisation of Sec. 2.4 runs first, since as long as one wing class appears under several spellings a stratum is not a well-defined set of flights.

Figure 2.11b–d gives three axes and two answers. Across the 6 wing classes the curves separate: the typical stratum departs from the pooled curve by 40 % and the extreme by 90 %, with the CIVL competition wings above the pooled curve throughout and the EN A wings well below it. Across the 5 orographic groups they separate further, by 62 % typically and 130 % at the extreme, with the Massif Central at half the pooled curve over the intermediate lags and the Alps above it. Across the 17 seasons they do not: the typical departure is 12 % and the largest 21 %, and the latter belongs to the smallest season in the archive.

So the ensemble may be pooled over time and may not be pooled over wing class or over terrain. Season is not a relevant stratum and the archive's two-decade span is not a confound. Class and geography are relevant strata, and the difference is itself a result rather than a nuisance: a competition wing covers ground faster than a recreational one, and thermals over a mountain range are anchored to slopes and organised along ridges whereas over flat terrain they drift with the wind, so the persistence of the glides and the spacing of the climbs need not be the same. The exponent of Chapter 3 is therefore reported by wing class as well as pooled, alongside cuts by native cadence, season and declared task (Table 3.3); the orographic cut is made here, on the ensemble MSD, which is the statistic it separates. A pooled exponent is read as an average over strata that are known to differ, not as a property of a homogeneous population.

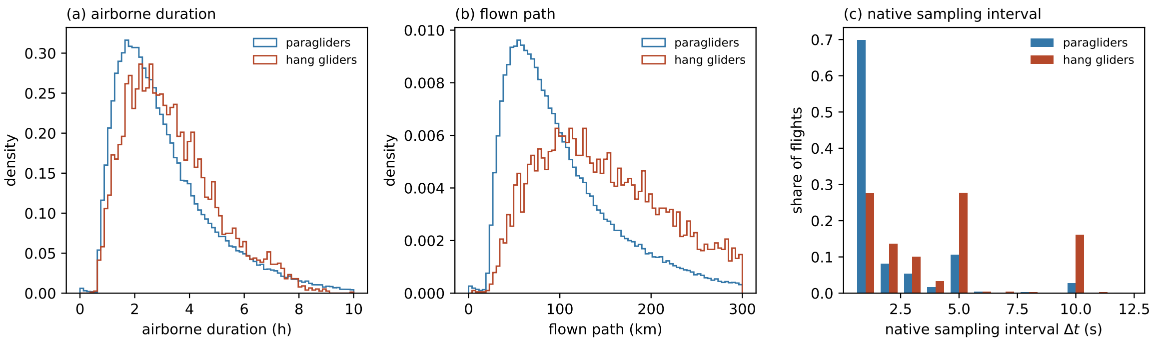

Figure 2.8. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt, per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.6 and 2.7. Computed by `scripts/reporting/generate_prelim_figure.py`.

2.9 Pipeline summary

Figure 2.12 closes the chapter with the whole processing chain in one map: every step the data undergo between the raw IGC tracks of Sec. 2.2 and the ensemble handed to the transport analysis, in the order the pipeline applies them. Table 2.8 pairs each stage with the level it acts at, its effect, and the adopted parameters; its last column leads back to the section where each choice is argued.

Two decisions organise the chain. First the order: every check that runs on the scalar record as logged — times, geodetic coordinates, altitude — comes first (stages 1–4), and the vector geometry is built only from what survives (stages 5–7); a coordinate conversion neither repairs a corrupt record nor has to be undone when one is dropped (Sec. 2.7). Second the level: each stage acts per fix, per flight or per segment, and never on more than one flight at a time. The point is not that pooling flights would be tempting—it would not—but that each stage's parameters are set from the flight it is acting on. The smoothing window comes from that flight's own Δt, the outlier scale from its own local MAD, the resampling grid from its own cadence; none is an archive-wide constant. What is shared across flights is only the thresholds of Table 2.5, and those are fixed a priori, not estimated from the ensemble—so no flight's cleaning can depend on which other flights happen to be in the dataset.

The parameters in the table are the adopted operating point of `configs/preprocessing.yaml`, typeset through macros that `generate_census_stats.py` regenerates from that same file (implementation details: Table 1.1), so the summary follows the configuration in force.

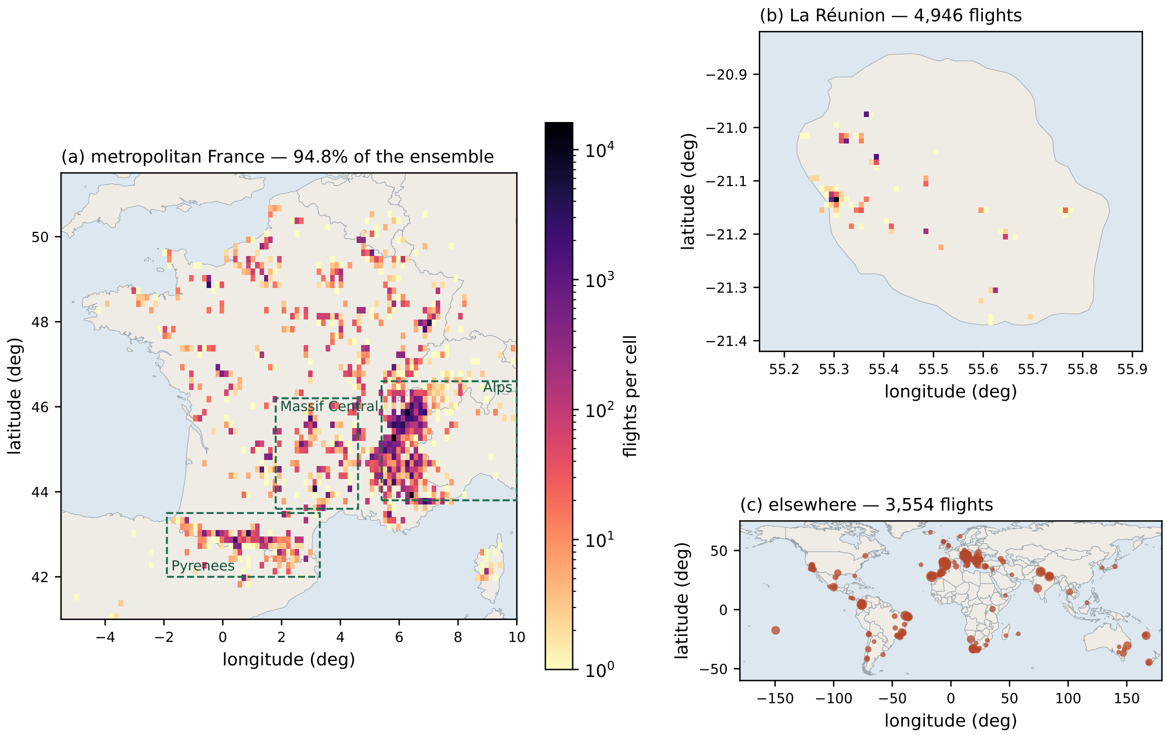

Figure 2.9. Take-off sites of the retained ensemble, on three frames. (a) Metropolitan France, density on a 0.15° cell and a logarithmic colour scale; the dashed boxes are the orographic groups used as strata in Fig. 2.11. (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count. Coastlines and borders are Natural Earth admin-0, cropped and committed by `scripts/reporting/build_basemap.py` so that the figure regenerates offline (implementation details: Sec. 2.8).

What the chain retains. Run over the whole archive, the cascade retains 155,788 of 186,052 paraglider flights (83.7 %) and 6,132 of 6,716 hang-glider flights (91.3 %). Table 2.7 says where the rest went. The numbers are a *cascade*, not a set of marginal effects: a flight counted against one criterion was still standing when that criterion tested it, so the table reads in pipeline order and the counts do not add up to what each cut would remove on its own (those are the `\StatScan*` numbers quoted in Secs. 2.7.4 and 2.7.6, measured on the raw population).

Inside the retained flights the cleaning is light, which is what the design asked for: it deletes 1.84 % of paraglider fixes and 0.75 % of hang-glider ones, and invalidates an altitude on a further 0.016 % and 0.009 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.97 % of fixes are flagged and *kept*. The flag is a per-fix record and a per-flight anomaly count, and what condemns a fix is the impossibility gate (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.32 % of grid points. A split is common but not the rule: 6.9 % of retained paraglider flights and 16.3 % of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.7 as Sec. 2.7.6 anticipated.

The analysis ensemble, defined. Everything downstream reads one object, and this is it. The *analysis ensemble* is the set of trajectories written to `fixes.parquet`: one row per grid point of every *retained* segment of every *retained* flight, carrying the elapsed time t in the parent clock and the local Cartesian coordinates (E, N, z) with their first two derivatives. A flight is retained when it passes stages 1–4 of Table 2.8 and at least one of its segments survives stage 6; a segment is retained when it is long enough and dense enough on its own uniform grid. That leaves 155,788 paraglider flights of 186,052 (83.7 %) in 179,395 segments, and 6,132

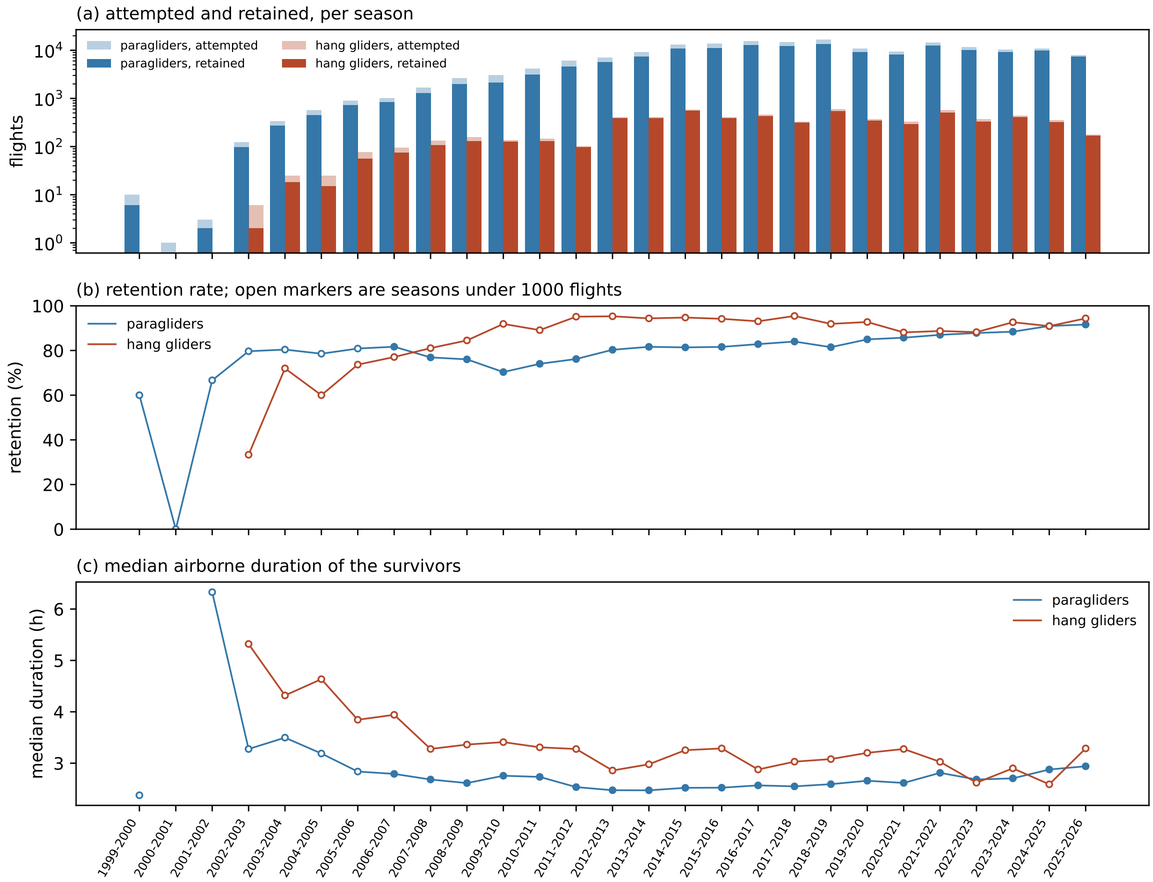

Figure 2.10. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights, whose rates are noise. Retention is flat across the substantial seasons, which is what says the cascade does not discard one era of the archive preferentially. Computed by `scripts/reporting/generate_dataset_stats.py`.

hang-glider flights of 6,716 (91.3 %) in 6,951. No consumer re-applies a retention rule: a flight or segment that should not be analysed has no rows to read. Every segment keeps its parent's origin and clock, so a position is always measured from that flight's take-off and a time is always elapsed since it, whether or not the record is continuous up to that point. Section 2.8 characterises this ensemble, Chapter 3 analyses it, and `docs/guide/data-on-disk.md` states which columns each estimator reads and what a segment boundary means to it.

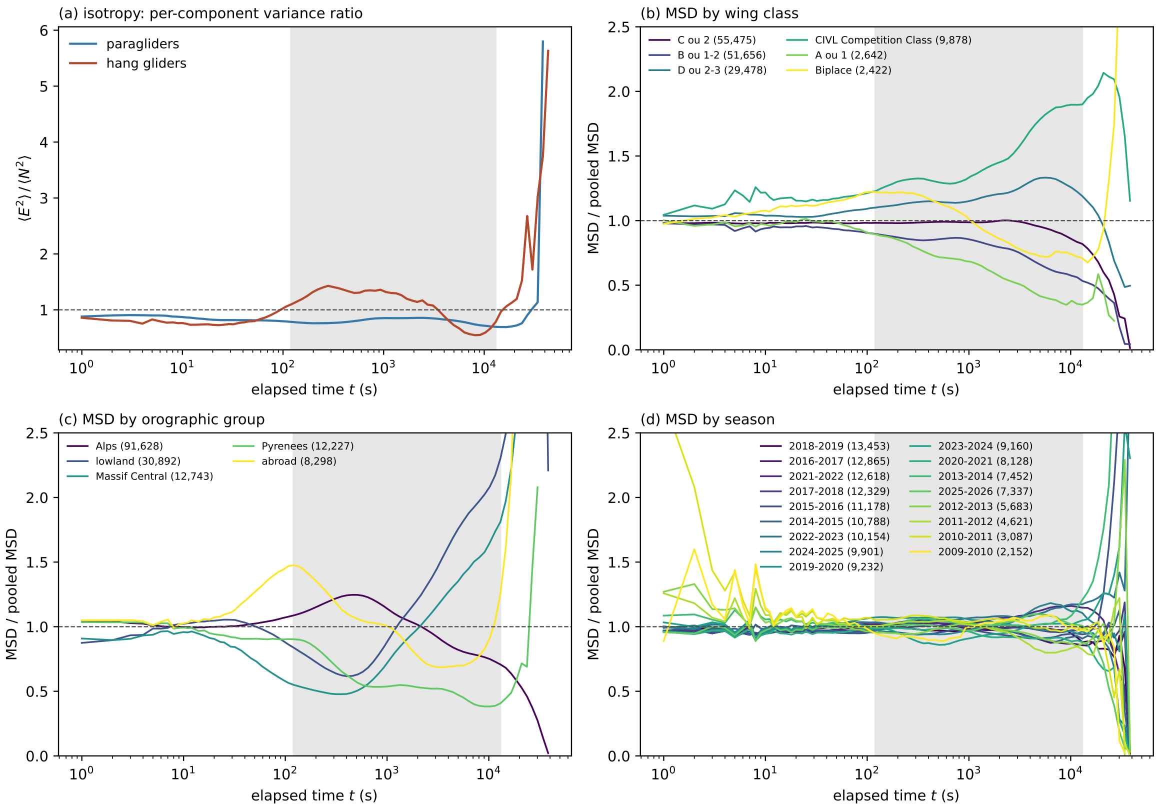

Figure 2.11. Whether the retained ensemble may be pooled. (a) The per-component variance ratio $\langle E^2\rangle/\langle N^2\rangle$ against elapsed time: unity is isotropy, and neither discipline reaches it over the shaded band, which is the lag range the transport analysis reads. (b–d) The ensemble MSD computed within each stratum and divided by the pooled curve, for paragliders, by wing class, by orographic group and by season; strata below 2,000 flights are not drawn. Class and terrain separate the curves, season does not. Computed by `scripts/reporting/generate_prelim_figure.py`.

Table 2.7. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* is how many flights had survived every earlier criterion when this one ran and *share* is the removal as a fraction of those, not of the archive: a criterion late in the cascade tests a population the earlier ones have already thinned, so a percentage of the archive would understate it. Read this way one cut dominates, and it is not one the text argues a threshold for—21,089 paraglider flights, 11.9 % of those still standing, leave nothing behind after resampling. The marginal effect of each cut on the raw population is a different quantity and is what the `\StatScan*` numbers of Secs. 2.7.4 and 2.7.6 report. Generated by `generate_dataset_stats.py` from the `flights_meta` table the pipeline itself writes.

	paragliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
unreadable track	186,052	27	0.01	186,025
never airborne	186,025	225	0.12	185,800
integrity gate	185,800	1,442	0.78	184,358
shorter than the duration cut	184,358	2,107	1.14	182,251
longer than the duration cut	182,251	3	0.00	182,248
path below the floor	182,248	182	0.10	182,066
altitude activity below the floor	182,066	4,960	2.72	177,106
out of reach of its own first fix	177,106	228	0.13	176,878
no segment survived resampling	176,878	21,089	11.92	155,789
no segment carries the smoothing window	155,789	1	0.00	155,788
retained				**155,788**

	hang gliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
never airborne	6,716	15	0.22	6,701
integrity gate	6,701	73	1.09	6,628
shorter than the duration cut	6,628	86	1.30	6,542
path below the floor	6,542	1	0.02	6,541
altitude activity below the floor	6,541	153	2.34	6,388
out of reach of its own first fix	6,388	7	0.11	6,381
no segment survived resampling	6,381	249	3.90	6,132
retained				**6,132**

Figure 2.12. The processing chain, from the raw archive to the analysis ensemble. Stage numbers match Table 2.8; the badge to the right of each stage is the level it acts at (per fix, per flight, per segment). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record that is about to be repaired or dropped.

	Stage	Level	Effect	Adopted parameters	Sec.
1	Choice of altitude channel	flight	adopt the barometric channel as z; fall back to GNSS when the sensor is absent or stuck	stuck sensor: whole-flight barometric range $< 30\,\mathrm{m}$	2.7.1
2	Fix-level cleaning	fix (gate: flight)	delete implausible fixes; drop the flight when the share of airborne fixes this stage removes, or leaves for the resampler to rebuild, exceeds the integrity gate	• $v_{xy} \leq 45/55\,\mathrm{m\,s^{-1}}$ (para/hang) • $\lvert v_z \rvert \leq 13\,\mathrm{m\,s^{-1}}$ • $z \in [-100, 6000]\,\mathrm{m}$ • Hampel: $\pm 20\,\mathrm{s}$, $k = 5$, $\varepsilon_{\min} = 15\,\mathrm{m}$ • frozen lock: $\delta_{xy} = 15\,\mathrm{m}$, $\delta_z = 5\,\mathrm{m}$, $\tau_{\mathrm{freeze}} = 60\,\mathrm{s}$ • integrity gate $\leq 10\,\%$	2.7.2
3	Trimming of ground phases	flight	cut ground records before take-off and after landing; excise interior landings, splitting the flight	airborne: $v_{xy} > 5\,\mathrm{m\,s^{-1}}$ sustained for $30\,\mathrm{s}$; interior stint $\geq 10\,\mathrm{min}$ with the barometric altitude flat to within $5\,\mathrm{m}$	2.7.3
4	Flight-level filtering	flight	drop flights outside the cross-country envelope	duration $\geq 40\,\mathrm{min}$ and $\leq 16\,\mathrm{h}$; path $\geq 20\,\mathrm{km}$; altitude range $\geq 600\,\mathrm{m}$	2.7.4
5	Local Cartesian coordinates	flight	map (φ, λ, h) to ENU about the first fix of the trimmed track; horizontal position from (E, N); z kept as measured	exact WGS84 transform; no thresholds	2.7.5
6	Uniform sampling	flight → segment	fill the native grid Δt, bridging short holes; split at long gaps; drop a segment that is too short, or too sparse—too many of its grid times having to be interpolated	• split at gap $> \min\{10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\Delta t)\}$ • interpolated fraction of a segment $\leq 10\,\%$ • segment duration $\geq 90\,\mathrm{s}$	2.7.6
7	Smoothing and differentiation	segment	Savitzky–Golay fit per component $(E,\ N,\ z)$: position, velocity, acceleration	• polynomial order $p = 3$ • window $w = \max\{\mathrm{odd}(\tau_c/\Delta t),\ 5\}$ samples • $\tau_c = 5\,\mathrm{s}$, measured (provisional until the spectra are re-read on the cleaned ensemble)	2.7.7

Table 2.8. The pipeline stage by stage: the level each stage acts at, its effect, and the adopted parameters. Every value is typeset through a macro regenerated from `configs/preprocessing.yaml` by `generate_census_stats.py` (implementation details: Table 1.1), so the summary follows the configuration in force. Stage numbers match Figure 2.12.

Chapter 3

Global transport

This chapter measures the transport of the retained ensemble without segmenting a single flight. What can be established that way is a *class* of transport: how far a wing gets in a given time, how that growth is distributed, how long the motion remembers its direction, and how many distinct regimes the answer needs. What cannot be established is which flight behaviour produces it, and that is Chapter 4.

The order is the order of dependence. Everything rests on a statistical contract stated first, because the archive's flights are not independent records and its curves are not independent points, and an exponent quoted against the wrong denominator is the most common way a result of this kind is wrong. The ensemble average about take-off is then shown to be a measurement of the population rather than of the motion, and retired as a source of exponents. What replaces it is a within-segment, trend-robust family of estimators, on which the question of how many regimes there are can be put properly. The rest of the chapter tries to break whatever survives.

3.1 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level.

Mean-squared displacement (MSD). With every flight starting at the origin ($\mathbf{r}(0) = \mathbf{0}$), the *ensemble-averaged* MSD is the mean squared horizontal distance reached after an elapsed time t,

$$\mathrm{MSD}(t) = \langle\, |\mathbf{r}(t)|^2 \rangle, \tag{3.1}$$

where $\langle\cdot\rangle$ is an average *over flights* (an ensemble average) at fixed t. Its growth law $\mathrm{MSD}(t) \sim t^\alpha$ fixes the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport. On this archive it is measured together with the time average below and read against it: the ensemble curve is synchronised at take-off and turns out to cross over where the population leaves its launch area, so what it yields is that timescale rather than a single α, and the exponent is read off the time average.

Time-averaged MSD (TA-MSD). A second MSD estimator is built entirely *within* a single flight. For flight k of duration T_k, one slides a window of width t along the trajectory $\mathbf{r}^{(k)}$ and averages the squared displacement over all starting times t_0 *inside that flight*:

$$\overline{\delta^2}^{(k)}(t) \;\equiv\; \frac{1}{T_k - t} \int_0^{T_k - t} \left|\mathbf{r}^{(k)}(t_0 + t) - \mathbf{r}^{(k)}(t_0)\right|^2 \mathrm{d}t_0. \tag{3.2}$$

The average over t_0 here is a *time* average internal to flight k; it has nothing to do with the ensemble average $\langle \cdots \rangle$ over different flights used in the MSD. Because a single-flight curve $\overline{\delta^2}^{(k)}(t)$ is noisy, the operationally useful quantity is its mean over all flights of comparable duration T:

$$\mathrm{TAMSD}(t,T) \;\equiv\; \left\langle \overline{\delta^2}^{(k)}(t) \right\rangle_{T_k \approx T}. \tag{3.3}$$

Ergodic case. For an ergodic process, $\mathrm{TAMSD}(t,T) \to \mathrm{MSD}(t)$ as $T \to \infty$, independently of T, and the individual values $\overline{\delta^2}^{(k)}(t)$ concentrate around the common limit (the distribution collapses to a delta function).

Non-ergodic case: two independent signatures [20].

(i) *Aging.* $\mathrm{TAMSD}(t,T)$ differs from $\mathrm{MSD}(t)$ and retains a systematic dependence on T: flights of different duration yield different curves. This is a strong indicator of non-ergodicity, but it is sufficient, not necessary – some non-ergodic processes show only a weak T-dependence.

(ii) *Non-self-averaging.* The distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories remains broad even as $T \to \infty$: individual flights give systematically different TA-MSD values and the distribution does not collapse. This is the more fundamental signature: it persists even when aging is weak.

We therefore monitor both $\mathrm{TAMSD}(t,T)$ and the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories at fixed t and T. The canonical illustration – a CTRW with power-law waiting times – is worked out analytically in Appendix 3.A, where both EA-MSD$\neq$TA-MSD and the T-dependence are derived explicitly.

Propagator. Since each flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, the position at time t and the displacement from the start coincide: $\mathbf{r}(t) = \mathbf{r}(t) - \mathbf{r}(0)$. We therefore define the propagator as the probability density of the position vector $\mathbf{x} \in \mathbb{R}^2$ at time t,

$$P(\mathbf{x},t)\,\mathrm{d}^2x = \Pr\!\big(\mathbf{r}(t) \in [\mathbf{x},\, \mathbf{x} + \mathrm{d}\mathbf{x}]\big). \tag{3.4}$$

For a *self-similar* process with exponent H the propagator obeys the scaling form

$$P(\mathbf{x},t) = t^{-2H}\, G\!\left(\frac{\mathbf{x}}{t^H}\right), \tag{3.5}$$

where $G : \mathbb{R}^2 \to \mathbb{R}_{>0}$ is a time-independent scaling function (normalised, $\int G\,\mathrm{d}^2u = 1$).

In practice one works with the **one-dimensional marginal** obtained by projecting onto a single Cartesian component x,

$$P(x,t) \equiv \int_{-\infty}^{+\infty} P((x,y),t)\,\mathrm{d}y. \tag{3.6}$$

Assuming the process is *isotropic*, $G(\mathbf{u}) = G(|\mathbf{u}|)$, the integral can be evaluated by the substitution $y = t^H v$:

$$P(x,t) = t^{-2H} \int_{-\infty}^{+\infty} G\!\left(\sqrt{\left(\tfrac{x}{t^H}\right)^2 + v^2}\right) t^H\,\mathrm{d}v = t^{-H}\, F\!\left(\frac{x}{t^H}\right), \tag{3.7}$$

where $F(u) \equiv \int_{-\infty}^{+\infty} G(\sqrt{u^2+v^2})\,\mathrm{d}v$ inherits the normalisation $\int F\,\mathrm{d}u = 1$ from G. The marginal therefore satisfies the same scaling form as the 2D propagator, with the same exponent H but in one dimension. The shape of F distinguishes transport regimes: a Gaussian F characterises ordinary diffusion, while heavy power-law or stretched-exponential tails signal anomalous transport.

The marginal $P(x, t)$ is the object actually estimated from data: use the East component, the North component, or both. Pooling them presupposes isotropy, which Sec. 2.8 tests and rejects on this ensemble, so the two marginals are analysed separately. The scaling collapse consists in plotting $t^H P(x, t)$ against x/t^H for several values of t: all curves should fall on the single function F.

Probability of return to the origin. The value of the propagator at its centre, $P_0(t) \equiv P(\mathbf{0}, t)$, i.e. the height of the peak. For a self-similar process it decays as a power law set by the same exponent, $P_0(t) \sim t^{-dH}$ in d dimensions (t^{-2H} in the plane, t^{-H} for a single component). It therefore yields H from the *bulk* of the distribution rather than from its tails, which is valuable when heavy tails make the variance – and hence the MSD – ill-defined; a change of slope in $\log P_0$ versus $\log t$ exposes a crossover between regimes, and a peak exponent that disagrees with the MSD exponent is itself informative, pointing to multi-scaling. (This is the route Mantegna and Stanley [21] used to read off the index of Lévy-like walks.)

Non-Gaussianity. How far the propagator departs from a Gaussian, measured by the non-Gaussian parameter (in the plane)

$$\alpha_2(t) = \frac{\langle |\mathbf{r}(t)|^4 \rangle}{2 \left\langle |\mathbf{r}(t)|^2 \right\rangle^2} - 1, \tag{3.8}$$

which vanishes for a Gaussian and is positive for heavier-than-Gaussian tails (the factor 2 is the two-dimensional Gaussian value of $\langle|\mathbf{r}|^4\rangle/\langle|\mathbf{r}|^2\rangle^2$). Evaluated on the propagator $P(\mathbf{x}, t)$ at each elapsed time t, it is a function of t (one number per time) tracing a curve $\alpha_2(t)$ that pinpoints the scales over which the displacement statistics are non-Gaussian (typically largest at intermediate times, where occasional long glides dominate, and decaying if a central-limit Gaussianisation sets in at long times). It summarises the shape of P without replacing it.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 + \tau) \rangle$ (often normalised by $C(0)$) correlates the velocity at a time t_0 with the velocity a delay τ later: it is large and positive when the motion still points the same way, so the scale over which $C(\tau)$ decays sets the memory time of the heading. It controls the MSD through the Green–Kubo relation (for a stationary velocity) $\mathrm{MSD}(t) = 2\int_0^t (t - \tau)\, C(\tau)\, \mathrm{d}\tau$: a slowly decaying C produces a faster-growing MSD. The functional form of $C(\tau)$ is the diagnostic: an integrable C relaxes to normal diffusion, a non-integrable (e.g. power-law) tail sustains persistent super-diffusion with no crossover, and a sign change reveals the anti-persistence expected from circling.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The first-passage time $T(L)$ is the first time the horizontal distance from take-off reaches a value L (the first-exit time, the first time the trajectory leaves a disc of radius L) [22]. For a self-similar transport process the characteristic time scales with the length as $\langle T(L) \rangle \sim L^{1/H}$, with dynamic exponent $z = 1/H$, so the slope of $\log\langle T\rangle$ against $\log L$ gives an estimate of H obtained in the space domain, independent of the MSD; the full distribution of $T(L)$ additionally exposes heavy tails coming from long trapping events.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Which of them this thesis measures. The list above is the programme the observable admits, not the contents of this chapter. Table 3.1 says which entries are carried through to a number here, which are answered in a different form than the entry proposes, and which are set up and left. The distinction matters because several of the unmeasured ones are the natural next measurement rather than an afterthought: the first-passage time would give H in the space domain and the peak of the propagator would give it from the bulk of the distribution, so each is an independent check on the exponent this chapter does report.

Table 3.1. The catalogue of Sec. 3.1 against what is measured. One entry is answered in a form it does not name: the scaling collapse of the propagator is tested through its moments rather than through histograms, since all moments scaling with one exponent is the same statement about F in a form that needs no binning.

observable	status	where
ensemble MSD	measured	Sec. 3.1
time-averaged MSD	measured	Sec. 3.1
velocity autocorrelation	measured	Sec. 3.5
propagator, scaling collapse	through its moments	Sec. 3.5
non-Gaussianity	measured	Sec. 3.5
turning angles	not measured	—
return to the origin $P_0(t)$	not measured	—
first-passage times	tried, not usable	(implementation details: Sec. 3)
speed distributions	not measured	—

The measured MSD. The first of these observables is now computed, on the analysis ensemble defined at the close of Sec. 2.9 and characterised in Sec. 2.8: 155,788 paraglider and 6,132 hang-glider flights, 179,395 and 6,951 retained segments. Figure 3.3 shows both estimators. Every flight enters at the elapsed times its retained segments cover, with no re-zeroing at a split (Sec. 2.7.6), and a flight contributes to a lag only when it holds a fix within half its own native interval of it—so a 10 s logger says nothing about a lag of 1 s rather than answering with its fix at $t = 0$.

Two estimators, two questions. The two averages are not two attempts at one quantity, and nothing below should be read as choosing between them. The ensemble average $\mathrm{MSD}(t) = \langle|\mathbf{r}(t)|^2\rangle$ is taken over flights at a fixed elapsed time since take-off, and answers how far from its launch site the population has got after flying for t. The time average $\overline{\delta^2(\tau)} = \langle|\mathbf{r}(t_0+\tau) - \mathbf{r}(t_0)|^2\rangle_{t_0}$ is taken over starting times inside one record, and answers how far a wing travels in a window of duration τ placed anywhere in its flight. Both are computed on the same ensemble, both are reported, and their difference is a measurement in its own right (paragraph *The two averages* below). What differs between them here is something narrower: whether a single exponent is an adequate summary of each curve.

How well one exponent summarises each curve. Least squares returns a slope whatever the curve underneath it does, so each fit is reported with the residual it leaves. Over the range each estimator is fitted on, the *ensemble* curve departs from its own fitted power law by 15.2 % rms for paragliders and 31.2 % for hang gliders; the *time-averaged* curve departs from its own by 3.8 % and 3.6 %. The local slope, panel (c), says the same in the other units: across the fitted decades the ensemble curve runs from 1.31 to 2.21 (paragliders) and from 1.10 to 2.52 (hang gliders), against 1.64–2.05 and 1.63–2.18 on the time-averaged one.

The size of a residual is only half of what decides whether an exponent describes a curve, and taken alone it is misleading. A Wald–Wolfowitz runs test on the *sign* of the residuals says the rest. Over its fitted range the time-averaged curve leaves 34 residuals in 4 sign runs where independent scatter would give 18 ($z = -4.9$), and the hang gliders the same ($z = -5.0$); the ensemble curve is worse still, $z = -5.8$ and -5.5. The residuals of both curves are arranged, not scattered: each is a systematic arch and the fitted line is a chord across it.

So neither estimator is a power law in the strict sense, and the difference between them is of degree. The time-averaged curve departs from its line by a factor of four to eight less and its slope wanders over a range a third as wide, which is enough to make an exponent a useful summary of it and not enough to make it a description. Both exponents below are therefore reported as *effective* exponents over a stated range, and the question of how many regimes each curve really holds is left open here; it is what the sections below take up with estimators built for it. The time-averaged values are $\alpha_{\mathrm{TA}} = 1.732$ ($H = 0.866$) for paragliders and 1.688 ($H = 0.844$) for hang gliders, over 121 s to 6,341 s and 121 s to 7,149 s. Both lie well above the Brownian $\alpha = 1$ and below the ballistic $\alpha = 2$, so on the time-averaged statistic cross-country soaring is strongly super-diffusive over the range where it is read. The straight line through the ensemble curve, $\alpha_{\mathrm{EA}} = 1.918$ and 1.896, is drawn in panel (a) and generated as `\StatMsd*Alpha`; it is used below to locate the crossover it averages over, and it is not a growth law the motion obeys. The ensemble curve's own result is stated in *Origin of the crossover*: it is not an exponent but a timescale.

Why neither figure above carries a ±. Both are quoted bare, and the omission is deliberate. The natural uncertainty to attach is a bootstrap over flights: 200 ensembles of the same size drawn with replacement, each refitted over the same range, and the standard deviation of their exponents reported. On this archive it returns ±0.001, which is not an uncertainty on these numbers but an artefact of two assumptions the archive breaks.

It treats 155,788 flights as independent records. They are not: two wings launched from one site on one day flew the same air, and the intraclass correlation of a per-flight scaling statistic at that level is 0.57 (Sec. 3.3). And it treats the sampling error as the whole error, where the dominant term is the movement of the exponent when the fitted range changes — fifty times larger, on the estimator where both have been measured. A curve that is a systematic arch has no single exponent for a sampling error to be an error *of*.

The exponents that carry an interval in this chapter are those of Sec. 3.3, where the resampling unit is justified by measurement and the range dependence is reported beside the sampling term.

The least-squares error on the fitted slope is the other number one might quote, and for each estimator it is the larger of the two: 0.006 on the paraglider time average, 0.017 on the paraglider ensemble fit — still an order of magnitude below the range dependence. That is not a disagreement about the same quantity but a measurement of a different one. Least squares asks how well a *straight line* describes the fitted points, treating each residual as an independent draw; the residuals here are neither independent—every lag is an average over the same flights—nor noise, since the local slope drifts across the fitted decades. So the least-squares error is dominated by the curve's departure from a pure power law, which is why it is nearly three times larger on the ensemble fit than on the time-averaged one and tracks the residuals of the previous paragraph rather than the sample size.

Which of the two is larger depends on the size of the ensemble, and that is what makes the distinction easy to get backwards. On synthetic ensembles of fractional Brownian motion with $\alpha = 1.4$ and 300 flights, the scatter of the fitted exponent across 40 independent ensembles is 0.025, the bootstrap reports 0.023, and the least-squares error reports 0.004,7—understating the sampling uncertainty 5.3 times. The archive has five hundred times as many flights, and the bootstrap falls with $1/\sqrt{N}$ accordingly, to 0.001; the least-squares error does not fall with it, because what it is measuring is not sampling at all. Both numbers are generated, the second as `\StatMsd*AlphaErrOls`, and the quoted $\pm$ is always the first.

The shape of the ensemble curve. The ensemble curve is not a power law with scatter on it; it has a definite and reproducible shape. Its local slope falls to a minimum of 1.31 near 137 s, rises to 2.21 near 1,504 s, and settles back towards 2 beyond it; the hang gliders do the same between 1.10 and 2.52. The maximum is the informative part, because it lies *above* the ballistic value. No sum of power laws with positive coefficients and exponents at or below 2 can produce a local slope above 2: for $f = \sum_i a_i t^{\alpha_i}$ with $a_i > 0$, $\mathrm{d} \log f / \mathrm{d} \log t = \sum_i a_i \alpha_i t^{\alpha_i} / \sum_i a_i t^{\alpha_i}$ is a weighted mean of the α_i and cannot exceed the largest of them. A slope above 2 requires the ballistic component to switch on at a finite time rather than to be present from $t = 0$.

Controls on that shape. Three mechanisms would produce the same shape without any of it belonging to the motion, and each is measured rather than argued away.

The first is the coverage rule against a changing population: the ensemble behind $\mathrm{MSD}(t)$ grows as slow loggers reach their own cadence and shrinks as flights land, so an average taken over a drifting population can bend. Splitting the archive by native cadence removes the drift entirely—within one cadence class every flight answers every lag above that cadence—and on the paraglider archive the shape is unchanged: across the 3 classes the local slope at a given lag spreads by 0.06, against the 0.91 the shape itself covers, so the drift accounts for under a tenth of it. The hang-glider classes hold one to two thousand flights each and their spread, 0.78 against a swing of 1.43, is not small enough to settle anything; that archive's version of this control is the fixed-duration one below, whose cohorts are larger.

The second is survivorship: flights that last longer are not a random sample, and if they also travel faster the curve steepens for a reason that is not motion. The fixed-duration cohorts of Sec. 3.1 control it—within a cohort of flights lasting at least T, every member is present at every lag up to T—and inside a cohort the same minimum and the same overshoot are there, the local slope spreading across cohorts by 0.11 and 0.21. On a synthetic ensemble in which every flight is exactly Brownian and only the *speed* is correlated with the duration, the pooled exponent reads 1.11 on the ensemble estimator and 1.14 on the time-averaged one while every cohort of both recovers $\alpha \simeq 1.0$: the measured spreads are an order of magnitude below what pure selection produces.

The third is a heading shared across the ensemble, which would make $\langle |\mathbf{r}(t)|^2 \rangle$ grow ballistically for a trivial reason. It is bounded by the part of the mean square that the mean vector carries, $|\langle \mathbf{r}(t) \rangle|^2 / \langle |\mathbf{r}(t)|^2 \rangle$, which over the fitted range never exceeds 1.8 % for paragliders and 10.0 % for hang gliders. The flights do not share a heading. They share only a starting point.

Origin of the crossover. A shared starting point is enough to produce it. The ensemble average is synchronised at take-off, and a wing does not leave its launch site immediately: it climbs out, often circling within a few kilometres of it, and only then commits to a course. Until it does, its distance from take-off is bounded by the size of the local soaring area however fast it is flying, which is what depresses the local slope at intermediate lags; once the population has committed, the distance grows with the flown speed, which is what returns the slope to 2. Because the commitment happens at a spread of times rather than at $t = 0$, the transition overshoots.

The time at which it happens is measurable and is not a free parameter. Taking the first crossing of 5 km from take-off as the departure onto the cross-country leg—well beyond a thermalling excursion, well inside the median flown path of 89 km—95.1 % of paraglider flights depart, with a median departure time of 1,911 s and an interquartile range of 1,334 s to 2,739 s. The local slope peaks at 1,504 s; for hang gliders the median departure is 1,695 s against a peak at 1,334 s. In both disciplines the overshoot sits on the same scale as the departure and slightly before it, which is where a transition driven by a spread of onset times should sit, the slope being steepest while most of the population is still crossing. The departure radius is chosen rather than fitted, so the agreement corroborates the account and does not measure it.

Two consequences follow. The ensemble MSD about take-off is a statement about the geometry of the flight relative to its launch site as much as about the transport, and over the decade where the overshoot sits it is mostly the former. And an exponent fitted across that crossover is a straight line through a bend: $\alpha_{\mathrm{EA}} = 1.918$ is what least squares returns over it, not a growth law the motion obeys.

The two averages, and the gap between them. Panel (b) is the same displacement statistic averaged differently: over the starting times *within* each segment rather than over flights at a fixed elapsed time, $\overline{\delta^2(\tau)} = \langle |\mathbf{r}(t_0+\tau) - \mathbf{r}(t_0)|^2 \rangle_{t_0}$, taken inside a segment and never across the gap that ends it (Sec. 2.7.6). The two answer different questions. The ensemble is synchronised at take-off, so a feature it shows at a fixed *elapsed time* is a property of that common origin. The time average slides its window along each record and sees a given *lag* wherever it occurs, with no privileged origin. Neither is a proxy for the other, and the difference between them is the measurement: a feature the ensemble shows and the time average does not belongs to the common origin, and a feature both show is a property of the motion. The crossover of the previous paragraphs is of the first kind, which is why it appears in panel (a) and not in panel (b)—and that absence is what identifies it, not a defect of either estimator.

A systematic gap between the two averages is one of the signatures of a non-ergodic, aging process [20], and there is a gap: fitted over their respective ranges the ensemble slope exceeds the time-averaged exponent by 0.187 for paragliders and 0.208 for hang gliders. That gap is not evidence of aging, and this observable as computed cannot be made into evidence of it. Reading a gap between two exponents presupposes that both are exponents, and the ensemble figure is a straight line through a curve whose local slope covers 0.91; changing the fitted range moves it by more than the gap. The comparison the aging signature calls for is between two power laws of the same process, and the ensemble estimator does not supply one.

What can be said without an exponent is weaker and is what the measurement supports: displacement measured *from take-off* grows faster, over the second half of the fitted range, than displacement over a window of the same duration placed anywhere in the flight. The account above attributes that to the launch geometry rather than to aging. Separating the two properly needs the phase segmentation of Sec. 4.1—an aging process should show its exponent drifting with the time already flown when the same statistic is computed on successive stretches of a flight, and a launch-geometry artefact should not—and the ergodicity-breaking parameter of Appendix 3.A, which compares the *distribution* of $\overline{\delta^2}^{(k)}$ across flights rather than two fitted slopes, and so does not need the ensemble curve to be a power law at all. Both are next steps, not results.

The short-lag end. Below about 100 s the ensemble carries visible bumps of a few per cent. They come from the coverage rule meeting a lag grid of consecutive integers: the clock is zeroed at take-off, so a class of loggers of interval Δt has its fixes at multiples of Δt *in phase across flights*, and at a lag that is not such a multiple every one of them answers with a fix up to $\Delta t/2$ away, so the error alternates with the lag instead of averaging out. Splitting the archive by cadence measures it: 2 s loggers alone alternate by 6.5 % between even and odd lags, 1 Hz loggers—which meet every integer lag exactly—by −0.05 %, and the mixture by 0.70 % (Sec. 3.1).

A second effect is present over the same lags. The smoothing of Sec. 2.7.7 evaluates its polynomial off-centre at the ends of a segment, and the table flags those samples as `edge`; the ensemble estimator, unlike the dataset verifier, does not qualify its samples by that flag, so they enter. They are under 1 % of the fixes and they carry almost every step the fix-level speed bound would refuse: 26,368 in-segment steps above it touch a reconstructed sample against 32 that do not, out of 1,363,818,736. Their effect on the curve is confined to the same short-lag region: excluding them moves the MSD by up to 13 % below 10 s and by under 0.2 % above 60 s.

Neither effect touches anything quoted. Both are confined below 100 s, the lower end of every fit range is 121 s, and `coverage_limited_range` places it there. Nothing below 100 s in Figure 3.3 should be read as motion.

The mean, and how far it sits from the median. The two part company as the lag grows—at 40 s the mean is the median, at 400 s three times it, and past 30,000 s more than ten times—and that is a property of the distribution rather than a fault. The MSD *is* the mean of $|\mathbf{r}(t)|^2$, and that distribution broadens enormously with the lag: its 10–90 band spans two orders of magnitude by 1,000 s. A mean over a broad, right-skewed distribution sits well above its median. Where it matters is at the very longest lags, where the ensemble collapses—2,319 flights at 30,000 s, against 154,840 at 1,700 s—so the mean becomes a statistic over a few long flights and its standard error rises to 10 %. That is the reason the fit stops where it does and not at the end of the plotted range.

What a split leaves in the absolute position. One limitation belongs with the ensemble estimator specifically, and it is bounded here rather than repaired. A segment keeps its parent's origin and clock (Sec. 2.7.6), which is right for a boundary at a *gap*: the record stops there, the wing does not, and the position after it is genuine. It is not right for a boundary at a *re-acquisition offset*, where the jump belongs to the instrument, so everything after it carries an unknown constant. That constant enters $\langle|\mathbf{r}(t)|^2\rangle$, which is built from absolute position, and cancels out of $\overline{\delta^2(\tau)}$, which is built from increments.

Its size is measured on the written table rather than assumed. Most boundaries are *gaps*: 8,927 paraglider and 320 hang-glider flights carry at least one, and the displacement across them is large—hundreds of metres in the median—because it is a wing flying across a hole in its own log. The two kinds are told apart by the speed bound itself: across a gap the displacement satisfies $|\Delta\mathbf{r}| \leq v_{xy}^{\max}\Delta t$ like any other motion, whereas a re-acquisition offset exists *because* that inequality failed. Read that way, only 45 paraglider flights (0.029 %) and 28 hang-glider flights (0.457 %) carry a boundary the bound cannot explain, with median offsets of 1,172 m and 1,554 m. A kilometre-scale constant contributes of order $10^6\,\mathrm{m}^2$ to $|\mathbf{r}|^2$ on the flights that carry one, against a mean of order $10^8\,\mathrm{m}^2$ over the fitted range and on a fraction of the ensemble measured in the thousandths—under a per cent of the mean even for hang gliders, where the affected share is the larger. The separation is computed by `verify_dataset.py` on the retained table, which is the only place it can be computed honestly: the per-flight `split_jump_max_m` the cleaning records is measured before trimming, so it counts jumps inside a ground phase that never reach the analysis.

The results, and their scope. The two estimators return two results, of different kinds. From the time average, an effective exponent: 1.732 and 1.688, transport well above Brownian, on a curve whose departure from a single power law is small in size and systematic in arrangement. From the ensemble average, a timescale rather than an exponent: displacement from take-off crosses over near 1,504 s for paragliders and 1,334 s for hang gliders, where the population leaves its launch area, and no single power law spans that crossover. The second is a property of the ensemble's common origin, which is the thing an ensemble average is sensitive to and a time average is blind to; reporting it is the point of computing both.

What neither settles is the ergodicity question, and for a specific reason: its usual form compares α_{EA} with α_{TA}, and that comparison needs both curves to be power laws. It is taken up again in the forms that do not (paragraph *The two averages*). Both estimators also mix the climbs, the glides and the searches into one number. Separating those is what the rest of this chapter is for.

3.2 An estimator the archive does not defeat

Section 3.1 leaves the displacement statistic in an awkward position. Both averages of it are curves with structure a single exponent does not describe, and the ensemble one is dominated over the decade that matters by where flights start rather than by how they move. Neither failure is a defect of the archive: they are properties of a quantity built from absolute displacement, measured on flights that share an origin and fly a declared course.

Three things contaminate that quantity here, and they are worth naming together because one estimator disposes of all three.

The first is each flight's own course. A wing that crosses country has a mean velocity, and a mean velocity contributes $|\mathbf{v}_d|^2\Delta^2$ to any mean square displacement whatever the underlying process. The obvious repair is to subtract it, and the obvious repair is a trap: $\mathbf{v}_d$ has to be estimated, the natural estimate $[\mathbf{r}(T)-\mathbf{r}(0)]/T$ uses the record's own endpoint, and subtracting it turns the series into a bridge whose increments must sum to zero by construction. On synthetic fractional Brownian motion of known exponent $\alpha = 1.50$ the operation returns 1.35, and the distortion has no safe range: the centred variation is already 2 % below the raw one at a lag of 0.05 % of the record and 76 % below at half of it (implementation details: Sec. 3).

The second is the declared task. The CFD scores closed courses, and 56 % of retained paraglider flights fly a triangle. A triangle returns towards its start, so the displacement saturates by construction, and it does so at lags set by the task rather than by the air.

The third is the population. Section 2.8 measured the strata of this archive as incompatible on the ensemble MSD, by 40 % across wing classes and 62 % across orographic groups. A superposition of ordinary processes with different persistence times imitates a single anomalous one, which is the most economical way this measurement could be wrong.

Filtering rather than subtracting. A finite difference of order p annihilates every polynomial of degree $p-1$ identically. Writing A_p for that difference taken at spacing Δ and summing over the two horizontal components, the *filtered variation*

$$V_p(\Delta) = \left\langle \left|A_p\,\mathbf{r}(\Delta)\right|^2 \right\rangle \tag{3.9}$$

scales as Δ^{2H} for a self-similar process of exponent H, at every order. What differs between orders is what each removes before measuring: $p = 1$ is the plain increment and removes a constant offset, $p = 2$ removes a constant velocity, $p = 3$ a constant acceleration. No drift is ever estimated, so none can be estimated badly.

The order scan is the measurement rather than a check on it. The difference $\alpha_1 - \alpha_2$ is the part of the apparent exponent that was course, read off directly; the agreement $\alpha_2 \simeq \alpha_3$ is the certificate that nothing polynomial is left to remove. On synthetic data with a course of $8\,\mathrm{m\,s^{-1}}$ imposed on a process of true exponent 1.50, order 1 returns 1.89 and orders 2 and 3 return 1.51 and 1.51.

Everything below is computed within a retained segment, never across the gap that ends one, and never on displacement from take-off. The estimator is `soaring.analysis.observables.variations`; the pass that applies it to the archive keeps one curve per flight, so every stratification in this section is a row selection rather than another traversal of the fix table (implementation details: Sec. 3).

3.3 The measurement

The fitted range is stated rather than discovered: 60 s to 2,000 s, which is 1.5 decades. Its floor sits above the smoothing scale of the slowest logger retained — five samples at 10 s is 50 s, so nothing below it is a statement about motion — and its ceiling below the lags at which flight-duration censoring begins to select the ensemble. Every exponent is quoted with the range it was fitted over and with how much it moves when that range changes.

Table 3.2. The order scan over 60 s to 2,000 s. Order 1 is the plain increment and still carries each flight's course; orders 2 and 3 annihilate a constant velocity and a constant acceleration, and agree with each other, which is what says nothing polynomial is left. The uncertainty is the combination of a day-and-site bootstrap and the movement of the exponent when the fitted range is halved; the two are given separately because they differ by a factor of fifty and only the second is worth arguing about.

	order p	α_p	sampling	range
	1	1.75 ± 0.06	0.001	0.062
paragliders	2	2.02 ± 0.10	0.001	0.104
	3	2.04 ± 0.14	0.001	0.140
	1	1.70 ± 0.06	0.002	0.063
hang gliders	2	1.94 ± 0.13	0.003	0.128
	3	1.97 ± 0.20	0.003	0.198

Table 3.2 gives the scan, and Figure 3.1 the curves behind it. Three things in them are the result.

Removing the course raises the exponent. α_1 and α_2 differ by -0.27 for paragliders and -0.24 for hang gliders, and the sign is the opposite of the one a residual drift would produce. A drift inflates a displacement statistic; what happens here is that the archive's closed tasks *suppress* it, because a triangle comes home. The plain increment sees that and reads an exponent that is too low; the second-order variation does not see it at all.

That reading is confirmed where it can be tested directly. Fitting closed and open tasks separately, the gap in the exponent is $+0.11$ at order 1 and -0.10 at order 2, and on a wider range the first grows to $+0.28$ while the second does not move. A difference of curvature does not care where the course ends.

Orders two and three agree. By 0.02 for paragliders and 0.02 for hang gliders. Nothing polynomial is left in the trajectory that the second difference has not already removed, so the exponent is not a residue of a trend of any degree.

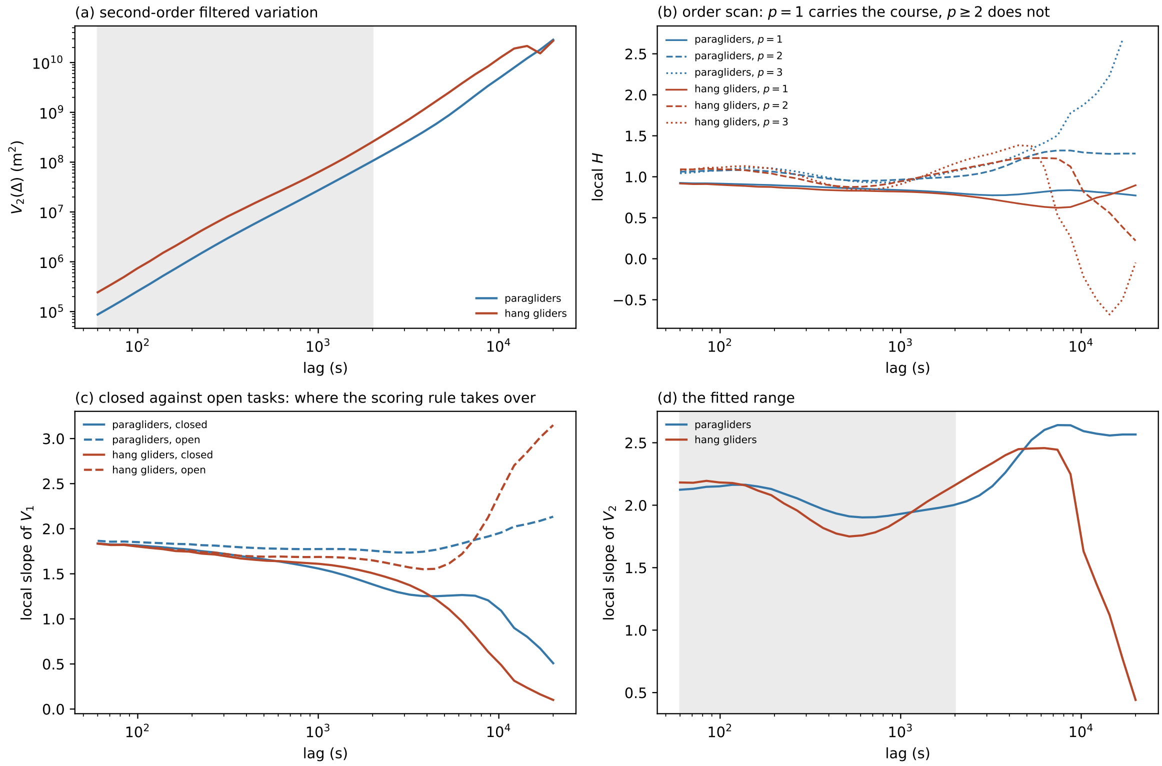

Figure 3.1. The filtered variation, on which the exponent is read. (a) $V_2(\Delta)$ against lag, log–log, with the fitted range shaded. (b) The local $H = \frac{1}{2}\, \mathrm{d}\log V_p / \mathrm{d}\log\Delta$ at the three filter orders: inside the fitted range $p = 1$ runs below $p = 2$ and $p = 3$, which is the flown course being read as a lower exponent, and the three separate beyond it. (c) The same local slope at $p = 1$, with the ensemble split by declared task; closed and open courses track each other to about 3,000 s and part company after it, which is where the scoring rule rather than the air governs the displacement. (d) The local slope of V_2, the quantity the fit averages over the shaded range. Computed by `scripts/reporting/generate_transport_figure.py`.

Order one reproduces the time-averaged MSD. It should: the first-order filtered variation *is* $\overline{\delta^2(\Delta)}$ under another name. That makes the comparison a check on the two implementations rather than on the physics, and it is worth stating because the two were written separately, traverse the fix table separately and are fitted over different ranges — 60 s to 2,000 s here against 121 s to 6,341 s in Sec. 3.1. They return $\alpha_1 = 1.75$ against $\alpha_{\mathrm{TA}} = 1.732$ for paragliders and 1.70 against 1.688 for hang gliders: closer, in both disciplines, than the movement either shows when its own range is halved.

The uncertainty is not the sampling error. The day-and-site bootstrap returns 0.001 on α_2; refitting each half of the range returns 0.104. The systematic is fifty times the statistical, and a number quoted with the second alone would carry four significant figures where its second is not determined. The choice of resampling unit is measured rather than asserted: the intraclass correlation of the per-flight variation is 0.00 at the flight level, 0.57 at day and site, 0.15 at day and 0.29 at pilot, so the resampling unit is the day and site, of which the archive holds 84,132.

3.4 Trying to break it

An exponent that survives only the analysis that produced it has not been measured. Four stratifications, each a row selection on the per-flight curves, ask whether the number belongs to the motion or to who is in the ensemble.

Table 3.3. What the exponent does when the ensemble is cut four ways, at order 2, over the same range. The comparison that matters is with Sec. 2.8, where the same strata departed from the pooled ensemble MSD by 40 % and 62 %: the heterogeneity that makes the ensemble estimator unusable does not reach this one.

cut	strata	α_2 range	spread
native cadence	6	2.01–2.03	0.02
wing class	6	1.97–2.03	0.06
season	17	2.00–2.03	0.03
declared task	2	—	−0.10

The pipeline is not being measured. The Savitzky–Golay window has a floor of five *samples* (Sec. 2.7.7), so a 10 s logger is smoothed over 50 s and a 1 Hz one over 5 s: a factor of ten in the scale below which the trajectory has been made smooth. Across the 6 cadence classes the exponent spreads by 0.02. Whatever the estimator is reading, it is not the filter.

The heterogeneity does not reach it. Across wing classes the spread is 0.06 and across seasons 0.03. This is the sharpest contrast the chapter offers: on the ensemble MSD of Sec. 2.8 the same wing classes departed from the pooled curve by 40 % and the same terrain groups by 62 %. A mixture of populations with different speeds and different courses is a mixture of *trends*, and a trend is what the second difference removes.

How many regimes the range supports. Fewer than the curve appears to show. Breakpoint selection on the second-order variation returns 1 break for paragliders and 2 for hang gliders, but the same procedure run on surrogates that are a single power law plus the measured sampling noise returns a break 16 % and 34 % of the time. Neither count is established.

The reason is the resolution of the data rather than the fitting. The residuals of any fit to these curves are correlated across lags, because every lag averages the same flights, so 22 lags carry no more than 2.0 independent points — the floor the estimator will report, reached in both disciplines. On that budget one breakpoint is at the edge of what is estimable and two are not, and the honest statement is a single effective exponent over a declared range — which is how Table 3.2 reports it.

3.5 Beyond the second moment

An exponent is a statement about one moment of one distribution, and two processes can agree on it and be different physics. A correlated Gaussian process and a Lévy walk both give $\langle|\Delta\mathbf{r}|^2\rangle \sim \Delta^{2H}$; what separates them is the rest of the spectrum. This section asks three questions the exponent cannot answer, and the first of them settles which of those two the archive is.

The whole spectrum of moments. Writing $\langle|\Delta\mathbf{r}(\Delta)|^q\rangle \sim \Delta^{q\nu(q)}$ and reading $q\nu(q)$ as a function of q: a self-similar monofractal process gives a straight line through the origin, $\nu(q)$ constant, while a Lévy walk gives a bilinear one whose knee sits at its tail index and whose right-hand branch has slope one, the ballistic front. The discrimination is visual rather than a delicate fit, which is what makes it worth a full traversal of the archive.

Over 60 s to 8,000 s and q from 0.50 to 4, the spectrum is straight. $\nu(q)$ varies by 0.019 across the whole range of q for paragliders and 0.009 for hang gliders, and the departure from a line through the origin is 0.0099 and 0.0056 in rms — an order of magnitude below what a knee would leave, and below the threshold at which a knee is declared at all (implementation details: Sec. 3). Neither archive shows strong anomalous diffusion.

That is a negative result and it is the informative kind. The Lévy walk is the model this thesis was framed around; on the whole trajectory, unsegmented, the archive does not support it. What it supports is a monofractal process: one exponent governing every moment.

The moments are estimable rather than merely computed. The share of each moment carried by the largest one per cent of its increments reaches 28 % at worst, and 16 % at $q = 4$; above a half a moment is one flight and is not an estimate, and nothing here is close.

Monofractal is not the same as Gaussian. The natural next step is to call the process a correlated Gaussian motion, and the same moments refuse it. One exponent for every moment is a statement about how the distribution *scales*; it says nothing about its *shape*, and the non-Gaussian parameter $\alpha_2 = \langle|\Delta\mathbf{r}|^4\rangle/2\langle|\Delta\mathbf{r}|^2\rangle^2 - 1$, which the fourth and second moments already carry, measures the shape directly. It vanishes for a Gaussian.

Over 60 s to 1,805 s it is +0.04 in the median for paragliders and +0.18 for hang gliders, flat across the range in both — it moves by +0.02 and −0.02 from one end to the other.

The calibration is what makes those numbers readable, and it has to be read at the right sample size. On an exact fractional Brownian motion the estimator returns 0.02 over the modest sample its test uses and collapses to within a hundredth of zero, sign included, when the sample grows — so the offset is finite-sample bias and not a floor. On a Lévy walk of tail index 1.5 it returns 0.55 and does not move (implementation details: Sec. 3). Against that pair, both archives are non-Gaussian and neither is close to Lévy; but they are not equally so, and the chapter should not average them. The hang gliders sit an order of magnitude above the Gaussian value and the paragliders a few times above it. Whatever makes these increments heavier-tailed than Gaussian acts about four times more strongly on the smaller, slower discipline.

Beyond the fitted range α_2 climbs, to +0.47 and +0.60 at the longest lags in panel (c). Those lags are where the declared task governs the trajectory and where the fourth moment is carried by the fewest flights, so the climb is reported and not quoted; the range over which every exponent in this chapter is read stops below it for the same reason.

How long a heading survives, and why that is consistent with the exponent. The velocity autocorrelation is positive throughout and decays without ever changing sign: $C = +0.37$ at the lower end of the grid and no sign change anywhere, where a change of sign would be the anti-persistence a circling wing produces. It is also the one observable in this chapter whose range is set by the pipeline rather than by the archive: the grid starts at 60 s because that is the smoothing scale of the slowest logger, and the correlation falls below $1/e$ by 74 s, within the first two lags of the grid.

That last fact and the exponent look incompatible, and it is worth showing that they are not. A memory already spent by 60 s would leave the motion diffusive over a range reaching to 2,000 s, whereas α_2 says the increments are still correlated across it. Green–Kubo reconciles them only through the shape of the decay: what matters is not when C becomes small but whether it becomes *integrable*, and a tail $C(\tau) \sim \tau^{-\gamma}$ with $\gamma < 1$ sustains $\langle|\Delta\mathbf{r}|^2\rangle \sim \Delta^{2-\gamma}$ however small C has already become. Over 60 s to 1,179 s the measured tail is a power law of $\gamma = 0.61$ for paragliders and 0.57 for hang gliders — non-integrable in both — so the memory is not spent at 60 s, only small.

The comparison is one-sided and is reported as such. The correlation is estimated per flight with that flight's own mean velocity removed, which pulls every lag down by about the record-mean of C and so steepens the tail, and the first decade of the Green–Kubo integral lies below the grid; both push γ up, so $2-\gamma$ is a floor rather than an estimate. That floor is 1.39 and 1.43, and the measured $\alpha_1 = 1.75$ and 1.70 clear it. A velocity channel whose memory decayed faster than this would have contradicted the position channel; it does not.

Whether there is a scale to separate. The persistence runs decompose each flight into non-overlapping stretches whose sinuosity — arc length over chord — stays under a threshold, and their durations are the geometric counterpart of a waiting time. Their tail index is what a Lévy walk would tie to the exponent, and it is not stable: scanning the threshold over $s_{\max} \in \{1.05, 1.15, 1.30\}$ moves β from 4.92 to 3.07 to 2.24, a spread of 2.69.

A run length that depends this strongly on the angle at which a turn is declared is a run length with no natural scale, which is a second and independent way of saying what the moment spectrum says: there is no distinguished flight-leg duration for a Lévy description to be built on. It is also a prediction about Chapter 4. A segmentation into climbs and glides declares a threshold of the same kind, and this measurement says the result will move with it.

The runs are decomposed without overlap on purpose. Sampling the run length at every instant returns the length-biased distribution, whose tail index is one below the truth; on synthetic runs of known index 2.0 the two estimators return 2.0 and 1.1 (implementation details: Sec. 3). At the values seen here that would not be a bias to correct but a different answer.

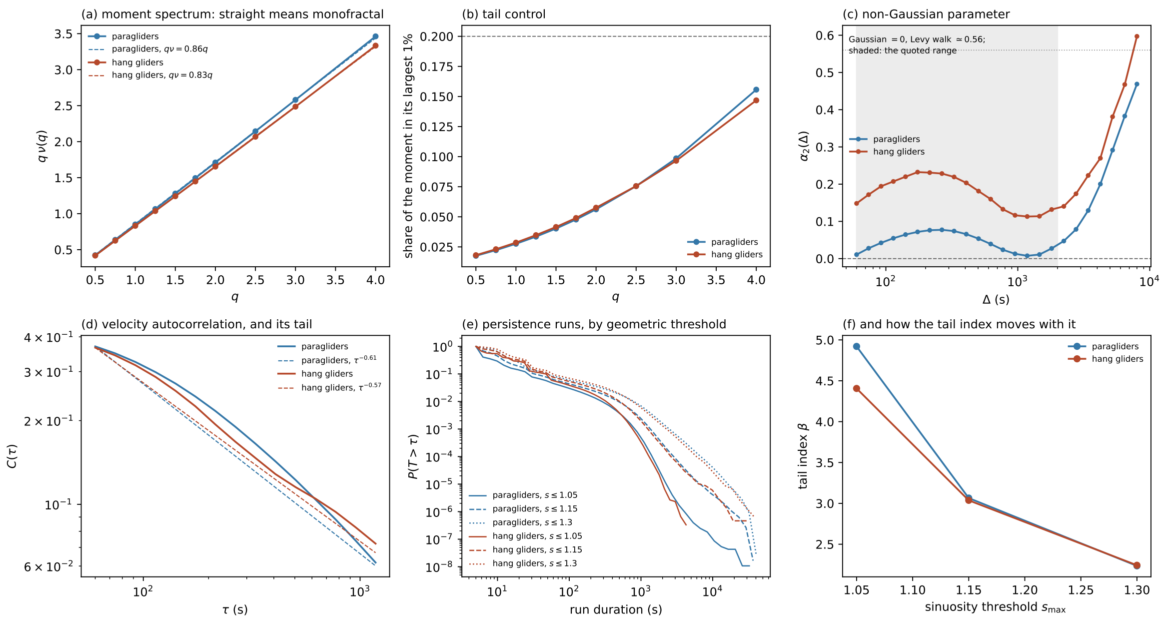

Figure 3.2. What the second moment cannot say. (a) The moment spectrum $q\nu(q)$ with the straight line $q\nu = \nu q$ dashed: straight means one exponent governs every moment, and a Lévy walk would bend here with a knee at its tail index. (b) The share of each moment carried by the largest 1 % of its increments; above the dashed line a moment is being set by a handful of flights. (c) The non-Gaussian parameter against lag, with the Gaussian value and the Lévy-walk calibration marked and the quoted range shaded; it is flat and clearly above zero inside that range and climbs outside it, where the declared task governs the trajectory. (d) The velocity autocorrelation, log–log with the fitted tail dashed: positive throughout, below $1/e$ within the first two lags of the grid — whose lower end is the smoothing scale rather than a property of the flying — and decaying too slowly to be integrable. (e) Survival of the persistence-run durations at three geometric thresholds; that they do not lie parallel is the statement that no run length is distinguished. (f) The same thing as one number: the fitted tail index against the threshold that defines a run. Computed by `scripts/reporting/generate_shape_figure.py`.

3.6 What this establishes

Over 60 s to 2,000 s, on an estimator that removes any polynomial trend and that the archive's cadence, wing class and season move by at most 0.06 in the exponent — the declared task by −0.10, the largest of the four — cross-country soaring gives $\alpha_2 = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders. The two disciplines agree within their uncertainties, which is not guaranteed — they differ in speed, in glide ratio and in where they fly.

Against the published value. Vilpellet et al. [7] report a global $H \approx 0.88$ for human soaring flight, on a different sample. The comparison is worth making carefully, because this chapter returns two numbers and only one of them is the same quantity. The plain increment, which is a displacement statistic with no trend removed, gives $H = 0.876$ for paragliders and 0.852 for hang gliders — the first of which rounds to the reported value, on an independent archive and at the precision that value is quoted to. The second-order variation gives 1.009 and 0.972, higher by about a tenth.

The gap is not a disagreement about the transport but the measurement of Sec. 3.3 seen from the other side: the difference between the two orders *is* the flown course, and on this archive the course *suppresses* displacement rather than inflating it, because the majority of the retained flights fly a closed triangle that comes home. A published exponent obtained without removing it is therefore expected to sit below one obtained with it, by about the amount seen. Which of the two answers a given question depends on the question — how far a wing gets is the first, what class of process the trajectory is is the second — and the reference figure is reproduced by the estimator that matches it.

An exponent of two is the boundary of the self-similar family: $\alpha = 2H$ with $H = 1$ is motion whose increments stay correlated at every lag in the range. It is worth being careful about what that does *not* mean, because the natural reading is wrong. It is not the signature of locally straight flight. A process observed inside its own persistence time is differentiable, and its second-order variation reads near three rather than two: measured on a persistent random walk of correlation time 3,000 s, the same estimator over the same lags returns 3.02, and on a straight line it returns identically zero. A measurement of two says the trajectory is persistent and still rough at every lag of the range — correlated, not smooth.

What the chapter cannot do is extend that range. Below it the trajectory has been smoothed by the pipeline; above it the declared task takes over, and the separation between closed and open courses grows from 0.03 in the local slope at 60 s to 1.13 at 12,000 s. The window in which this archive speaks about transport rather than about the scoring rule is 1.5 decades wide, and the pre-asymptotic character of what is measured in it follows from the system rather than from the method: a soaring day ends, so there is a physical cutoff and no amount of data moves it.

That is the limit of what the whole trajectory can say. It gives one exponent, over one window, for a motion that mixes climbing, gliding and searching into a single number. Chapter 4 is about separating them.

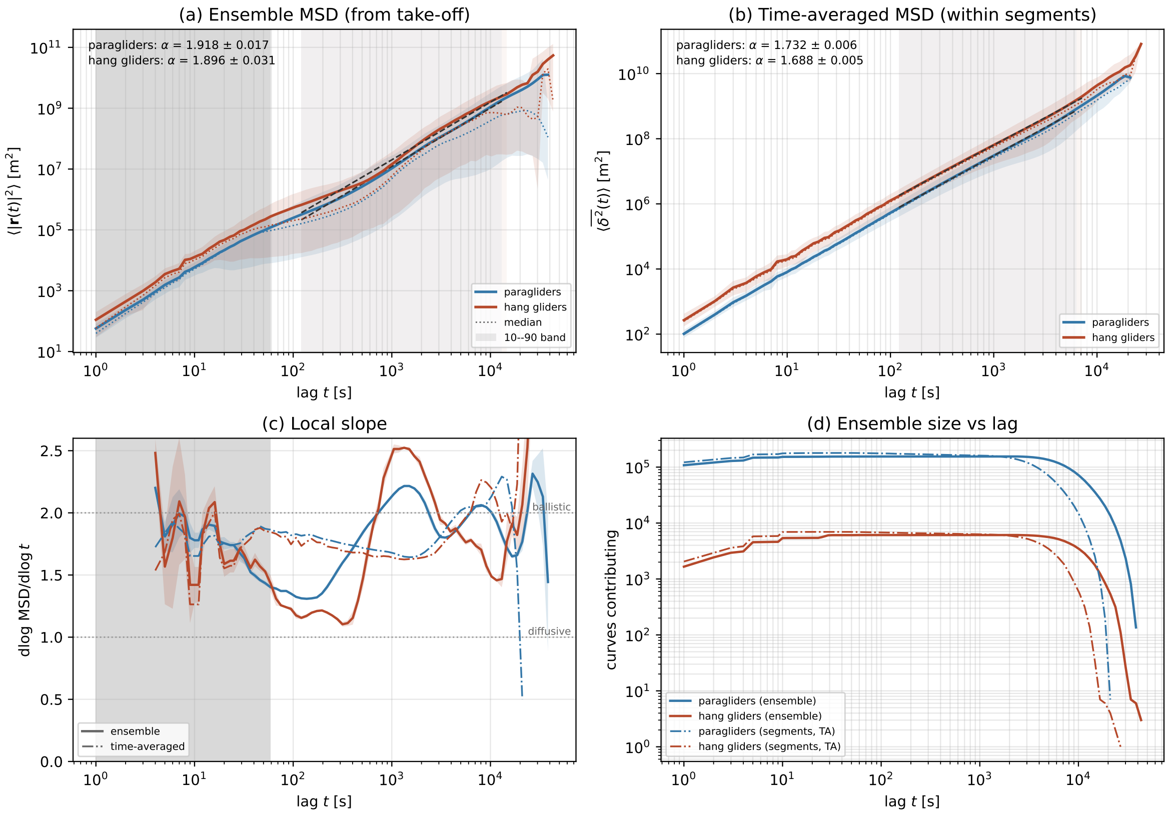

Figure 3.3. Mean-squared displacement of the retained ensemble (155,788 paraglider and 6,132 hang-glider flights), by two averages. (a) The *ensemble* MSD $\langle|\mathbf{r}(t)|^2\rangle$ against elapsed flight time, log–log, with the fitted straight line dashed, its range shaded, and the 10 % to 90 % band across flights with the median dotted. (b) The *time-averaged* MSD $\overline{\delta^2(\tau)}$, averaged over starting times within each segment and cut at half a segment's length. (c) The local logarithmic slope $\mathrm{d}\log \mathrm{MSD}/\mathrm{d}\log t$ of both, over a window of ± 0.15 decades, with the ballistic and diffusive values marked. (d) The flights, or segments, contributing at each lag, on which the upper end of each fit range is set; the rising limb is shaded in (a) and (c), where the average is still taken over a growing population. Computed by `scripts/reporting/generate_msd_figure.py` from the `fixes` table; curves written alongside as `msd_curve.csv`.

Chapter 4

Flight phases, and what they explain

Chapter 3 determines which class of transport the archive shows. It cannot say which flight behaviour generates it, because every observable there is an average over climbs, glides and searches together. Separating them needs a segmentation of each flight into its phases, and that is the work this chapter plans.

The distinction is worth stating plainly, because it sets what a segmentation is allowed to claim. Pre-segmentation determines *which* class of transport is present; segmentation explains *which mechanism* produces it. A segmentation explains an exponent, it does not measure one — and if the two disagree, the disagreement is evidence about the segmentation rather than about the transport.

None of this chapter is implemented. Each section states what is to be built and what it would settle.

4.1 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [7], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed "step".

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermalling). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized "trap".

That work performs the segmentation with a hidden Markov model [23] whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.2 Phase-resolved observables

The segmentation gives access to a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 3.1, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c, the search duration τ_s, and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction matters: Vilpellet et al. [7] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$, the quantity that determines whether trapping drives sub-diffusion, would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz, mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g, with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ–τ_g scatter, the conditional mean $\langle \ell \mid \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 3.1. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states $\{\text{transition}, \text{search}, \text{climb}\}$, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \Pr(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight ($\mathrm{T} \to \mathrm{S} \to \mathrm{C} \to \mathrm{T}$) gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide.

Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. These measurements determine whether a memoryless model is admissible.

4.3 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [7]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 2), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the $\sim$30 s thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios ($\approx$ 8.5, 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx$60% transition, $\approx$35% climb, the remainder searching). The global exponent is not on this list: it needs no segmentation, and Sec. 3.6 has already put this archive's value beside the reported $H \approx 0.88$.

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.4 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.5 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

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Appendices

Each appendix is numbered after the body chapter that first cites it: Appendices 2.A and 2.B belong to Chapter 2, Appendix 3.A to Chapter 3. They follow one another in the order in which the text first cites them.

Appendix 2.A

Power spectral density and Welch's method

This appendix explains the tool used in Chapter 2 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method* [24]. No prior familiarity with spectral estimation is assumed.

The power spectral density

Consider a signal sampled at a fixed rate, here an altitude series $z_n = z(n\,\Delta t)$, $n = 0, \ldots, N-1$, with sampling interval Δt (so the sampling frequency is $f_s = 1/\Delta t$). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** describes how the power (the variance) of the signal is distributed across frequencies. A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD $S(f)$ is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f)\,\mathrm{d}f \;=\; \mathrm{Var}(z), \tag{2.A.1}$$

with units of (metres)^2 per hertz. The upper limit $f_s/2$ is the *Nyquist frequency*: a series sampled every Δt cannot resolve anything faster than one oscillation per two samples. This separates a slowly drifting but locally clean channel (power concentrated at low f) from one with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

The periodogram

The naive estimator of $S(f)$ is the **periodogram**: take the discrete Fourier transform $\hat{z}(f) = \sum_n z_n\, e^{-2\pi i f n \Delta t}$ and form $|\hat{z}(f)|^2$ (suitably normalised). It is asymptotically unbiased but *not consistent*: its variance does not shrink as the record gets longer, because a longer signal gives more frequency points, each as noisy as before. Plotted directly it is a spiky curve, too noisy for a reliable comparison of two channels.

Welch's method

Welch's method turns the noisy periodogram into a usable estimate by *averaging*. It proceeds in three steps:

(i) **Segment.** Split the record into several shorter, overlapping segments.

(ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.

(iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging K segments cuts the variance of the estimate by roughly a factor K, at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals; it is the standard, robust spectral estimator. Each segment is typically detrended before transforming (e.g. by removing a linear fit), so that the estimate reflects the fluctuations about the trend, where the sensor noise lives, rather than the trend itself (implementation details: Sec. 2.7.1).

The spectra of the two altitude channels, and the choice of channel they motivate, are discussed in Sec. 2.7.1; this appendix covers only the method.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.6), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a, flattening f, $b = a(1-f)$ the semi-minor axis, $e^2 = f(2-f) = 1 - b^2/a^2$ the first eccentricity squared [14].

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O: the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X–Z plane by λ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p\cos\lambda, \qquad Y = p\sin\lambda, \qquad Z = Z. \tag{2.B.1}$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h.

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \tag{2.B.2}$$

The geodetic latitude is defined through the surface *normal* (Figure 2.4). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2,\ Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan\varphi = \frac{Z_s/b^2}{p_s/a^2} \qquad \Longrightarrow \qquad Z_s = \frac{b^2}{a^2}\,p_s\tan\varphi = (1-e^2)\,p_s\tan\varphi. \tag{2.B.3}$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1-e^2)$,

$$\frac{p_s^2}{a^2}\Big(1 + (1-e^2)\tan^2\varphi\Big) = 1 \quad \Longrightarrow \quad p_s^2 = \frac{a^2\cos^2\varphi}{\cos^2\varphi + (1-e^2)\sin^2\varphi} = \frac{a^2\cos^2\varphi}{1 - e^2\sin^2\varphi}, \tag{2.B.4}$$

that is,

$$p_s = N(\varphi)\cos\varphi, \qquad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2\sin^2\varphi}}, \tag{2.B.5}$$

and, back through (2.B.3),

$$Z_s = (1 - e^2)\, N(\varphi) \sin\varphi. \tag{2.B.6}$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a short computation shows it is precisely the distance, along the normal, from the surface point to the polar axis (the segment Q–P of Figure 2.4): the normal direction in the meridian plane is $(\cos\varphi, \sin\varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N\cos\varphi = 0$ by (2.B.5).

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \tag{2.B.7}$$

Adding $h\,\hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h)\cos\varphi\cos\lambda, \qquad Y = (N + h)\cos\varphi\sin\lambda, \qquad Z = \big(N(1 - e^2) + h\big)\sin\varphi, \tag{2.B.8}$$

which is Eq. (2.6). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $(N(1 - e^2) + h)$ is the ellipsoid's flattening at work: on a sphere $(e = 0)$ both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan\psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2)\sin\varphi}{N\cos\varphi} = (1 - e^2)\tan\varphi, \tag{2.B.9}$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where tan is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2. Using $\tan(\varphi - \Delta) \simeq \tan\varphi - \Delta\,(1 + \tan^2\varphi)$ in (2.B.9),

$$\tan\varphi - \Delta\,(1 + \tan^2\varphi) \simeq (1 - e^2)\tan\varphi \quad \Longrightarrow \quad \Delta \simeq \frac{e^2\tan\varphi}{1 + \tan^2\varphi} = e^2 \sin\varphi\cos\varphi, \tag{2.B.10}$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2}\,\sin 2\varphi, \tag{2.B.11}$$

which is the expression quoted in Sec. 2.7.5. It is largest where $\sin 2\varphi = 1$, i.e. exactly at $\varphi = 45°$, and there equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192° \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus\,(e^2/2) \approx 21$ km at mid-latitudes—the $\sim$20 km quoted in the body. The error vanishes at the equator and at the poles and is symmetric about 45°.

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing to the pole; Up, along the ellipsoid normal (2.B.7). East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial\mathbf{X}/\partial\lambda}{\|\partial\mathbf{X}/\partial\lambda\|} = (-\sin\lambda_0,\ \cos\lambda_0,\ 0), \tag{2.B.12}$$

Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0), and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin\varphi_0 \cos\lambda_0,\ -\sin\varphi_0 \sin\lambda_0,\ \cos\varphi_0). \tag{2.B.13}$$

The ENU components of a displacement $\Delta\mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E \cdot \Delta\mathbf{X}$ and likewise for N and U; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix}}_{R} \Delta\mathbf{X}, \tag{2.B.14}$$

whose rows are orthonormal by construction, so R is a rotation ($R^{\mathsf{T}} R = \mathbb{1}$) [18]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.6), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Working in a fixed tangent frame replaces distances measured *along* the curved surface by straight-line (chord) distances in the plane. For two surface points separated by an arc d on a sphere of radius $R_\oplus$, the chord is

$$c = 2R_\oplus \sin\frac{d}{2R_\oplus} = d - \frac{d^3}{24\,R_\oplus^2} + O(d^5), \tag{2.B.15}$$

so the horizontal distance error of the tangent-plane picture is $d^3/(24R_\oplus^2)$: about 1 cm at $d = 20$ km and 13 cm at $d = 50$ km — far below the metre-scale GPS noise on the same displacement, which is the comparison made in Sec. 2.7.5. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2/(2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the barometric altitude itself (Sec. 2.7.5), and the U component serves only the geometric construction.

Appendix 3.A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T.

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2 \rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha\, \tau_0^\alpha}{\tau^{1+\alpha}}, \qquad \tau \to \infty, \quad 0 < \alpha < 1. \tag{3.A.1}$$

The mean waiting time diverges ($\langle \tau \rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t, $T_n = \sum_{i=1}^{n} \tau_i$ the time of the n-th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \tag{3.A.2}$$

the position at time t.

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2 \, \langle n(t) \rangle. \tag{3.A.3}$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ. Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau)\, \mathrm{d}\tau \simeq 1 - \Gamma(1-\alpha)\, (\tau_0 s)^\alpha + \cdots$ for $s \to 0$ (a consequence of ψ having a power-law tail with exponent $1 + \alpha$ and amplitude $\alpha \tau_0^\alpha$), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\Gamma(1-\alpha)\, \Gamma(1+\alpha)\, \tau_0^\alpha} = \frac{\sin \pi\alpha}{\pi\alpha} \Big(\frac{t}{\tau_0} \Big)^\alpha, \qquad t \to \infty. \tag{3.A.4}$$

Hence the **EA-MSD** grows sub-linearly,

$$\mathrm{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\Gamma(1-\alpha)\, \Gamma(1+\alpha)\, \tau_0^\alpha}\, t^\alpha \equiv K_\alpha\, t^\alpha, \qquad 0 < \alpha < 1. \tag{3.A.5}$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T, the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t;T) = \frac{1}{T-t}\int_0^{T-t} \left[x(t_0+t) - x(t_0)\right]^2 \mathrm{d}t_0. \tag{3.A.6}$$

Since $\langle [x(t_0+t) - x(t_0)]^2 \rangle = \sigma^2 \langle n(t_0+t) - n(t_0) \rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t. The integral $\int_0^{T-t}[n(t_0+t) - n(t_0)]\,\mathrm{d}t_0$ counts, for each step k that lands at time $T_k \in [0,T]$, the length of the t-wide sliding window that contains T_k. For T_k deep in the interior ($t \le T_k \le T-t$) this length is exactly t; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)]\,\mathrm{d}t_0 \approx t\,n(T), \qquad t \ll T, \tag{3.A.7}$$

and

$$\overline{\delta^2}(t;T) \approx \frac{\sigma^2\, t\, n(T)}{T}. \tag{3.A.8}$$

Taking the ensemble average and using $\langle n(T) \rangle \sim K_\alpha T^\alpha/\sigma^2$,

$$\mathrm{TAMSD}(t,T) \equiv \langle \overline{\delta^2}(t;T) \rangle \approx \sigma^2\, \frac{\langle n(T) \rangle}{T}\, t \sim K_\alpha\, T^{\alpha-1}\, t. \tag{3.A.9}$$

Comparing (3.A.5) and (3.A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TA-MSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t.
- The TA-MSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T, so longer trajectories yield a smaller TA-MSD at the same lag. This T-dependence is the signature of *aging*: the increment statistics depend on how long the process has already been observed.
- The ratio $\mathrm{TAMSD}(t,T)/\mathrm{MSD}(t) \sim (t/T)^{1-\alpha} \to 0$ as $T \to \infty$ at fixed t.

Distribution of the TA-MSD: non-self-averaging

Equation (3.A.8) shows that $\overline{\delta^2}(t;T) \approx \sigma^2 t\, n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0,T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T) \rangle} \tag{3.A.10}$$

converges in distribution as $T \to \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution* M_α, whose moments are $\langle \xi^n \rangle = n!\,\Gamma(1+\alpha)^n/\Gamma(1+n\alpha)$ [20]. This is a broad, non-Gaussian distribution on $(0,\infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t;T)$ even in the limit $T \to \infty$. Hence the TA-MSD is *non-self-averaging*: the value from a single trajectory is not a reliable estimate of any ensemble quantity, however long the trajectory.

Quantitatively, the ergodicity-breaking parameter

$$\mathrm{EB}(t,T) \equiv \frac{\langle [\overline{\delta^2}(t;T)]^2 \rangle - \langle \overline{\delta^2}(t;T) \rangle^2}{\langle \overline{\delta^2}(t;T) \rangle^2} = \mathrm{Var}(\xi) \xrightarrow{T\to\infty} \frac{2\,\Gamma(1+\alpha)^2}{\Gamma(1+2\alpha)} - 1 \;>\; 0 \tag{3.A.11}$$

tends to a finite positive constant rather than to zero [20]: the scatter of $\overline{\delta^2}$ across trajectories survives even in the limit of infinite observation time (for $\alpha = 1$ the constant is 0 and ergodicity is recovered). Monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories, and not only its mean, therefore diagnoses non-ergodicity directly: the distribution collapses with growing T for a self-averaging ergodic process and converges to a non-trivial limit for a non-ergodic one.

For comparison, in an ordinary random walk ($\alpha = 1$, finite mean waiting time) the ratio $n(T)/\langle n(T)\rangle$ tends to 1 in probability as $T \to \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t;T) \to \mathrm{MSD}(t)$ almost surely, and the process is ergodic.

Implementation and Computational Details

Chapter 1

Overview

This part collects the implementation- and computation-level detail behind the choices made in the main text: the exact code module and script that produced each figure or table, the configuration values in force, sample sizes and computation times, and the status of each piece (implemented and tested, designed but not yet built, or planned). It is organised as a document within the document: its own chapters, numbered independently of the appendices, mirror the body chapter by chapter. It is not meant to be read start to end – the body links to the relevant place at the point where a detail was trimmed out of the text, via a cross-reference such as "(implementation details: Sec. 3.5.2)".

Contents of this part

Table 1.1 is a quick index of which script reproduces which auto-generated figure or table (all paths relative to `scripts/reporting/`); the chapters that follow go into the reasoning and numbers behind each one.

Figure/Table	Script	Notes
Stats macros; Tables 2.3, 2.4	`generate_stats.py`	Reads committed `data/*/seasons_index.csv`; instant; part of the pre-commit hook.
The three processed tables (Sec. 2.2)	`../preprocess.py`	Runs stages (i)–(vii) over an archive; ∼80 min for both disciplines on 8 workers. `../verify_dataset.py` checks the result against the invariants of Ch. 2.
Figure 3.3; `\StatMsd*`	`generate_msd_figure.py`	Streams `fixes.parquet` once, through `soaring.analysis.derived.stream_flights`, and accumulates in that single pass both estimators and the three fixed-duration cohorts (Sec. 3.1); ∼20 min on the paraglider archive. The cohorts cost nothing beyond three more accumulators, because the expensive step is the read. `--redraw` re-renders the figure from the written `msd_curve.csv` in a second, without touching the archive: nothing there recomputes an estimator, so a change to the drawing need not cost another pass over 10^9 rows.
`\StatPipe*`, the per-stage shares quoted around Table 2.7	`generate_pipeline_census.py`	Queries `flights_meta.parquet`; instant.
Table 2.7 itself; `\StatData*`, `\DataCascade*`	`generate_dataset_stats.py`	Reads `flights_meta.parquet` and the catalogue; seconds.
Census macros (`\StatScan...`)	`generate_census_stats.py`	Reads the cached track scans on the SSD (never rescans); instant; best-effort in the pre-commit hook, so the committed macros refresh whenever the SSD is mounted. Also emits the flight-level filtering census (`LongEnough`/`Overlong`/`ShortPath` families, Sec. 2.7.4) and the missing-barometric-fix statistics (`BaroMiss`, Sec. 2.7.1), against the thresholds in `configs/preprocessing.yaml`, and re-exports those thresholds as the `\Preproc*` macros quoted by the pipeline map (Table 2.8).
— (all of the above, in order)	`../regenerate.sh`	Runs every row of this table that depends on the processed dataset, in the one order that is correct, and refuses to start while `preprocess.py` is still writing. The ordering is a real constraint rather than a convenience: the driver rewrites `<data_root>/derived/` in place and completes `flights_meta` only after its last flight, so a generator run against a live archive reads a partial fix table beside the previous run's metadata and produces numbers that are wrong and look right. That happened once. The macro contract below is checked before the build, because an undefined generated macro inside a `\SI` is a fatal LaTeX error rather than a warning.

Chapter 2

The dataset

Mirrors Chapter 2 section by section, opening with the data acquisition (Sec. 2.1) and including the pre-processing and preliminary- characterization sections (Secs. 2.7, 2.8) – the cleaning and first-look steps are part of describing the dataset, not of the planned analysis in Chapter 3.

Table 2.1 collects the implementation status of every step discussed below; the sections point to it instead of repeating status remarks.

Pipeline step	Status	Where
Data acquisition (fetch, download, verify)	implemented	Sec. 2.1
IGC parsing and format validity	implemented, unit-tested	Sec. 2.3
Catalog and season index (`build-catalog`)	implemented	Sec. 2.4
Census scan, diagnostic figures and macros	implemented	Sec. 2.7, Table 1.1
Glider-label normalisation	designed	Sec. 2.5
Altitude-channel choice (`alt_source`)	implemented, unit-tested	Sec. 2.7.1
Fix-level cleaning (three detectors)	implemented, unit-tested	Sec. 2.7.2
Validation battery for the cleaning	planned, follows the cleaning	Sec. 2.7.2
Trimming of ground phases, interior guard	implemented, unit-tested	Sec. 2.7.3
Flight-level filtering	implemented, unit-tested	Sec. 2.7.4
ENU conversion	implemented, unit-tested	Sec. 2.7.5
Uniform sampling and segmentation	implemented, unit-tested	Sec. 2.7.6
Smoothing and differentiation	implemented; timescales measured	Sec. 2.7.7
Preliminary characterization	implemented	Sec. 2.8
MSD audit (residuals, controls, isotropy)	implemented	Sec. 3.1
Ensemble MSD (Sec. 3.1)	implemented, unit-tested	Sec. 3.1

Table 2.1. Implementation status of the pipeline steps, collecting the status remarks of this appendix in one place. *Implemented* code is committed and in use; *designed* steps have their rules and thresholds fixed (and, where noted, their parameters already in the configuration) but no running routine yet; *planned* steps follow the designed ones.

2.1 Data acquisition

Mirrors Sec. 2.1.

2.1.1 Pipeline and tooling

Package `soaring.acquisition.ffvl`, exposed as two CLI entry points, `soaring-para` and `soaring-delta`, sharing subcommands: `fetch-xml` (archive season XMLs), `download` (fetch IGC files), `build-catalog` (rebuild catalog/index), `status` (per-season coverage), `verify` (post-hoc integrity scan, Sec. 2.1.3). One package serves both disciplines because the two FFVL sites expose an identical URL structure and the same machine-readable XML export format; only the host and the data root differ between the two CLIs.

Resumable downloads. Files already present on disk are skipped, so an interrupted job resumes freely without re-fetching.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an `A` header record followed by at least one `B` fix record); HTML error pages and truncated responses are rejected and retried.

Rate-limited networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests keep the load on the server low; a browser TLS fingerprint is impersonated to pass the site's Cloudflare challenge.

2.1.2 Reproducibility

Destination directory (`data_root`) set per machine via an environment variable – `SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders – or a config file shipping a placeholder (`configs/para_download.yaml/ configs/delta_download.yaml`); every command validates the setting at start-upand aborts with an explicit message if the target is missing.

2.1.3 Integrity

The `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors.

2.2 Where the data lives

Everything is on one external disk and nothing is in the repository, so a reader who has the disk can locate any number this thesis quotes. There is one root per discipline, identically laid out, found through an environment variable rather than a hard-coded path (`SOARING_PARA_DATA_ROOT`, `SOARING_DELTA_DATA_ROOT`); the repository keeps only the small versioned `data/<discipline>/seasons_index.csv`, so the season tables of Sec. 2.5 build without the disk mounted.

Path	**Size**	**What it is**
raw/ — immutable, never modified after download		
`raw/igc/<season>/`	186,052 files	the track files, one directory per season, named `<date>_<flight_id>.igc`
`raw/raw_xml/<year>.xml`	27 files	the season listings exactly as the site served them: the provenance of the catalog
catalog/ — regenerable from raw/, metadata only		
`catalog/catalog.csv`	87 MB	one row per listed flight, 23 columns (Sec. 2.3)
`catalog/seasons_index.csv`	2 kB	one row per season: listed, with a track, downloaded
logs/ — what the acquisition did		
`logs/download.log`, `logs/failures.csv`	80 kB	the run log and the retry registry, one row per track that could not be fetched
derived/ — regenerable from raw/, everything the analysis makes		
`derived/track_scan.parquet`	8.5 MB	the census cache: one scalar row per parsed flight (Sec. 2.7)
`derived/fixes.parquet`	43 GB	the processed trajectories: 1.36×10^9 rows for paragliders
`derived/segments.parquet`	4 MB	one row per segment the splitting produced, retained or not
`derived/flights_meta.parquet`	10 MB	one row per flight *attempted*, dropped ones included

The three levels are a rule and not a habit: `raw/` is never written to after the download, `catalog/` is a function of `raw/`, and `derived/` is a function of both. Deleting `derived/` costs computation and never data. It also means the two census families cannot be confused: `track_scan.parquet` describes *parsed* tracks and is what the `\StatScan*` macros query, because a cut must be audited on the population it acts on; `flights_meta.parquet` describes what the pipeline *did*, and is what the `\StatPipe*` macros query.

The trajectory table. `fixes.parquet` carries one row per grid point of every retained segment, keyed `(source,flight_id,segment_id)`. Printed as it is on disk (the hang-glider root; `scripts/reporting/show_dataset.py` produces this view):

```
34,389,245 rows in 43 row groups, 1,156.0 MB on disk, ZSTD

dtypes:
  source   str       t  float32      v_E  float32      interpolated     bool
  flight_id str      E  float32      v_N  float32      edge             bool
  segment_id int16 N  float32      v_z  float32      hampel_flagged   bool
                     z  float32      a_*  float32      alt_invalidated  bool

head(4), columns 1-9:
    source flight_id  segment_id    t          E           N           z        v_E       v_N
hangglider       975           0  0.0  -2.644085   -4.948205 1458.585693 13.275144  1.215108
hangglider       975           0 10.0 111.172691  -33.921207 1460.657104  9.778760 -5.975531
hangglider       975           0 20.0 198.742096 -103.775276 1468.514282  8.025670 -6.961107
hangglider       975           0 30.0 279.220947 -138.911942 1463.914307  7.008325 -2.623670

head(4), columns 10-17:
      v_z       a_E       a_N       a_z  interpolated  edge  hampel_flagged  alt_invalidated
-0.532143 -0.436803 -1.029317  0.192857         False  True           False            False
 0.721429 -0.262474 -0.408811  0.057857         False  True           False            False
 0.625000 -0.088144  0.211696 -0.077143         False False           False            False
-1.466667  0.072790  0.463077 -0.047143         False False            True            False
```

The four rows are the first four grid points of a 10 s logger: t starts at zero, E and N within metres of it, z at its measured 1,458 m. Coordinates are `float32`, a part in 10^7 on a coordinate of tens of kilometres and orders below the GPS noise on it. The file is written in batches of 400 flights, and an analysis streams it rather than loading 43 GB—which is how the mean-squared displacement of Sec. 3.1 is computed.

A row group is not a batch, and the difference cost a measurement. The writer hands Parquet 400 flights at a time; Parquet then cuts what it is handed into row groups of its own default size, which is why the header above reads 43 row groups for an archive written in 16 batches. A reader that takes a row group for a unit of flights therefore meets, at each boundary, one flight arriving as two fragments—and processes it as two flights. The ensemble MSD counted 157,134 paraglider flights where the pipeline had written 156,017, 0.7 % too many; worse, the time-averaged MSD is computed *within* a segment and cut at half its length, so a straddling segment stopped answering at half of its longer fragment instead of half of itself, and the longest segments—the ones that carry the long-lag tail, and the ones likeliest to straddle—were the ones truncated. The reading was thus biased by the writer's buffering rather than by the data. Every consumer now goes through one reader, `soaring.analysis.derived.stream_flights`, which holds back the last flight of each row group and prepends it to the next, so what it yields is always a whole flight; it raises rather than continues if the writer's ordering ever stops holding. The dataset verifier reads through it too, since it had the same blind spot in a worse place: the step *joining* two fragments was never differenced, so it was never checked, and a check with a blind spot reports success over it.

The three timing conventions a reader of this table has to know are argued elsewhere in the chapter and worth restating together: t is zero at the first fix of free flight *of the parent flight*, so a segment never restarts it (Sec. 2.7.6); E and N are zero at that same fix (Sec. 2.7.5); and z is the adopted altitude at its measured value, never re-zeroed. The only times in another clock are `ground_phase_start_s` and `ground_phase_end_s` in `flights_meta`, which are in the recorded clock of the IGC file because the integrity gate counts its removals inside that window.

The segment table. One row per segment the splitting produced, retained or not, and the reason for each drop:

source	flight_id	segment_id	t_start	t_end	n_fix	n_fix_raw	frac_interpolated
hangglider	975	0	0.0	20640.0	2065	2064	0.001453
hangglider	1032	0	0.0	14175.0	0	691	0.001479

censored_start	censored_end	kept	drop_reason
False	False	True	<NA>
False	False	False	channel_not_reconstructable

`n_fix` counts the rows the segment contributed to `fixes` (zero when dropped) against `n_fix_raw`, its measured fixes. `segment_id` is assigned at split time and stays stable, so a gap in the numbering is itself the record of a drop.

The flight table. One row per flight *attempted*, forty-one columns, read down rather than across—one retained flight beside one the flight-level filter rejected:

	a retained flight	a dropped one
source	hangglider	hangglider
flight_id	975	830
pipeline_version	1.1.0	1.1.0
drop_stage	NaN	flight_filter
drop_reason	NaN	duration_below_minimum
alt_source	baro	gnss
n_fix_raw	2099	349

n_fix_clean	2069	348
n_removed_frozen	30	0
n_flagged_kept	40	8
n_boundaried	0	0
integrity_fraction	0.0	0.004587
ground_phase_start_s	300.0	485.0
duration_flight_s	20645.0	1085.0
path_km	283.408578	6.044969
lat0	43.812717	NaN
dt_native_s	10.0	NaN
n_segments_kept	1.0	NaN
savgol_window_horiz	5.0	NaN

(abridged: the table carries 47 columns in all.) A dropped flight keeps everything the stages *before* the verdict had already measured—flight 830 was judged on a duration of 1,085 s, and its cleaning counters remain auditable even though none of its fixes reached the trajectory table. That is what makes the removals a census rather than an absence.

What is not stored. Anything that is pure algebra of the stored columns—the speed $|\mathbf{v}|$, the heading, the angular velocity, the turn radius, the glide ratio, the mechanical energy—is computed at analysis time and never materialised. Only the outputs of a non-trivial parametric step are kept, which is what makes the table's size a function of the archive rather than of the questions asked of it.

Checking it. `scripts/verify_dataset.py` reads the three tables back and tests, on the data itself, the properties this chapter claims: completeness within a segment, a strictly increasing and uniform time base, no impossible step between two interior samples of a segment (Sec. 2.7.2), the origin at $r(0) = 0$, reachability—$|\mathbf{r}(t)| \le v_{xy}^{\max} t$ for every sample—and referential integrity between the tables. Each check names the section it comes from, so a failure says which claim is false rather than only that something is wrong.

Reachability is the one check stated without reference to anything the pipeline did: it follows from the frame and the speed bound alone, because the flight began at the origin. That is what makes it able to catch a defect no per-step test can see. A flight whose *first* fix is corrupt has every step plausible—it simply sits, in the archive, 4,500 km from where it says it began, with the whole trajectory referred to that point (Sec. 2.7.2).

The qualification in the third check is not a hedge but the statement being precise. The bound is a claim about *measured* motion, and two kinds of sample in the table are not measurements. At a segment edge the smoothing polynomial is evaluated off-centre (Sec. 2.7.7), so its position carries more variance than a centred one and on a slow logger can imply a step over the bound where the underlying fixes do not. Across a bridged gap the monotone cubic is a reconstruction, and its slope may exceed the secant it spans—a hole crossed at $42\,\mathrm{m\,s^{-1}}$, itself under the bound and so legitimately bridged, leaves interpolated points implying nearly $60\,\mathrm{m\,s^{-1}}$. Both kinds are flagged in the table for exactly this reason (`edge`, `interpolated`), so the check reads the invariant between samples that are measured and centred, and *reports* the excess on the others. That reported number is one of the acceptance criteria Sec. 2.7.7 asks for: measured, rather than assumed away.

The script found two defects this text would otherwise have asserted without warrant. A randomised robustness suite (`test_pipeline_robustness.py`, which combines injected spikes, null-island runs, frozen locks, time defects, gaps and offsets at random and asserts only these invariants) found two more, and both were faults in the *statement* rather than in the pipeline: the invariant had been written without either qualification above. A third, the corrupt opening block of Sec. 2.7.2, was found by neither: it was found by the ensemble MSD of Chapter 3 being wrong, which is the expensive way to find a bug, and the reachability check above is what makes it cheap next time. It also found one in the cleaning—a track crossing the antimeridian is read as motionless, because the great-circle distance is correctly aware of the wrap and therefore reads the two numerical extremes of a straddling bounding box as metres apart. No flight in the present archive reaches within ten degrees of that meridian; the further sources of Sec. 2.1 may.

2.3 The IGC tracklog format

The parser (`soaring.analysis.igc.parse_igc`) checks, at decode time and independently of any flight-dynamics judgement: that the time is a valid 24-hour `hh:mm:ss`; that the arc-minutes field of latitude and longitude is `< 60` (the `DDMM.mmm` encoding is undefined otherwise); that the hemisphere letter is one of the two valid characters; and that the decoded latitude/longitude fall within $[-90, 90]/[-180, 180]$ degrees. A record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field. This format-validity check and the fix-level plausibility checks are distinct stages, and both are implemented and unit-tested (Sec. 2.7.2, Table 2.1).

2.4 Flight metadata and the catalog

Both `catalog.csv` and `seasons_index.csv` are regenerated on demand (Sec. 2.3) by the acquisition CLI's `build-catalog` subcommand (Sec. 2.1.1) – never hand-edited.

2.5 Glider categories

The placeholder class is the literal "0" in `aile_class`; where possible the true class is recovered from the raw `aile` (wing-model) field. The normalisation runs at the fix-level/pre-processing stage (Sec. 2.7.2) and is defined in Sec. 2.4. Status: Table 2.1.

2.6 Coverage and statistics

The season tables of Sec. 2.5 are typeset from the versioned `seasons_index.csv` (Sec. 2.4); the trajectory-level census numbers quoted from Sec. 2.7 onward come from a different pipeline, the cached track scan described at the head of Sec. 2.7.

2.7 Trajectory pre-processing

Mirrors Sec. 2.7 and its subsections below. Parsing: each IGC file is decoded by the `soaring.analysis.igc` parser, which reads the `B` records at their fixed character positions (Sec. 2.3) and returns both altitude channels alongside time and geodetic coordinates. The Welch/PSD procedure used to compare altitude channels is stated once, in Sec. 2.7.1; the smoothing calibration (Sec. 2.7.7) reuses it, applied to the horizontal components and to the flight's adopted altitude channel.

The census scan and its cache. One full parse of the archive backs every trajectory-census number and figure in this appendix and in Chapter 2: the `load_or_scan_tracks` scan reads every parsed track (186,025 paraglider, 6,716 hang-glider) once and caches a per-flight summary table at `<data_root>/derived/track_scan.parquet`, one cache per discipline.

That cache is worth describing, since every census number in the chapter is a query against it. It is a table with *one row per parsed flight* and one column per summary quantity, all produced by `track_stats` in `soaring.analysis.census` as it walks a flight's fixes once:

- `duration_s`, `n_fix`, `dt_s` — the recorded span, the number of `B` records, and the median inter-fix interval;
- `path_km`, `extent_km` — the summed great-circle steps and the farthest any fix lies from the first;
- `max_gap_ratio`, `missing_fraction` — the largest inter-fix gap in units of `dt_s`, and the share of a uniform grid at that interval with no fix;
- `max_vxy_mps`, `max_vz_mps` — the extreme inter-fix speeds;
- `baro_alt_min_m`, `baro_alt_max_m`, `baro_present_frac` — the barometric range and how much of the flight the channel covers.

Nothing in it is a trajectory: it is deliberately one scalar row per flight, which is what makes it small enough to hold in memory for the whole archive and to re-query in seconds. Its purpose is exactly that—to decouple the cost of *reading* the archive (hours, once) from the cost of *asking* it a question (instant, as often as a threshold changes). Every `\StatScan*` macro, and the flight-level and sampling diagnostics of Figs. 2.3 and 2.6, are functions of these columns alone.

What the census is not. It is measured on the **raw** archive, one parse of the `B` records with no cleaning, no trimming and no filtering—which is the only thing it could be, since its purpose is to choose the thresholds that cleaning, trimming and filtering then apply, and a threshold chosen on the population that survives it is chosen circularly. So every `\StatScan*` number in this thesis is a statement about the archive *as recorded*: a duration is the recorded span and not the airborne one, a path includes whatever a corrupt fix contributed, and a retained-fraction curve is what a cut would keep if applied alone to the raw population. The retained ensemble has its own numbers and its own family, `\StatPipe*`, computed from `flights_meta.parquet` after all seven stages have run (Table 2.7); where a quantity appears in both, the two are not expected to agree, and the difference is precisely what the pipeline did. The two families are never mixed in one statement.

`generate_census_stats.py` recomputes the quoted macros from that cache against the thresholds currently in `configs/preprocessing.yaml`, so a threshold change propagates to every quoted count by re-running one script, with no rescan. The cache is refreshed when the archive changes—which for the FFVL collections means once a season closes, and more substantially when a new source is added (the sailplanes of Sec. 2.1 above all); between refreshes the cached counts can trail the season tables' download counts, and the two are not expected to agree to the flight. The same scan feeds Figures 2.3, 2.6 and 2.7, regenerated together by one run of `generate_preproc_figure.py` (Table 1.1); the fix-level diagnostics of Figure 2.2 are the one exception, drawn from a separate sampled pass (Sec. 2.7.2).

2.7.1 Choice of altitude channel

The `alt_source` field. The channel a flight uses is stored as an `alt_source` column (`baro` or `gnss`)— two-valued and categorical, so its storage cost is negligible —in the processed dataset's per-flight table, alongside the duration and the raw fix count—the number of `B` records the IGC file yields at parse time, before any cleaning, since the channel choice precedes the cleaning in the pipeline order.

Welch/PSD procedure and its sample (Figure 2.1a–c). The sample is a seeded random subsample of 3,000 flights per discipline.The size is set by convergence of the ensemble spectrum, not by the spectral shape: with a few hundred flights the high-frequency part of the median barometric curve is still visibly ragged (under-averaged), and it smooths out only with a few thousand flights, so 3,000 buys a stable band at modest cost. This is an assertion the reader cannot presently check, and it should not stay one: the convergence figure—the same median spectrum drawn at a few hundred, a thousand and 3,000 flights, on one pair of axes—is to be produced and shown here, so that the choice of sample size rests on a picture rather than on a claim. Panel (a)/(b) use one representative flight (the one with the most fixes among the 3,000 sampled flights of that discipline); its measured offset (panel b) is annotated on the panel itself, not restated here.

IGC loggers record one fix per second nominally, but timestamps are not guaranteed to be perfectly regular (a missed fix, a logger writing at 1.05 s instead of 1.00 s); Welch's method (Appendix 2.A) requires a uniformly sampled signal, so the following runs on every flight before the transform.

(i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample (the *mode*) is taken as the common target Δt (in practice $\Delta t = 1\,\mathrm{s}$ for both disciplines, giving $f_s = 1\,\mathrm{Hz}$ and $f_{\mathrm{Nyq}} = 0.5\,\mathrm{Hz}$).

(ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal Δt are retained, so pooled spectra share one frequency axis. Mixing cadences here would not be admissible: a spectrum is defined on a frequency axis that runs to $1/(2\Delta t)$, so a 1 s and a 9 s logger do not even share a Nyquist frequency, and averaging their periodograms per frequency bin would compare different bands. That is the whole reason for this cut—and it is a cut on the *diagnostic* sample only. The pipeline proper keeps every flight at its native cadence (Sec. 2.7.6); it is only this spectral estimate, which pools flights, that needs them commensurate. Flights with fewer than $N_{\mathrm{seg}} = 256$ fixes ($\sim$4 min at $\Delta t = 1\,\mathrm{s}$) are also excluded (a series shorter than one Welch block of $N_{\mathrm{seg}} = 256$ samples cannot contribute a periodogram, Appendix 2.A). Of the 3,000 sampled flights per discipline, this leaves 1,641 paragliders but 727 hang gliders: paragliders are predominantly 1 s loggers (69 %, Sec. 2.7.6), so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5, $\geq$8 s, Figure 2.7) and only their 1 s-logger minority survives this cut.

(iii) **Uniform resampling.** The altitude series $z(t)$ is linearly interpolated onto a grid $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \ldots$ spanning the flight. Linear and not cubic, because the resampling must not disturb the band the estimate is about to read. Interpolation acts on a spectrum as a filter: linear interpolation attenuates high frequencies smoothly and monotonically, whereas a higher-order interpolant has a transfer function that can exceed unity near the Nyquist frequency, adding power that was never in the signal. Since the quantity being measured *is* the flat floor at high frequency, an interpolant that can lift it would bias the very number in question; one that only attenuates cannot manufacture a floor, and the timestamps are near-uniform to begin with, so the attenuation it does apply is slight.

(iv) **Welch on the resampled series.** `scipy.signal.welch` is called with $f_s = 1/\Delta t$, $N_{\mathrm{seg}} = 256$, 50% overlap, a Hann window of the same $N_{\mathrm{seg}} = 256$ samples (the window spans one Welch segment by construction—256 s at the 1 s cadence, giving a frequency resolution of about 4 mHz), and linear detrending per segment.

(v) **Robust ensemble estimator.** The per-flight PSDs are kept individually (per discipline and channel) and reduced across flights by the per-frequency *median*, with a shaded 10 to 90 percentile band; panel (c) then pools the two disciplines, whose medians coincide. The median, not the arithmetic mean, because the per-flight spectra are heavy-tailed: a mean swings by one to two orders of magnitude between samples – for the hang-glider barometric floor, from 0.8 to $93\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$ between an $n = 114$ and an $n = 727$ draw

– whereas the median is stable. Because Δt is the same for every retained flight by construction, the figure's Nyquist dotted line is a single fixed value.

Reading the band (panel c).

Why the medians coincide. For the typical flight both channels are dominated at high frequency by the metre resolution to which the IGC format rounds every altitude: the white-noise floor of a 1 m quantiser, $\frac{1}{12}/f_{\mathrm{Nyq}} \approx 0.17\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$,[1] sits just below where the measured medians level off ($\approx 0.3\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$).

Where the channels part. In the tail: at the 90th percentile the high-frequency GNSS floor is 4× the barometric one for paragliders and 108× for hang gliders, and 13 % to 20 % of flights carry a GNSS floor above three times their own barometric one. What sits in that tail is receiver tracking noise scaled by the dilution of precision, since a raised *floor* is by definition a white contribution; multipath, being correlated over minutes (Sec. 2.7.1), does not show up there. Why the hang-glider fleet is worse in this respect is not settled by these spectra alone—older or lower-grade receivers are the obvious candidate—and the figure is not claimed to answer it.

Consequence. This heterogeneity, rather than a uniform gap, motivates the choice of the barometric channel (Sec. 2.7.1).

Census (panel d). Panel (d) is a full census, not a sample: the barometric-presence fraction is read off the shared track scan (Sec. 2.7), already paid for by the flight-level diagnostics.

"Present" needs a number, and the number is a decision. The scan records `baro_present_frac`, the share of a flight's fixes carrying a non-zero pressure altitude, and a flight counts as barometric when that share reaches `BARO_PRESENT_MIN` = 95 % (defined once, in `soaring.analysis.altitude_noise`, and imported by both the cleaning and the census so the three cannot drift). The threshold is high deliberately. Presence is bimodal—the census below shows the channel is either essentially absent or essentially complete—so any value in the middle of the range separates the same two populations; but a mid-scale cut would also admit a flight missing, say, a third of its pressure altitudes, and such a flight is not one the vertical analysis should treat as barometric: its v_z would rest on a channel with holes, reconstructed by interpolation over stretches long enough to erase transitions. Setting the cut at 95 % sends those flights to the GNSS fallback, where their altitude is at least uniformly sourced. The number quoted in the body (Sec. 2.7.1) is the same one.

Presence is whole-file, checked on the same census: when the channel is absent it reads exactly zero for 99.3 % (paragliders) / 100 % (hang gliders) of those flights, and when present it covers every fix for 96 % / 98 % of them; the rest miss individual fixes, dropped one by one (Sec. 2.7.2), and no more than 5 % of the flight by construction, since below that the channel is not counted as present at all.

The bimodality asserted above is what these numbers measure. Where the channel is absent it is *exactly* zero for 99.3 % of those flights, and where it is present it is complete for 96 %; the population in between is thin. Raising the threshold from 0.5 to 0.95 reclassified 241 paraglider flights and no hang-glider ones—0.13 % of the archive—which is the direct measurement of how thin. Those flights are also where the worst partial coverage sat: the largest number of missing barometric fixes on a flight still counted as barometric fell from 14,878 to 1,397 when the threshold moved, so most of what looked like a pathological tail was

[1] Rounding to a step q adds an error that is well modelled as uniform on $[-q/2, q/2]$ and independent between samples, of variance $q^2/12$. A white sequence of variance σ^2 has a flat one-sided power spectral density $\sigma^2/f_{\mathrm{Nyq}}$, since integrating it over $[0, f_{\mathrm{Nyq}}]$ must return the variance. With $q = 1$ m and $f_{\mathrm{Nyq}} = 0.5$ Hz this is $(1/12)/0.5 \approx 0.17\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$.

a classification artefact rather than a sensor one. The liveness threshold is `baro_min_range_m` (key `alt_channel`; value in Table 2.8); rule and rationale in Sec. 2.7.1.

2.7.2 Fix-level cleaning

This section gives the working definitions the narrative (Sec. 2.7.2) states in words: the three detectors, their parameters, the order they run in, and how the whole step is validated. In the order of Table 2.6, the three families map onto a robust local test (a short window of context), a structural rule (a whole run), and an absolute bound (no context). Status: Table 2.1.

The detectors run in a fixed order, which the subsections below follow:

(1) format validity, at decode time (Sec. 2.3);

(2) time defects: backward steps, then duplicates;

(3) position outliers (the impossibility gate; the Hampel flag is recorded, not required—Sec. 2.7.2);

(4) frozen-lock runs;

(5) altitude: absolute bounds and altitude outliers.

The order matters because a speed or a residual is only meaningful once the time base is monotone and free of duplicates. Merging duplicates is componentwise and unweighted: fixes sharing one UTC second collapse to a single fix at that second, carrying the mean of their latitudes and longitudes and the mean of each altitude channel over the members that have one; the `V` flag of the merged fix is the logical *or* of the members', so a degraded member cannot be averaged away. The position-outlier and frozen-lock passes are one left-to-right scan that carries the last accepted fix as an anchor: it never re-walks the whole track, and it absorbs a run by extending the anchor across the flagged block – tracking the block's running bounding box to test its bounding diameter against δ_{xy} – so the cost is $O(N)$ in the number of fixes.

Time-base defects

The parser unwraps the UTC-midnight roll-over — a drop of ∼86,400 s advances the day count — and *leaves* residual backward jitter in place (Sec. 2.3). That is deliberate, and it is the one behaviour the redesign changed: the parser used to flatten the jitter with a running maximum, which is a repair, and repairing it there destroys the evidence this stage has to act on. A backward step that is *not* a roll-over is now removed rather than absorbed – the fix with the corrupt time is deleted as a node, just as a fix with a corrupt position is. A corrupted timestamp is recorded as a defect rather than silently repaired. Which fix carries the corrupt time is decided by minimal removal: delete the smallest set whose removal leaves a strictly increasing clock (the complement of the longest increasing subsequence), so a single fix with a forward-jumped clock is itself removed, rather than condemning every later fix against a running maximum. This requires moving the clamp out of the parser: `parse_igc` keeps the roll-over unwrap (a format concern, needing the raw time of day) but stops flattening backward jitter with the running maximum – flattened, the evidence the cleaning must act on no longer exists downstream.

Fixes sharing one UTC second are collapsed to a single representative at that second's centroid position. This replaces the provisional “keep the first” of the earlier design: keeping the first privileges an arbitrary member and, for a genuine ≥ 2 Hz logger, would silently halve the cadence, whereas the centroid is order-independent and the sub-second resolution it discards is discarded anyway once the flight is resampled to a common Δt (Sec. 2.7.6). One dilution case is accepted: if one member of a duplicate pair is itself a position spike, the centroid halves its amplitude before the outlier pass (which runs after the time defects) sees it;

a large spike is still flagged at half size, and one small enough to slip under $k\sigma_k$ after halving is within what the smoother absorbs.

Robust local-outlier test and action

The test is Eqs. (2.2)–(2.3) of Sec. 2.7.2, evaluated on a *time* window of half-width w (not a sample count), so it adapts to the native cadence and to irregular sampling; working defaults in Table 2.2. The window must hold at least a handful of fixes (order five); a sparser neighbourhood falls back to the absolute bounds alone.[2]

Three properties distinguish the test from the bare speed bound:

- *The floor $\varepsilon_{\min}$ is not cosmetic.* On a very smooth stretch $\sigma_k \to 0$, so without it any sub-metre wiggle would exceed $k\sigma_k$ and the test would chase GPS noise where there is no outlier.
- *A burst is one block.* A short burst of consecutive spikes inflates the windowed MAD only mildly, so it is still flagged, and it is removed as the block whose bracketing neighbours rejoin within scale (the attribution argument is the removal-policy paragraph of Sec. 2.7.2).
- *It does not clip real manoeuvres.* A genuine sharp turn is drawn by consecutive, mutually consistent fixes, each well predicted by its neighbours and so of small residual; only an isolated point off the local trend scores large. The one failure mode, the under-sampled tight turn (Sec. 2.7.2), is measured directly by the injected-defect test (check (1) below).

The flag itself is context-sensitive – a $30\,\mathrm{m\,s^{-1}}$ step out of a $12\,\mathrm{m\,s^{-1}}$ glide is flagged while the same step passes unremarked where it is ordinary – but a flag alone does not delete (next paragraph): a flagged-but-physically-possible fix is *kept*, its flag recorded as a per-flight anomaly count alongside the integrity-gate counters. The test runs on the horizontal position *alone*. An earlier draft of this appendix had it running on the adopted altitude channel as well, which the three-part argument of Sec. 2.7.2 supersedes: on z an absolute bound is sufficient where on xy it is not, a local test on z would flag the thermal entries the segmentation is built on, and a false vertical flag is expensive exactly where it is most likely. The vertical is therefore cleaned by its absolute bounds and by nothing else.

From flag to action. The identifier only *flags*; it never repairs. A flagged horizontal fix is *deleted* as a node – not overwritten with the local median – with all reconstruction deferred to the single resampling step (Sec. 2.7.6; the one-policy rationale is Sec. 2.7.2). This is the *Hampel identifier*, the detection half of the eponymous filter, not the median-substituting filter itself: the local median enters only as the reference for the residual r_k, never as an output value.

Position. Deletion is the impossibility gate: a fix goes when it is unreachable from the last accepted fix at the absolute v_{xy} bound *and* the removal of the block it belongs to lets the track rejoin within that bound. The flag is neither necessary nor sufficient—why it cannot be required is Sec. 2.7.2—and the genuine under-sampled tight turn is kept because it flies at an ordinary speed, not because anything exempts it.

Byte-identical, on the parsed table. The frozen-lock witness of Eq. (2.4) is specified on the raw `B` records, but the parser returns decoded floats and keeps no raw text. No raw text is needed: the `DDMM.mmm` field is a fixed-width integer count of thousandths of

[2] The residual and run-diameter distances are the haversine great-circle distance of Eq. (2.1); a vectorised off-the-shelf implementation exists as `sklearn.metrics.pairwise.haversine_distances`, should the hand-rolled one need cross-checking.

an arc-minute, so the decode is injective and two records are byte-identical in latitude and longitude *iff* their decoded values compare exactly equal. The test is therefore `lat[i]==lat[i+1]andlon[i]==lon[i+1]` on the parsed columns, with exact equality and no tolerance — a tolerance would turn the conclusive signature into the jittering-freeze case it is meant to exclude.

Discontinuity. When no bounded block restores continuity – a position discontinuity, such as a re-acquisition offset, where every removal still leaves an impossible step – the scan stops deleting and marks a split at the step, handled like a long gap (Sec. 2.7.6): both sides are genuine, only the transition is unknown.

How long a block may run. “Bounded” needs a number, and it is the Hampel half-width w: a block that has not rejoined within 20 s is called a discontinuity instead. The value is reused rather than introduced, and it also fixes the scale at which the two cases part—an offset smaller than $v_{xy}^{\max} w \approx 900\,\mathrm{m}$ becomes reachable from the anchor before the block expires and is removed as an excursion rather than split. That is a real limit of the rule and a bounded one: at most one block of genuine fixes, at a discontinuity, against the alternative of splitting a flight in two for every isolated spike.

The postcondition, and its counter. After the scan, any step between two *surviving* fixes that still breaks the bound is made a segment boundary (Sec. 2.7.2). The number of boundaries this adds is recorded per flight as `n_boundaried` in `flights_meta`, alongside the other cleaning counters: it is the audit of a rule that would otherwise repair without saying so, and it is expected to be small—a large value would indict the scan rather than the data.

Altitude. No flag reaches it: the vertical is cleaned by the absolute bounds of Table 2.6 alone, which mark the value missing and never touch the fix(a deferral: Sec. 2.7.2). Both bounds act on the channel the flight *adopted*, whichever it is. The $|v_z|$ bound was placed on barometric distributions, but what it expresses is a climb/sink envelope of the aircraft, which does not change with the instrument reading it; leaving a GNSS-fallback flight with no vertical cleaner at all — the consequence of dropping the local test — would be the worse reading. It fires more often there, and the rate is recorded per `alt_source` so the audit can see by how much.

Isolated, not sustained. The $|v_z|$ bound marks an altitude missing when the record jumps away from it and back — both adjoining steps past the bound, with opposite signs. This is check (5) below made operational: a *run* of super-threshold steps of one sign is a sustained manoeuvre, a spiral dive holding $15\,\mathrm{m\,s^{-1}}$ to $20\,\mathrm{m\,s^{-1}}$ of real sink, and it is counted and left alone rather than censored.

The first fix. An anchor-carrying scan never questions its own first anchor, and no rule here supplies one: the discontinuity is left as a discontinuity, for the reason Sec. 2.7.2 gives. The guarantee is enforced one stage later, by the reach criterion of Sec. 2.7.4, and checked on the written table (Sec. 2.2).

This is the *default to keep* rule of Sec. 2.7.2: removal requires positive, corroborated evidence.

Frozen-lock detection

The rule, its witnesses and its margins are Eq. (2.4) and Sec. 2.7.2; parameter values in Table 2.2. Three points are implementation-level.

- *Diameter, not step.* The run's bounding diameter is the discriminating statistic because a step-based test would fire on every 1 Hz climb: the fix-to-fix step of a circling wing, $v_{xy}\Delta t \approx 10\,\mathrm{m}$, is of the order of δ_{xy} itself.
- *Robust witness.* Δz_{baro} is evaluated between medians of the run's ends, so an altitude spike inside the run – not yet cleaned, since the altitude pass comes last – cannot fake a climb.
- *Bookkeeping.* On a barometric flight the `V` flag and the zeroed GNSS altitude are recorded as corroborating diagnostics only; they fill the witness slot on GNSS-fallback flights alone (Eq. (2.4)).

A run the rule condemns is marked as a data gap; the split it induces is realised at the uniform-sampling stage (Sec. 2.7.6), on the same footing as a native long gap.

Absolute bounds and gates

The six numbers of Table 2.6 are the `FixLevelThresholds` dataclass, which `load_preproc_config` reads from `configs/preprocessing.yaml` under the key `fix_level`; nothing is hard-coded in Python. They are the context-free floor of the design, and are placed on the real per-fix distributions (Figure 2.2) rather than guessed. The horizontal-speed bound is set where the bin-to-bin ratio of a finely binned ($2.5\,\mathrm{m\,s^{-1}}$) histogram settles at ≈ 1, i.e. where the decay has actually stopped, not where it first visibly slows ($\approx 40\,\mathrm{m\,s^{-1}}$ for hang gliders, versus the adopted $55\,\mathrm{m\,s^{-1}}$), since a shallow but real decreasing stretch can still be genuine rare dynamics. The Hampel and frozen-lock parameters live under the same key; Table 2.2 collects them.

Flight-level integrity gate. The pass records how much it changed each flight (or segment) – `n_fix_raw`, `n_fix_clean` and `frac_interpolated` in `flights_meta`. A flight whose deleted-plus-reconstructed fraction exceeds f (10 %, Table 2.2) is dropped whole; the counting window (airborne only) and the rationale are Sec. 2.7.2, whose threshold sweep validates the adopted value rather than setting it. This is the cleaning counterpart of the sampling-regularity cut (Sec. 2.7.6).

Diagnostics and validation

Diagnostic sample (Figure 2.2). Computed by `fix_level_distributions` and rendered by `make_fixlevel_diagnostics_figure` (Table 1.1). Panels (a)–(c) pool a seeded random sample of 15,000 paraglider flights (126,723,499 consecutive-fix pairs, ∼73 s to compute) and the complete 6,716-flight hang-glider population (a full census at this size, 37,915,334 pairs, ∼25 s) – under two minutes total. The large sample is deliberate: resolving the sparse artefact tail well enough to tell it from real dynamics needs far more points than a headline statistic would. The x-range in (a)/(b) extends to each discipline's own 99.99th percentile rather than cropping just past the bound; panel (c) keeps a narrower range, its cut sitting far out in an otherwise tight, unimodal distribution. Panel (b) bins one integer $\mathrm{m\,s^{-1}}$ per bin because barometric altitude is logged to the metre: differencing at a 1 s cadence returns integer rates, and a finer grid would alias them into a spurious comb.[3]

Validating the cleaning. The primary validation is the two-stage procedure of Sec. 2.7.2: freeze the working values a priori, then audit the per-rule removal fractions over the archive

[3]Not every flight samples at 1 s (Figure 2.7), and differencing over a longer native step returns finer values (multiples of $0.2\,\mathrm{m\,s^{-1}}$ at 5 s) that partly fill the gaps once cadences are pooled: among 1 s-native fixes about 99% land within $0.05\,\mathrm{m\,s^{-1}}$ of an integer, against only 34–42% at every other cadence.

Parameter	Role, and the scale that sets it
Robust local-outlier test (Hampel)	
w	half-width of the local time window; a few native intervals
k	flag threshold, in robust standard deviations σ_k
$\varepsilon_{\min}$	absolute residual floor; a few times the horizontal GPS noise, so a smooth stretch ($\sigma_k \to 0$) is not chased
Frozen-lock rule (Eq. 2.4)	
δ_{xy}	run-diameter tolerance; a few times the GPS noise, below the $2R \approx 30\,\mathrm{m}$ to $100\,\mathrm{m}$ of a circling climb
δ_z	barometric-flatness witness; a few times the sensor noise and quantisation, $\ll$ the $\gtrsim 60\,\mathrm{m}$ a real climb gains over τ_{freeze}
τ_{freeze}	minimum span of a cut run; two thermalling periods, so a slow, tight climb is never mistaken for a freeze
Flight-level integrity gate	
f	maximum deleted-plus-reconstructed fraction; past it the flight is being rebuilt, not cleaned, and is dropped whole

Table 2.2. Fix-level cleaning: what each parameter of the robust and structural detectors does, and the physical scale that fixes it. The *values* are deliberately not repeated here—they are in Table 2.5, with the rest of the pipeline's working values, so that there is one place to read a number and one place to change it. They are fixed a priori and checked a posteriori on the per-rule removal fractions (Sec. 2.7.2), with the injected-defect calibration as the direct measurement of the error rates. All live as config keys under `fix_level` in `configs/preprocessing.yaml`, alongside the absolute bounds of Table 2.6.

and sweep each threshold around its working value. The checks below are the fuller battery behind it, run as the implementation matures.

(1) *Injected defects.* The synthetic-defect procedure of Sec. 2.7.2, item (d), gives the detection floor – the smallest spike caught – and, where the removal-fraction audit flags a rule, recalibrates k, w, δ_{xy} and τ_{freeze} to a target rate: the only direct measurement of the cleaning's false negatives.

(2) *Downstream invariance.* Recompute the transport estimators – the displacement-scaling exponent and the waiting-time-tail exponent – over a grid of cleaning strengths, i.e. re-running the cleaning with all thresholds tightened or loosened together by a common factor around the working point, and take the operating point on the plateau where they stop moving; movement under stricter cleaning is the signature of a cut biting into signal.

(3) *Detector nesting.* The fixes flagged by the absolute bound should nest inside those flagged by the local test (the impossible is a subset of the locally anomalous); a large disagreement points to a defect class neither was designed for – a cheap guard.

(4) *Per-phase removal rates.* The error this pipeline most wants to avoid is losing a slow-phase fix. Once the segmentation is in place (Ch. 3), the removal and split rates are measured *per phase* and required to be flat across transition, search and climb rather than enriched in the two slow phases; and synthetic tight climbs – circles at flying speed, over a range of radii and wind drift – planted in otherwise clean tracks confirm directly that the frozen-lock detector never splits a real climb. This is the false-*positive* complement of check (1), and its natural readout is the tail of $\psi(\tau_w)$: a cut biting into climb or search would thin that tail before it moved anything else.

(5) *Isolation of the vertical cuts.* The fixes cut by $|v_z| > 13\,\mathrm{m\,s^{-1}}$ must be *isolated*, not

consecutive: a sustained spiral dive holds $15\,\mathrm{m\,s^{-1}}$ to $20\,\mathrm{m\,s^{-1}}$ of real sink, so a run-shaped excess is the signature of a genuine manoeuvre – calling for a raised bound or a coherent-run exemption rather than censoring – whereas an isolated super-threshold difference, bracketed by ordinary rates, is a sensor error and safe to cut.

2.7.3 Trimming of ground phases

The two thresholds of Eq. (2.5) (v_0, T_0) are the `TrimmingThresholds` dataclass, loaded from `configs/preprocessing.yaml` (key `trimming`); values in Table 2.8.

The interior-ground guard of Sec. 2.7.3 keeps `interior_ground_s` (T_{ground}, 600 s) and `ground_flatness_m` (5 m) in the same group. The flatness is tested on the *detrended* altitude, as Sec. 2.7.3 requires: fit a straight line to the adopted channel over the candidate stint by least squares, subtract it, and compare the residual's central span—its 5th to 95th percentile—against `ground_flatness_m`. The linear term is the barometric drift, which over a stint of T_{ground} is already of the order of the tolerance; the residual is what is left of genuine vertical motion, and the central span rather than the full range because a range grows with the length of the stint and a central span does not (Sec. 2.7.3). No new config key is needed.

Building it made one thing explicit that the rule as stated leaves open. Detrending cannot by itself separate a pressure drift from a steady climb, because both are linear and the fit removes either completely: a wing thermalling at $1\,\mathrm{m\,s^{-1}}$ in still air holds a ground speed near zero, gains 600 m over a T_{ground} stint, and leaves a detrended residual of *zero* — so the flatness condition alone would excise it as a mid-flight landing, which is precisely the failure this guard exists to avoid. What separates the two is the *rate*: drift runs at tens of metres per hour, a soaring climb two orders of magnitude faster. The fitted slope is therefore bounded as well, at $\delta_z/\tau_{\mathrm{freeze}}$ — the frozen-lock rule's own statement of what counts as motionless, read as a rate, so the guard borrows a number the configuration already fixes instead of introducing one. Below T_{ground} the same predicate is stored as a list of suspect intervals per flight, which the segmentation later maps onto the phases they touch (consumption of the flag: Sec. 2.7.3).

2.7.4 Flight-level filtering

The three cuts of Sec. 2.7.4 are the `FlightLevelThresholds` dataclass (key `flight_level` in `configs/preprocessing.yaml`); values in Table 2.8. The census measurements behind the path floor and the upper duration bound, and the totals of Figure 2.3, are quoted through macros computed from the shared track scan (Sec. 2.7); the headline duration and path medians are quoted in Sec. 2.8.

The plausibility and reach bounds of Sec. 2.7.4 are not further `flight_level` keys: both read `max_horizontal_speed_mps` from the `fix_level` block, so the same number governs a single step, a whole-flight average and a whole-flight displacement, and the three cannot drift apart. Both are evaluated on the trimmed track, after the per-block sums of the duration and the path, and they are the last of the whole-flight cuts because each is a statement about a record the earlier ones have already accepted. The extent they need is returned by `flight_quantities` alongside the duration and the path, from the same haversine call, so it costs nothing: the maximum of a vector the function already forms. Unlike the duration and the path it is *not* summed per block—a displacement from a fixed origin has nothing to sum—which is why a split does not weaken it.

The chain behind every number quoted in Sec. 2.7.4 is short enough to name in full, so that a count can be traced to the code that produced it:

- `soaring.analysis.igc.parse_igc` decodes one IGC file into a table of fixes;
- `soaring.analysis.census.track_stats` reduces those fixes to the one summary row described in Sec. 2.7, and `scan_tracks` / `load_or_scan_tracks` do that for every flight

of a discipline and cache the result;

- `fraction_retained` and `retention_curve` in the same module evaluate a single cut against a column of that table—the first at one threshold, the second along a grid of them, which is what the bottom row of Fig. 2.3 plots;
- `scripts/reporting/generate_census_stats.py` calls the same functions with the thresholds read from `configs/preprocessing.yaml` and writes the `\StatScan*` macros the text quotes.

Two consequences follow, and both matter for reading the numbers. The counts are *marginal*: each is the effect of one criterion applied alone to the whole parsed population, not of the cascade. And they are pre-cleaning: the scan runs on parsed tracks, so a count in Sec. 2.7.4 is what the criterion removes from the parsed archive and not from the cleaned one. That is deliberate and is argued for above (*What the census is not*): a threshold chosen on the population the cleaning has already reshaped would be chosen against its own effect. The post-cleaning counts are the cascade of Sec. 2.9, and the two answer different questions.

2.7.5 From geographic to local Cartesian coordinates

No implementation content: this step is pure coordinate geometry with no configurable parameters, and both of its figures (Figure 2.4, Figure 2.5) are hand-drawn TikZ diagrams, not data plots. The mathematics behind Eqs. (2.6) and the ENU rotation is derived in Appendix 2.B.

2.7.6 Uniform sampling within a flight

The sampling thresholds (key `sampling`, the `SamplingThresholds` dataclass; values in Table 2.8) act as follows.

Split bound. $g_{\max} = \min\left(\texttt{max\_gap\_factor} \times \Delta t,\ \max(\texttt{max\_gap\_seconds},\ 2\,\Delta t)\right)$, pinned by the constraint chain of Sec. 2.7.6. Its audit is a per-flight count – the census scan records `max_gap_ratio` and `dt_s`, so the absolute gap is their product – and joins Figure 2.6 when regenerated.

Missing fraction. Acts per *segment* (rule: Sec. 2.7.6); the per-flight value in the census scan is the pre-split diagnostic, not the operational gate.

Segment gate. `min_segment_duration_s` (90 s). It only needs to support the smoothing window and the within-segment statistics: a few smoothing timescales, never less than $w\,\Delta t$; the only gate applied at segment level (Sec. 2.7.6).

Measured vs interpolated. A grid point counts as measured when a fix lies within half a native step of it, interpolated otherwise; the same tolerance decides whether a flight is uniform as recorded (no interpolated points) or was resampled (`was_resampled`).

The bound is on the *time base* and is the same for every channel: an order-of-magnitude smaller v_z buys no wider gap for the altitude, since the interpolation error scales with curvature, not speed, and a wide altitude gap would erase the climb–glide transitions that v_z is there to resolve (Sec. 2.7.6).

Segment handling.

Origin and clock. A split assigns a `segment_id` and nothing else (semantics: Sec. 2.7.6).

Tables. One row per fix in `fixes`, keyed (`flight_id`, `segment_id`); per-segment records (start/end, fix count, `frac_interpolated`, censoring flags) in a `segments` table beside `flights_meta`. No separate files; the ensemble MSD at lag t reads $\mathbf{r}(t)$ wherever some segment covers t.

Censoring. Any phase duration truncated at a segment boundary is flagged censored, for the $\psi(\tau)$ fits to exclude.

Per-channel fill. Across a short gap the horizontal position is interpolated by monotone piecewise cubics (`scipy.interpolate.PchipInterpolator`) and the adopted altitude channel linearly (its within-phase smoothness makes the choice among low-order methods immaterial), realising the completeness invariant of Secs. 2.7.2 and 2.7.6. Each filled grid point is flagged in an `interpolated` boolean column of `fixes`, and `frac_interpolated` records the per-segment fraction for the integrity gate.

Figures 2.6 and 2.7 are rendered by `make_gap_diagnostics_figure` and `make_sampling_figure` from the shared track scan (Sec. 2.7).

Gap-ratio discreteness (Figure 2.6). Checked directly: only 86 %/63 % (para/hang) of largest-gap-ratio values are exact integers, the rest a genuinely continuous scatter; the integer peaks visible for hang gliders sit around $k = 3, 6, 10, 15$. Panels (a)/(b) of the figure therefore use linear axes: a logarithmic axis would compress precisely this small-integer structure, on which the adopted cut sits.

Native-rate discreteness (Figure 2.7). Checked directly: Δt lands on an exact whole second for 99.9 % of paragliders and 100 % of hang gliders. Per-rate breakdown: paragliders sit mostly (69 %) at 1 s; hang gliders spread across the rates below.

Native rate	1 s	2 s	5 s	10 s	other
Hang gliders	30 %	13 %	27 %	15 %	remainder

2.7.7 Smoothing and differentiation

Three decisions are pinned for the implementation (status: Table 2.1):

- the filter runs per segment (never across a boundary) with `scipy.signal.savgol_filter` in `mode='interp'`, so the terminal half-windows are fitted on the edge window rather than padded, and edge samples inherit the segment-boundary caution of Sec. 2.7.6;
- the window is floored at $w = 5$, the smallest a cubic fit accepts (slow-logger consequences: Sec. 2.7.7, item iii);
- the polynomial order and the three smoothing timescales are the `SavgolParams` dataclass in `configs/preprocessing.yaml` (key `savgol`): `tau_c_horizontal_s` for (E, N), `tau_c_vertical_baro_s` and `tau_c_vertical_gnss_s` for the vertical channel (the flight's adopted altitude, Sec. 2.7.1); values in Table 2.8.

The measured timescales. The knees were measured with `estimate_savgol_timescales.py`, which applies the Welch procedure of Sec. 2.7.1 to E, N (local equirectangular metres) and to the altitude channel each flight actually uses, on a seeded sample (908 horizontal flight-components, 719 barometric and 189 GNSS-fallback flights at 1 Hz). The knee rule: the floor is the median PSD over 0.35 Hz to 0.5 Hz; f_c is the lowest frequency whose median PSD falls below three times it. The measured floors sit at ≈ 0.60 and $\approx 0.26\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$ against the theoretical quantisation floors $q^2/12/f_{\mathrm{Nyq}}$ of 0.57 and 0.17; the three knees coincide at

$f_c \approx 0.2\,\text{Hz}$, hence $\tau_c = 5\,\text{s}$ for every channel (Sec. 2.7.7, item (iv), for the reading), and even the 90th-percentile spectra move the knee by less than a second. The two vertical keys stay, conditioned on `alt_source`, for the noisy-minority refinement; the validation of Sec. 2.7.7 (flat residual spectrum, stability under one window step) is what confirms the values. The filter runs on the resampled grid, so the small interpolated fraction (`frac_interpolated`) is already in place when it reads its window.

2.8 Preliminary characterization

`scripts/reporting/generate_prelim_figure.py` produces both figures of Sec. 2.8 and the `\StatPrelim*` macros. It reads three things: the per-flight summary and the per-lag positions `audit_msd.py` wrote (Sec. 3.1), `flights_meta.parquet` for the take-off coordinates, and the season catalogue for `wing_class` and `season`. The join on `flight_id` resolves for 99.99 % of retained flights.

Reading the audit arrays rather than the fix table is what makes the stratification affordable: a stratified MSD is then a row selection on a matrix already in memory, so the three stratifications and the isotropy check together cost one pass that has already been paid, instead of one 43 GB scan each.

The basemap. The take-off panels are drawn on Natural Earth admin-0 country polygons (public domain), which give the coastline and the borders in one layer and carry the overseas departments inside the France feature—which is what puts La Réunion on the map at all. The geometry is not fetched at draw time and Cartopy is not a dependency: `scripts/reporting/build_basemap.py` downloads it once, keeps the rings that intersect each panel, decimates them (0.01° for France, 0.001° for La Réunion, 0.25° for the world frame) and writes `data/basemap.json`, which is committed at 0.76 MB. The figure then needs nothing but Matplotlib and regenerates from a clean checkout with no network. Rings are kept whole rather than clipped to the frame—Matplotlib clips to the axes anyway, and clipping a polygon properly would need the geometry library the arrangement exists to avoid. Re-run the builder only to change a panel's extent.

The three frames. Metropolitan France $\lambda \in [-5.5, 10]$, $\varphi \in [41, 51.5]$; La Réunion $\lambda \in [55.15, 55.92]$, $\varphi \in [-21.42, -20.82]$; the world panel for everything outside both. The split is the archive's, not a choice: 4,939 paraglider flights launch on La Réunion and 3,359 outside France altogether, and a single frame that held them would render metropolitan France four pixels wide. Each panel sets its aspect to $1/\cos\varphi$ at its own mid-latitude, so the coastline has its true shape rather than a stretched one.

The orographic boxes. The strata are longitude/latitude boxes rather than polygons traced from the map: a box that is written down can be checked against a gazetteer, and the test the strata support is whether the curves separate, not where exactly the Alps end. Alps $\lambda \in [5.4, 10.0]$, $\varphi \in [43.8, 46.6]$; Pyrenees $\lambda \in [-1.9, 3.3]$, $\varphi \in [42.0, 43.5]$; Massif Central $\lambda \in [1.8, 4.6]$, $\varphi \in [43.6, 46.2]$; anything else inside $\lambda \in [-6, 10]$, $\varphi \in [41, 52]$ is "lowland" and anything outside it is "abroad". The boxes are coarse and overlap along their edges, with the first match winning in the order listed; they are a stratification, not a geography, and the test they support is whether the curves separate rather than where exactly the Alps end.

The stratum floor. A stratum below 2,000 flights is pooled into the residual group rather than drawn. Below that the curve's own sampling error at the long-lag end is comparable to the separations being tested for, and a panel that draws it invites the reader to compare noise.

The floor is why the hang-glider archive, at 6,132 flights, contributes only to the isotropy panel: one wing class and one orographic group clear it, so the stratified panels would compare a stratum against itself.

The isotropy test. The variance ratio is computed on the same per-lag position matrix, column by column, over the flights covering that lag. The Kolmogorov–Smirnov comparison runs at four lags spanning the fit range, on the signed components rather than their magnitudes, so that a difference in *spread* and a difference in *shape* both register. Only the distance is quoted. At 155,788 flights the two-sample p-value underflows to zero for any departure worth naming, so it separates nothing; the distance is the effect size and is what the text reads.

Stratum departure. Each stratum's curve is divided by the pooled curve over the fit range and the largest $|\cdot - 1|$ recorded. Two summaries are generated: the maximum over strata (`\StatPrelim*MaxDeparturePct`) and the median over strata (`*TypicalDeparturePct`). The maximum alone reads as a statement about the smallest and noisiest stratum, which is what it usually is; the median is what distinguishes an axis along which the ensemble genuinely splits from one along which a single small group wanders. Both are quoted in Sec. 2.8.

Chapter 3

Global transport

This chapter mirrors Chapter 3. Built and recorded here: the two mean-squared-displacement estimators and their audit, the filtered-variation family that replaces them as a source of exponents, the moment spectrum, the non-Gaussian parameter, the velocity autocorrelation and its tail, and the persistence runs. Three observables the body catalogues are defined and not measured, and Table 3.1 says which.

3.1 Observables on the whole trajectory

The mean-squared displacement of Sec. 3.1 is implemented and unit-tested, as are four more of that section's observables; the shape ones are documented below and the three that are not measured are named in Table 3.1. The estimators live in `soaring.analysis.observables.transport` and the figure, the curve table and the `\StatMsd*` macros come from one pass of `generate_msd_figure.py` (Table 1.1). A second pass, `audit_msd.py`, keeps what the first one averages away, and `audit_msd_report.py` reduces it into the `\StatAudit*` macros; both are described at the end of this section.

3.1.1 The two estimators

`MSDAccumulator` builds the ensemble average and `TAMSDAccumulator` the time average. Both accumulate by streaming rather than by loading: the `fixes` table runs to $\sim 1.4 \times 10^9$ rows over the two archives, and neither estimator needs more than one flight in memory at a time.

Coverage. A flight answers a lag t with the fix nearest to it, and only if two conditions hold: that fix lies within $\max(0.5\,\mathrm{s}, \Delta t/2)$ of t, and $t \geq \Delta t$, where Δt is the median inter-fix time of the flight's own concatenated segments. The second condition is the one that is easy to omit and expensive to omit. Without it a 5 s logger asked for a lag of 1 s answers with its fix at $t = 0$—a displacement of exactly zero—and a slow-cadence minority drags the short-lag MSD down until the grid crosses their step. The same rule appears in the time-averaged estimator, which refuses any lag below its segment's own step. A lag falling inside a gap between two retained segments is answered by neither side, since the split bound (Sec. 2.7.6) makes every gap wider than the tolerance.

The lag grid. `log_lag_grid` spaces 90 points geometrically from 1 s to 43,200 s, the duration bound of the flight-level filter, and snaps them to the second. The snapping is what makes the short end usable: unsnapped, lags of 1.00, 1.13 and 1.27 seconds all resolve to the same fix at a 1 Hz cadence, and the curve there is a staircase of repeated values whose local slope alternates between zero and a spurious spike. Snapping merges them instead, which is why the grid delivers 80 distinct lags rather than 90, and why the lags below 8 s are consecutive integers. At long lags the rounding is a part in 10^4. The consecutive-integer run is also what lets the coverage artefact of Sec. 3.1 have a period; that is the cost of the snapping, it is measured in Sec. 3.1 by splitting the archive by cadence, and it sits an order of magnitude below the lower end of every fit range.

The time average. `time_averaged_msd` is computed within a segment, never across the gap that ends it. The direct double loop is $O(n^2)$ and hopeless at 10^4 samples per segment; writing $D_j = |\mathbf{r}_j|^2$, the identity $\overline{\delta^2}(k) = \langle D_j + D_{j+k}\rangle_j - 2\langle \mathbf{r}_j \cdot \mathbf{r}_{j+k}\rangle_j$ turns the second term into an autocorrelation, which the FFT gives in $O(n \log n)$. The curve is cut at half a segment's length, beyond which the average over starting times has too few of them to be one.

3.1.2 The fit, and what is quoted from it

The range. The lower end is an argument, not a consequence: at 120 s it sits above the thermalling period, so what the slope measures is transport rather than the circling superimposed on it, and above the whole region the short-lag artefacts occupy. The upper end is a consequence. `coverage_limited_range` walks the count-per-lag curve and cuts where it has fallen to `min_coverage` = 0.25 of its value at the lower end; past that the curve is an average over a survivorship-selected minority. Applied to the four curves this gives the ranges quoted with each exponent.

The two errors. `fit_msd_exponent` fits unweighted in log space, which gives every decade the same say; a fit weighted by the much larger absolute errors at long lags would be dominated by the last decade alone. It returns the least-squares error on the slope, which is written out as `\StatMsd*AlphaErrOls` and is *not* the quoted uncertainty. `bootstrap_alpha_error` resamples 200 ensembles of flights with replacement, refits each over the same range, and reports the standard deviation as `\StatMsd*AlphaErr`. Neither figure is attached to an exponent in the body: both MSD exponents are quoted bare, because the bootstrap treats clustered flights as independent records and because the range dependence is fifty times larger where both have been measured. Sec. 3.1 makes that argument and gives the synthetic check that establishes what each of the two errors measures; the exponents that do carry an interval are those of Sec. 3.3.

The cohorts. Three fixed-duration cohorts at T = 1, 2 and 4 h. Within a cohort every member is present at every lag up to T, so the population is constant by construction and any remaining growth is motion; each is therefore fitted from the range's lower end up to its own T, which is as far as the guarantee reaches. The time-averaged cohorts are keyed on segment span and capped at $T/2$, for the same reason the time-averaged curve is. A cohort is compared against the *pooled curve refitted over the cohort's own lags*, never against the pooled exponent over the pooled range: the local slope is not constant across those two spans, so the naive comparison charges the shape of the curve to the change of population. Both readings are generated (`\StatMsd*CohortSpread` and `*CohortSpreadNaive`), and on the ensemble estimator they differ by a factor of five, which is the size of the mistake avoided.

The local slope, and why it is a regression. The panel that carries the most interpretation is the one whose estimator is easiest to get wrong. The local slope is $\mathrm{d}\log \mathrm{MSD}/\mathrm{d}\log t$, and the obvious way to compute it—a centred difference between neighbouring lags—divides by the grid spacing, which on the logarithmic grid used here is about 0.05 decades. The division multiplies whatever scatter the curve carries by $1/(2 \times 0.05) \approx 4$: a point-to-point scatter of a few tenths of a per cent in the MSD, which is about what the ensemble's own standard error allows, becomes a slope that jitters by 0.05 and reads as noise. The panel then testifies against a curve that is not in fact noisy.

`soaring.analysis.observables.transport.local_slope` therefore works on a window: at each lag it takes the points within 0.15 decades on either side—five to six at this grid spacing—and reports the *median of the slopes of all pairs* of them, a Theil–Sen estimator. The choice is for resistance to a single displaced lag rather than to a heavy-tailed residual, which is the failure this curve actually has: a coverage artefact of a few per cent at one lag tilts a least-squares window and moves a median by nothing. The uncertainty reported with it, and drawn as a band, is the interquartile range of those pairwise slopes divided by 1.35 and by the square root of the window's size—the window's own scatter about its line, expressed as an error on the slope. Over the fitted range it is 0.017, against a point-to-point variation of 0.052: the curve moves more between neighbouring lags than the scatter within a window accounts for, which is the quantitative statement that what the panel shows is structure and not noise. The half-width is a stated compromise rather than a tuned one: narrower and the amplification returns, wider and a genuine bend is flattened into the straight line the panel exists to test for; 0.15 decades resolves a crossover of half a decade. It is not a smoother in disguise. The residual wiggle of this archive is correlated across some four grid points, so it survives the window—which is the right outcome for something that is not noise, and the reason the window is reported with the figure rather than chosen to make the curve look better.

3.1.3 The audit pass

The ensemble curve turned out not to be a power law (Sec. 3.1), and none of the questions that decides—is the shape the coverage rule, the survivorship, a shared heading, or the motion—can be asked of an averaged curve after the fact. Each is a different reduction of the same per-flight quantity, and that quantity is gone by the time the average exists.

`audit_msd.py` therefore makes a second streaming pass and writes what the first one discards: one row per flight and one column per lag, holding each flight's E and N at that lag under the estimator's own coverage rule, `NaN` where it does not cover it, plus a per-flight summary (duration, path, native step, mean-velocity components, largest in-segment step speed, coordinate extent). The coverage rule is *copied* into that script rather than imported, so that a change to the estimator surfaces as a disagreement between the figure and the audit instead of being tracked silently. The arrays run to a few hundred megabytes per discipline and are analysis products, not thesis ones: they live outside the repository, under `AUDIT_DIR`.

`audit_msd_report.py` reduces them, together with the curve table `msd_curve.csv` and the segment and flight tables, into `\StatAudit*`. Everything the audit asserts in the body reaches it through those macros. What it computes: the rms residual each estimator leaves against the power law fitted to it, in dex and as a percentage; the range of the local slope over the fitted decades; the mean vector's share of the mean square, $|\langle \mathbf{r} \rangle|^2 / \langle |\mathbf{r}|^2 \rangle$; the local slope recomputed on sub-ensembles of a single native cadence, where the population cannot change, and on fixed-duration cohorts, where it cannot be selected; the first-crossing time of 5 km from take-off, per flight; the per-component variance ratio and the Kolmogorov–Smirnov distance between the E and N marginals; and the pre-processing losses of Table 2.7 decomposed by the rule that caused them.

Two of its numbers are worth naming here because they are checks on the pipeline rather than on the estimator. The retained tables carry 0 non-finite coordinates and 0 non-monotone clocks, which is what Sec. 2.2 asserts and this pass confirms independently. And the flights whose first retained segment did not survive resampling start their record at $t > 0$ rather than at the origin—1.4 % of paraglider and 11.7 % of hang-glider flights. Their position is still measured from their own take-off, since the ENU frame is anchored there and a dropped segment does not move it; they simply do not answer the lags before their record starts. For the rest, the position at $t = 0$ is the origin to within 0.09 m in the median, the smoothing's displacement of the first sample.

3.1.4 The shape observables, and what calibrates them

The non-Gaussian parameter. Computed from the moment spectrum already accumulated, as $\alpha_2 = \langle|\Delta\mathbf{r}|^4\rangle/2\langle|\Delta\mathbf{r}|^2\rangle^2 - 1$ at each lag, on order-1 increments. Two calibrations make the archive's value readable and both are pinned by `tests/analysis/observables/test_moments.py`, at the sample size the test uses: an exact fractional Brownian motion returns 0.02 in the median over a ten-point grid spanning the quoted range, and a Lévy walk of tail index 1.5 returns 0.55. The Gaussian 0.02 is estimation bias and not shape — a fourth moment over 10^4 increments is not yet its own expectation — and sixteen times the data collapses it to within a hundredth of zero, sign included. The same test pins that too, since it is what makes the archive's own value a departure rather than an offset. It is quoted over 60 s to 2,000 s rather than over the whole grid; above that the declared task governs the trajectory and the fourth moment rests on the fewest flights, and the curve climbs to +0.47 there.

The tail of the velocity autocorrelation. Green–Kubo is used in its scaling form, $C(\tau) \sim \tau^{-\gamma}$ with $\gamma < 1$ giving $\langle|\Delta\mathbf{r}|^2\rangle \sim \Delta^{2-\gamma}$, and not as the integral. Two reasons, both properties of the archive: the integral runs from zero and C is estimable only above the smoothing scale of the slowest logger, so its first decade is missing; and C is estimated per flight with that flight's own mean velocity removed, which pulls every lag down by about the record-mean of C and so steepens the tail. Both push γ up, so $2 - \gamma$ is a floor and `vacf_tail_exponent` documents it as one. The one-sidedness is itself tested, on an Ornstein–Uhlenbeck velocity whose true correlation is known: the estimate with the mean removed is always the steeper.

The filter kernel, against a closed form. For any process with stationary increments and $S(\Delta) = \sigma^2\Delta^{2H}$, the two orders are related exactly by $V_2 = 4S(\Delta) - S(2\Delta) = S(\Delta)\,(4 - 2^{2H})$. That identity pins the kernel in a way a second convolution cannot: the order-2 kernel is symmetric, so an independent `np.convolve` route performs the same three products in the same order and agrees to the last bit whether or not the kernel is right — which is what a recomputation of the paraglider exponent by that route in fact found. The estimator also has to be unbiased where the archive sits, at H near 1, which is the boundary of the family: on exact fractional Brownian motion of $H = 0.99$ it returns 1.98.

When a knee is declared. A bilinear fit to $q\nu(q)$ has two more parameters than a straight line through the origin, always reduces the residual, and on a grid of ten moment orders buys the reduction cheaply — fitted to a fractional Brownian motion, which has no knee, BIC prefers the bilinear form while the straight line already fits to an rms of 0.003. So BIC alone is not the rule. `bilinear_fit` declares a knee only when three conditions hold together: BIC prefers the bilinear form, *and* the straight line leaves a residual above `min_departure` = 0.02, *and* the knee sits inside the q grid rather than on its edge. On the synthetic pair those three give no knee for fractional Brownian motion and a knee at 1.57 with a right-hand slope of 0.96 for a Lévy walk of index 1.5. The archive's departure from a line is 0.0099, an order of magnitude below the threshold.

Why the drift is filtered and never subtracted. `centring_bias` measures the alternative rather than arguing against it: fractional Brownian motion of known exponent is generated, the mean-squared displacement is computed raw and after subtracting each realisation's own net velocity, and the two are compared lag by lag. Subtracting a velocity estimated from the record's endpoint turns the series into a bridge whose increments must sum to zero, so the centred curve is pulled down at every lag and hardest at the long ones — 2 % at a lag of 0.05 % of the record and 76 % at half of it, with no range in between where the distortion is negligible. On $\alpha = 1.50$ the operation returns 1.35. The filtered variation of `soaring.analysis.observables.variations` estimates no drift at all, so none can be estimated badly, and the order scan is what reads off how much drift there was.

The variation pass, and why stratifying is free. `measure_variations.py` traverses the fix table once and keeps, per flight and per filter order, one curve over the lag grid together with the flight's cadence, wing class, season and declared task. Every stratification of Sec. 3.4 is then a row selection on that table rather than another pass over 1.4×10^9 rows, which is what makes four cuts affordable where one would otherwise be a day's work.

The persistence runs, and the bias that decomposition avoids. `persistence_runs` walks the record greedily from its start, extending a run while the sinuosity — arc length over chord — stays under the threshold and beginning the next run where the last one ended, with a floor of five samples. The stopping rule is *first violation*: the condition is not monotone in the endpoint, since a stretch can come back inside the threshold after leaving it, so the rule is stated rather than implied. No sample belongs to two runs, and that is the point. Sampling the run length at every instant instead returns each run in proportion to its own length — the length-biased distribution, whose tail index is one below the truth. `inspection_biased_runs` implements the wrong way on purpose so the size of the error can be measured: on synthetic runs of known index 2.0 the two estimators return 2.0 and 1.1. At the values this archive shows that is not a bias to correct afterwards but a different answer. The first implementation of the biased estimator measured the residual life instead of the containing run, which is a third quantity again; a test caught it.

First-passage times, and why they are not reported. The audit arrays keep each flight's position at every lag, so the first-passage time $T(L)$ — the first time a flight is L from take-off — costs no traversal, and $\langle T(L)\rangle \sim L^{1/H}$ would give the exponent in the space domain, independently of any moment fitted over lags. It was computed and is not reported, for a reason worth recording so it is not attempted again. Read naively it returns $\alpha = 2.46$, which no process with stationary increments can give. The cause is right censoring: at 34 km only a quarter of paraglider flights arrive and the missing three quarters are the slow ones. Treating non-arrival as censored and taking a Kaplan–Meier median instead returns 1.95 for paragliders and 2.03 for hang gliders, both inside the paraglider interval — but refitting over sub-ranges of radius moves the paraglider value between 1.51 and 2.03, a spread five times the uncertainty it was meant to check. The reason is the one Sec. 3.1 gives for the ensemble MSD. $T(L)$ is measured from take-off, so it shares that estimator's origin: at small L the wing is still climbing out inside its launch area, and at large L the censored median extrapolates past the duration of most flights. It is the same crossover seen along the space axis instead of the time axis, and it is not an independent check on anything.

3.1.5 Known limitations, with their size

Four, each bounded rather than repaired.

Edge samples enter. The smoothing evaluates its polynomial off-centre at the ends of a segment and the table flags those samples `edge` (Sec. 2.7.7). `verify_dataset.py` qualifies its invariants by that flag—which is why its speed invariant holds—and the MSD estimators do not, and read every fix. Those samples are under 1 % of the table and account for 26,368 of the $26{,}368 + 32$ in-segment steps above the fix-level speed bound, 0.001,9 % of all steps. The worst step between two *measured* samples is $49\,\mathrm{m\,s^{-1}}$ against a bound of $45\,\mathrm{m\,s^{-1}}$: an excess of under a tenth, which the verifier reports and does not fail on, its threshold for a data defect being twice the bound. Excluding the flagged samples moves the ensemble MSD by up to 13 % below 10 s and by under 0.2 % above 60 s, so the fit ranges are unaffected; excluding them would also cost every flight its shortest lags, since the first samples of a record are exactly the flagged ones.

Absolute position carries re-acquisition offsets. Bounded in Sec. 3.1 at under a per cent of the mean over the fitted range, on 0.029 % of paraglider flights.

The archive holds a few duplicate flights. The same physical flight uploaded twice, detected by take-off coordinates, duration and path length agreeing to four decimal places, one second and ten metres: 186 paraglider flights in 93 groups, 0.119 % of the archive, and 29 hang-glider flights in 14 groups, 0.473 %. Ensemble averaging assumes independent draws, and at this rate the violation is immaterial; it is reported so that it is not assumed to be zero.

The last lag of the grid is empty. 43,200 s is the flight-level duration bound, so no retained flight reaches it and the row is written with a null MSD and a zero count. It is dropped by every fit and every plot through the finiteness mask, and appears in `msd_curve.csv` for grid completeness.

Reading the table. Through `soaring.analysis.derived.stream_flights`, never by iterating Parquet row groups: a row group is a unit of storage and cuts across flights, so the direct reading counted 0.7 % of flights twice and truncated the longest segments, which are precisely the ones the long-lag end of the time-averaged curve rests on.

Chapter 4

Flight phases

This chapter mirrors Chapter 4. None of that code exists yet, so each entry is a placeholder rather than a populated record; each will gain its own implementation content — module names, sample sizes, reproduction commands — as the code is written.

4.1 Segmentation into flight phases

Not yet implemented; see Sec. 4.1.

4.2 Phase-resolved observables

Not yet implemented; see Sec. 4.2.

4.3 Reproducing the reference results

Not yet implemented; see Sec. 4.3.

4.4 Modelling and simulation

Not yet implemented; see Sec. 4.4.

4.5 Tooling

Largely underway already: see Sec. 4.5 and the reproducibility map, Table 1.1.